AS AMENDED
See Page No : 2, 3, 9, 47-49, 102-105

FINAL
TRIM AND STABILITY BOOKLET
OF
ARCADIA ZARAH
(OFFICIAL NO. MAR-2403-D)

Rev.	Description	Date	Prepared	Review

Issued for : Approval

Client:

K.B. SHIPPING & CO.(JAMNAGAR)

Builder:

CHISTI MARINE ENGINEERING WORKS

Project:

PONTOON
(YARD NO.CHISTI/023)

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GENERAL PARTICULARS

Name of the Vessel	ARCADIA ZARAH
Client	K.B. SHIPPING & CO. (JAMNAGAR)
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/023
Type of Vessel	PONTOON
Official No.	MAR-2403-D
Port of Registry	MORMUGAO
Class Notation	卐SU PONTOON, DECK STRENGTHENED FOR 10T/m^2 FOR CRANE OPERATION

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	2.200 M
Draft (Ext.)	2.212 M
Displacement(Ext.)	1569.15 T
Lightship	373.37 T
Deadweight	1195.78 T
Gross Tonnage	564
Net Tonnage	169

NOTE :

During unmanned tow minimum forward draft of 2.0 m to be maintained at all times for compliance with slamming requirements.(Refer "Strctural plan" approved on 05 jul 2022 for details)

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.

2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

3) The minimum range of stability should be:
   For L <= 100m: 20^0
   For L >= 150m: 15^0
   For intermediate length by interpolation

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.

2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.

3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.

4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.

5. In this booklet hydrostatic particulars and cross curves are given at following trims.

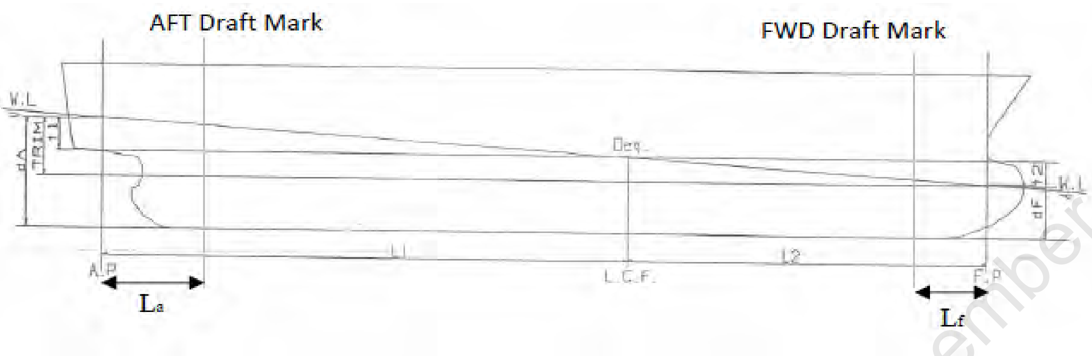
   -0.50m,-0.25m, 0.00m,0.25m, 0.50m

   These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

a) Longitudinal Centre of gravity of the ship calculated as follows:

$$LCG = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

b) Longitudinal centre of buoyancy, LCB

c) Displacement, Δ

d) Moment to change trim, MCT1cm

e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$Trim = \frac{\Delta * (LCB-LCG)}{MCT1cm * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{LCF * Trim}{LBP.} + T\ (extreme)$$

$$T_{FP} = T_{AP} - Trim$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{Trim * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{Trim * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

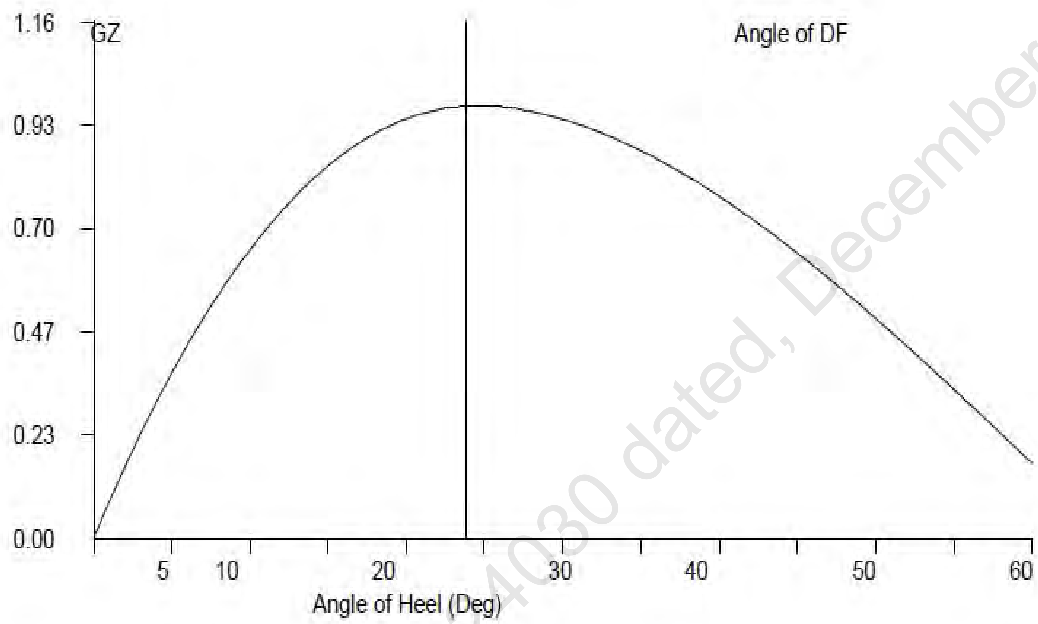
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

1) GZ = KN - KGo Sin (theta)
2) Area under 0-10 = 10*0.01745*[GZ0 + GZ10] / 2
3) Area under 10-20 = 10*0.01745*[GZ10 + GZ20] / 2
4) Area under 20-30 = 10*0.01745*[GZ20 + GZ30] / 2
5) Area under 30-40 = 10*0.01745*[GZ30 + GZ40] / 2
6) Area under 40-50 = 10*0.01745*[GZ40 + GZ50] / 2
7) Area under 50-60 = 10*0.01745*[GZ50 + GZ60] / 2
8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples

1 Loading the Vessel

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 5: Lightship + Crane + Ballast + Deck Cargo

Displacement	1569.15	Tonnes
LCG	24.993	M Fwd of AP
VCG (KG)	3.461	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

Free Surface moment for a Tank = Moment of Inertia (M^4) for a Tank x Density of Liquid in Tank

KGo (Fluid) = KG (Solid) + Free Surface Correction.

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

Trim = (LCB-LCG) X displacement / (MCT x 100) + Trim*

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
03_PFWBTK.C	132.30	1.000	132.30	16.000	2116.80	0.000	0.00	1.500	198.45	91.87
05_PFWBTK.C	132.30	1.000	132.30	34.000	4498.20	0.000	0.00	1.500	198.45	91.87
TOTAL TANK			264.60	25.000	6615.00	0.000	0.00	1.500	396.90	183.74

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			373.37	24.970	9,323.05	0.000	0.00	3.000	1,120.11
CRANE			250.00	25.000	6,250.00	0.000	0.00	6.120	1,530.00
DECK CARGO			681.18	25.000	17,029.50	0.000	0.00	3.500	2,384.13
TOTAL FIXED WEIGHT			1,304.55	24.991	32,602.55	0.000	0.00	3.859	5,034.24

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			264.600	25.000	6615.000	0.000	0.000	1.500	396.900
TOTAL FIXED WEIGHT			1304.550	24.991	32602.549	0.000	0.000	3.859	5034.240
TOTAL WEIGHT			1569.150	24.993	39217.549	0.000	0.000	3.461	5431.140

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Free Surface Correction for VCG = FSM/Displacement

	0.117	m
VCGc	3.578	m

From the MaxVCG table,

Displacement	1569.15	T
Allowable Max VCG	5.093	m

VCGc < Maxvcg PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	1569.150	T
Draft ,Tm @ 25.0f =	2.212	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	32.380	T-m/cm
KMT =	10.344	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.003 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.0 (MS),Tmid	2.212	m
Draft @ 50.0f (FP),Tfwd	=Tm - Trim/2	
	2.210	m
Draft@0(AP),Taft	=Tm + Trim/2	
	2.214	m

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	32.380	T-m	Draft (Mean)	2.212 m
LCB	25.000	m	Draft AP (USK)	2.214 m
			Draft FP (USK)	2.210 m
Metacentric Height				
KMt	10.344	m		
KG	3.461	m		
KG0	3.578	m		
GMt	6.766	m	**Trim**	0.003 m

The condition shown above is first worked
using Even Keel hydrostatic perticulars
Program then interpolates the hydrostatic perticulars
from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			**Drafts & Trims**	
MCT	31.300	T-m	Draft (Mean)	2.212 m
LCB	24.993	m	Draft AP (USK)	2.214 m
			Draft FP (USK)	2.210 m
Metacentric Height				
KMt	10.344	m		
KG	3.461	m		
KG0	3.578	m		
GMt	6.766	m	**Trim**	0.004 m

Trim is calculaed as follows:
Trim = (LCB - LCG) X displacement / (MCT X 100) + Trim*
Where Trim* is trim at which hydrostatics were calculated.

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Intact Stability (Primary Criterion)

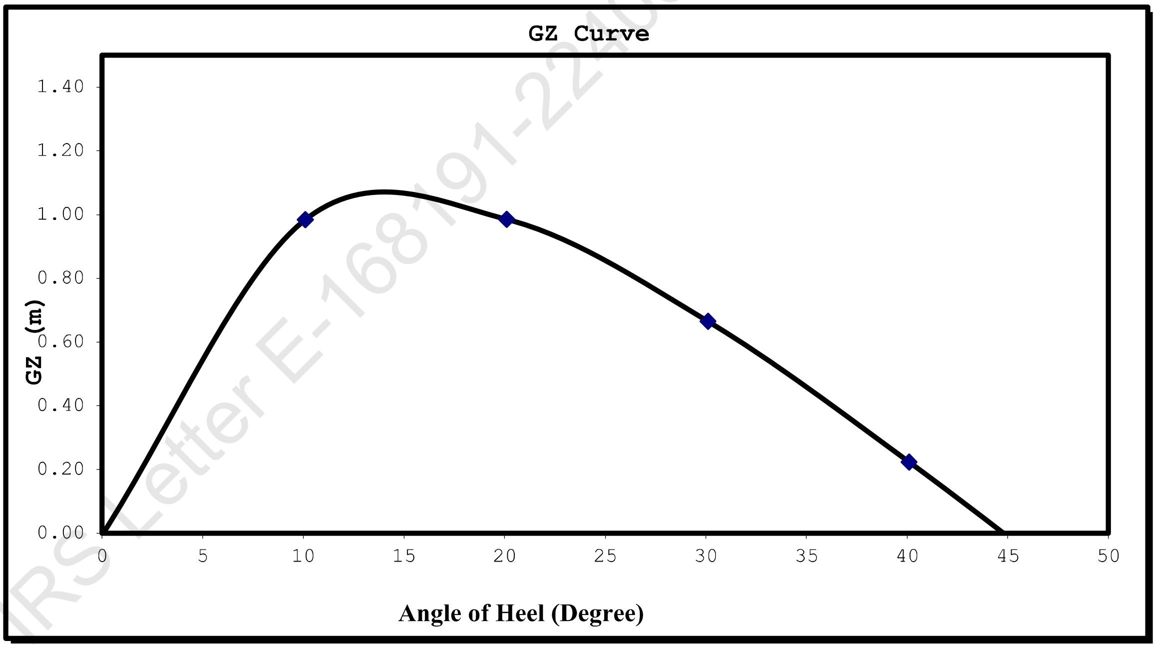
			Minimum Required
Displacement	1569.15	T	
KG0	3.578	M	
GMt	6.766	M	0.150 M

Residual Arm Curve

Angle (Deg)		0	10	20	30	40	50
KN	(M)	-0.006	1.611	2.215	2.459	2.528	2.487
GZ	(M)	-0.006	0.984	0.985	0.665	0.223	-0.258

	Computed values		Minimum Required	
Residual Area from abs 0 deg to MAxRA	0.1781	m-rad	0.080	m-rad
Angle from Equilibrium to RAZero or flood	44.69	m-rad	20.00	Deg.
Angle from Equilibrium to 50% DK Imm Angle	3.03	m-rad	0.00	Deg.

Plot GZ values against the angles of heel as above.
At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates.Join the point to the origin.
This is the Tangent to the GZ Curve at the Origin.Find the Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding
If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	2.212	M
LCF from AP	25.000	M
LCB from AP	24.993	M
MCT	31.300	T-M/Cm
Trim	0.004	M
Draft (AP)	2.214	M
Draft (FP)	2.210	M
KG	3.461	M
KGo	3.578	M
GoMt	6.766	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (Ext) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			373.370	24.970	9,323.049	0.000	0.000	3.000	1,120.110
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK									
TOTAL FIXED WEIGHT									
TOTAL WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

m

VCGc m

From the MaxVCG table,

Displacement T

Allowable Max VCG m

VCGc < Maxvcg

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft ,Tm @ 25.0f = m

LCB = m

LCF = m

MCT = T-m/cm

KMT = m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)

m,(+) is aft trim,(-) is fwd trim

Determining the Forward perendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.0 (MS),Tmid m

Draft @ 50.0f (FP),Tfwd =Tm - Trim/2

m

Draft@0(AP),Taft =Tm + Trim/2

m

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floatation

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

MULTIPLY BY	TO CONVERT FROM	TO OBTAIN	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram’s	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	TO OBTAIN	**TO CONVERT FROM**	**MULTIPLY**

Relation between Weight and volume

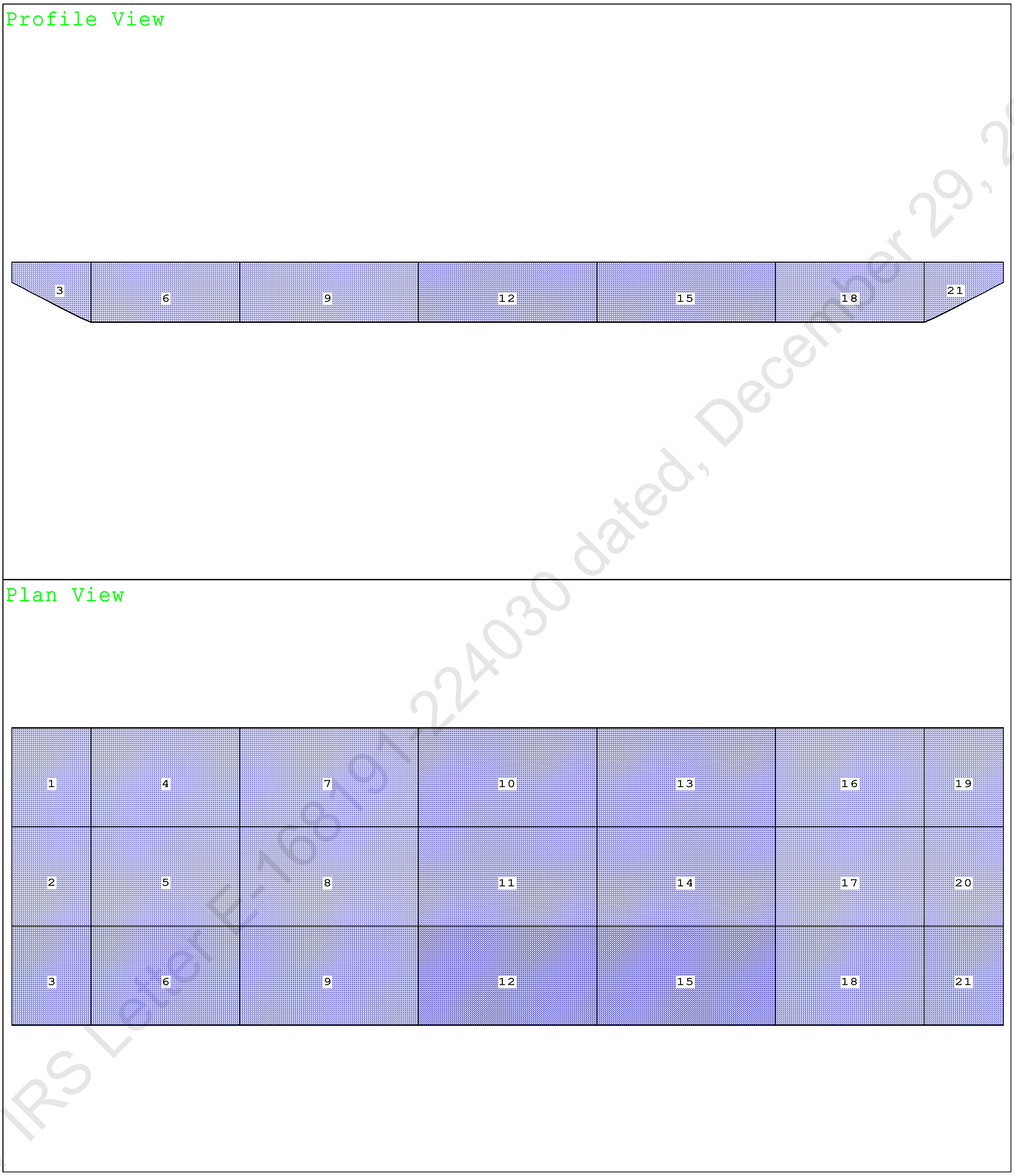
1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill off this opening is a `potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
   If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

   As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

CAPACITY PLAN AND TABLE



Tanks

1	01_VOIDTK.P.100%	FRESH WATER
2	01_VOIDTK.C.100%	FRESH WATER
3	01_VOIDTK.S.100%	FRESH WATER
4	02_VOIDTK.P.100%	FRESH WATER
5	02_VOIDTK.C.100%	FRESH WATER
6	02_VOIDTK.S.100%	FRESH WATER
7	03_VOIDTK.P.100%	FRESH WATER
8	03_PFWBTK.C.100%	FRESH WATER
9	03_VOIDTK.S.100%	FRESH WATER
10	04_VOIDTK.P.100%	FRESH WATER
11	04_VOIDTK.C.100%	FRESH WATER
12	04_VOIDTK.S.100%	FRESH WATER
13	05_VOIDTK.P.100%	FRESH WATER
14	05_PFWBTK.C.100%	FRESH WATER
15	05_VOIDTK.S.100%	FRESH WATER
16	06_VOIDTK.P.100%	FRESH WATER
17	06_VOIDTK.C.100%	FRESH WATER
18	06_VOIDTK.S.100%	FRESH WATER
19	07_VOIDTK.P.100%	FRESH WATER
20	07_VOIDTK.C.100%	FRESH WATER
21	07_VOIDTK.S.100%	FRESH WATER

TANK STATUS
Trim: zero, Heel: zero

Part	Cu.M.	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
01_VOIDTK.P	39.1	1.000	39.13	2.339f	4.971p	1.914	40.83*
01_VOIDTK.C	39.6	1.000	39.61	2.335f	0.000	1.904	40.83*
01_VOIDTK.S	39.1	1.000	39.13	2.339f	4.971s	1.914	40.83*
02_VOIDTK.P	109.3	1.000	109.34	7.750f	4.981p	1.511	76.56*
02_VOIDTK.C	110.2	1.000	110.25	7.750f	0.000	1.500	76.56*
02_VOIDTK.S	109.3	1.000	109.34	7.750f	4.981s	1.511	76.56*
03_VOIDTK.P	131.2	1.000	131.21	16.000f	4.981p	1.511	91.88*
03_PFWBTK.C	132.3	1.000	132.30	16.000f	0.000	1.500	91.87*
03_VOIDTK.S	131.2	1.000	131.21	16.000f	4.981s	1.511	91.88*
04_VOIDTK.P	131.2	1.000	131.21	25.000f	4.981p	1.511	91.88*
04_VOIDTK.C	132.3	1.000	132.30	25.000f	0.000	1.500	91.87*
04_VOIDTK.S	131.2	1.000	131.21	25.000f	4.981s	1.511	91.88*
05_VOIDTK.P	131.2	1.000	131.21	34.000f	4.981p	1.511	91.88*
05_PFWBTK.C	132.3	1.000	132.30	34.000f	0.000	1.500	91.87*
05_VOIDTK.S	131.2	1.000	131.21	34.000f	4.981s	1.511	91.88*
06_VOIDTK.P	109.3	1.000	109.34	42.250f	4.981p	1.511	76.56*
06_VOIDTK.C	110.2	1.000	110.25	42.250f	0.000	1.500	76.56*
06_VOIDTK.S	109.3	1.000	109.34	42.250f	4.981s	1.511	76.56*
07_VOIDTK.P	39.1	1.000	39.13	47.661f	4.971p	1.914	40.83*
07_VOIDTK.C	39.6	1.000	39.61	47.665f	0.000	1.904	40.83*
07_VOIDTK.S	39.1	1.000	39.13	47.661f	4.971s	1.914	40.83*
Total Tanks--------->			2,077.75	25.000f	0.000	1.553	1531.25

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

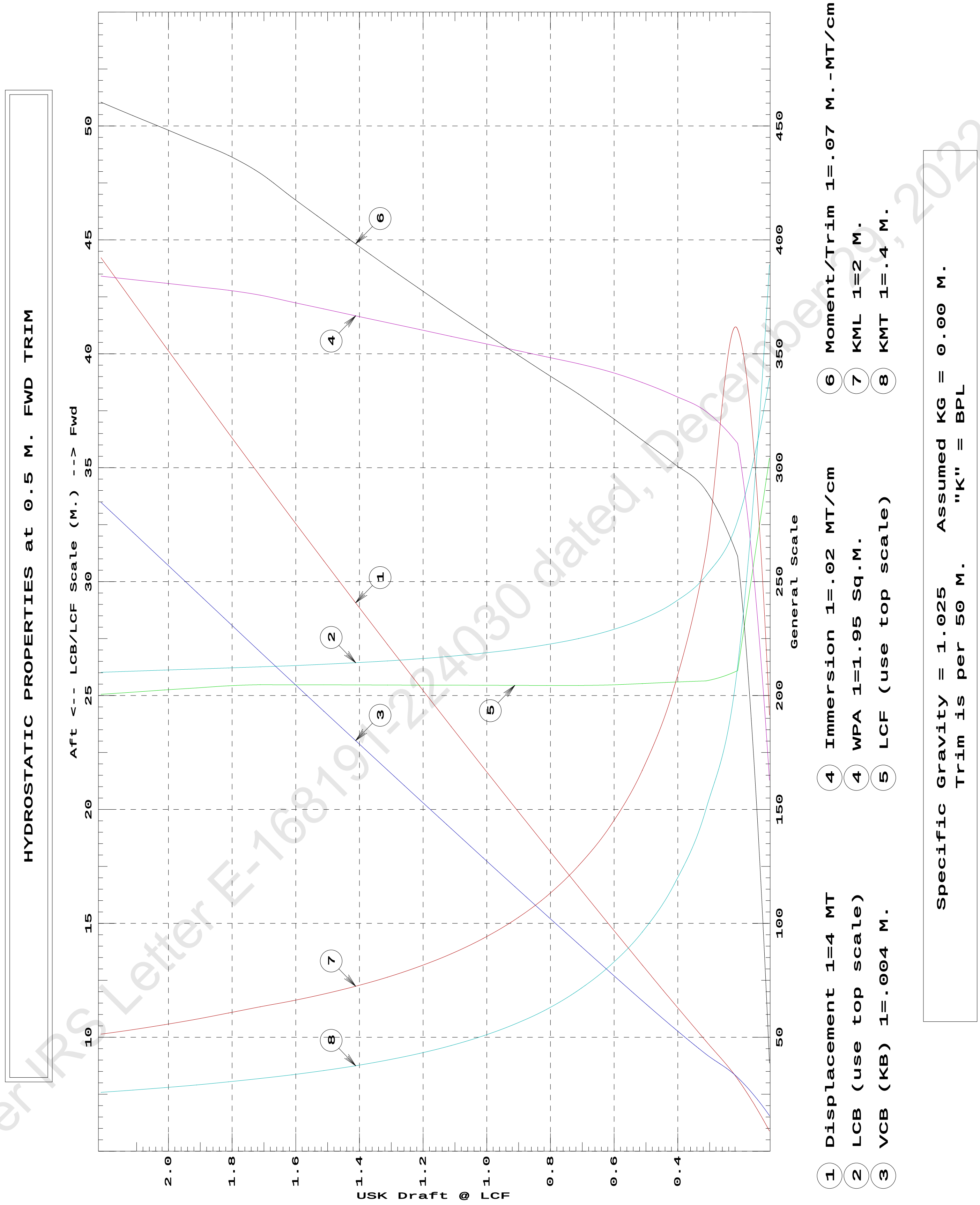
HYDROSTATIC PARTICULARS

HYDROSTATIC PROPERTIES
Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	35.31	38.933f	0.062	3.26	35.422f	2.70	382.72	155.640
0.212	126.67	32.680f	0.129	6.21	26.091f	18.28	721.37	85.218
0.312	193.60	30.258f	0.171	6.49	25.644f	20.31	524.43	59.944
0.412	259.63	29.079f	0.216	6.63	25.587f	21.12	406.76	46.782
0.512	326.98	28.354f	0.264	6.75	25.525f	21.85	334.10	38.458
0.612	395.34	27.859f	0.314	6.84	25.465f	22.58	285.53	32.636
0.712	464.24	27.501f	0.364	6.91	25.439f	23.27	250.56	28.275
0.812	533.64	27.233f	0.414	6.97	25.443f	23.89	223.80	24.906
0.912	603.65	27.026f	0.465	7.03	25.446f	24.52	203.11	22.305
1.012	674.25	26.861f	0.515	7.09	25.451f	25.17	186.64	20.238
1.112	745.46	26.726f	0.566	7.15	25.454f	25.83	173.21	18.554
1.212	817.28	26.614f	0.618	7.21	25.457f	26.50	162.12	17.166
1.312	889.68	26.520f	0.669	7.27	25.462f	27.18	152.75	15.996
1.412	962.69	26.440f	0.721	7.33	25.465f	27.88	144.78	15.003
1.512	1,036.38	26.371f	0.773	7.39	25.470f	28.59	137.93	14.149
1.612	1,110.58	26.311f	0.825	7.45	25.473f	29.31	131.94	13.408
1.712	1,185.40	26.258f	0.877	7.51	25.476f	30.05	126.74	12.763
1.812	1,261.19	26.210f	0.929	7.56	25.427f	30.60	121.30	12.168
1.912	1,337.62	26.163f	0.982	7.59	25.335f	31.00	115.89	11.627
2.012	1,414.37	26.115f	1.034	7.62	25.243f	31.42	111.05	11.147
2.112	1,491.44	26.068f	1.087	7.65	25.148f	31.82	106.66	10.716
2.212	1,568.83	26.020f	1.139	7.68	25.053f	32.23	102.71	10.332

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=2 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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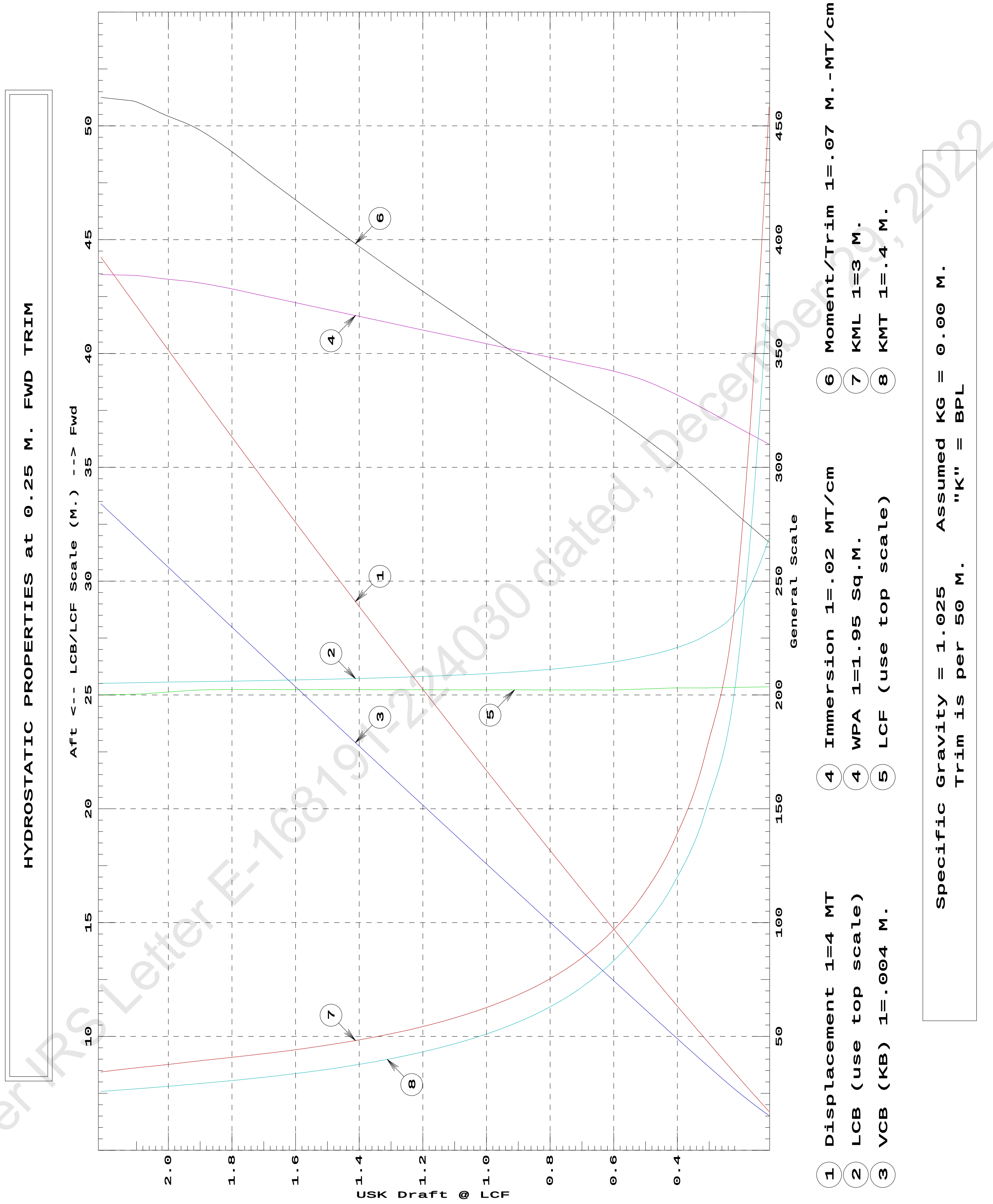
GHS 16.50

HYDROSTATIC PROPERTIES
Trim: Fwd 0.250/50.000, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	67.99	31.878f	0.061	6.20	25.355f	18.70	1375.02	154.077
0.212	130.85	28.739f	0.104	6.35	25.333f	19.54	746.69	84.364
0.312	195.25	27.612f	0.152	6.51	25.309f	20.41	522.70	59.562
0.412	261.01	27.032f	0.202	6.65	25.309f	21.22	406.53	46.823
0.512	328.33	26.675f	0.253	6.77	25.258f	21.96	334.41	38.687
0.612	396.62	26.426f	0.304	6.85	25.218f	22.65	285.59	32.693
0.712	465.44	26.248f	0.355	6.91	25.219f	23.26	249.88	28.193
0.812	534.86	26.114f	0.406	6.97	25.221f	23.89	223.30	24.847
0.912	604.89	26.011f	0.458	7.03	25.223f	24.52	202.66	22.253
1.012	675.51	25.929f	0.509	7.09	25.225f	25.16	186.26	20.193
1.112	746.74	25.862f	0.561	7.15	25.227f	25.83	172.93	18.521
1.212	818.57	25.806f	0.613	7.21	25.229f	26.49	161.82	17.132
1.312	891.00	25.759f	0.664	7.27	25.230f	27.18	152.52	15.970
1.412	964.03	25.719f	0.716	7.33	25.233f	27.88	144.57	14.979
1.512	1,037.66	25.685f	0.768	7.39	25.234f	28.58	137.73	14.127
1.612	1,111.90	25.655f	0.821	7.45	25.237f	29.31	131.80	13.391
1.712	1,186.73	25.629f	0.873	7.51	25.238f	30.04	126.56	12.744
1.812	1,262.18	25.605f	0.925	7.57	25.240f	30.79	121.97	12.177
1.912	1,338.33	25.584f	0.978	7.62	25.209f	31.43	117.41	11.662
2.012	1,415.08	25.561f	1.031	7.66	25.117f	31.84	112.52	11.180
2.112	1,492.21	25.536f	1.083	7.69	25.024f	32.27	108.13	10.753
2.212	1,569.14	25.510f	1.136	7.69	25.009f	32.37	103.16	10.342

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.


HYDROSTATIC PROPERTIES at 0.25 M. FWD TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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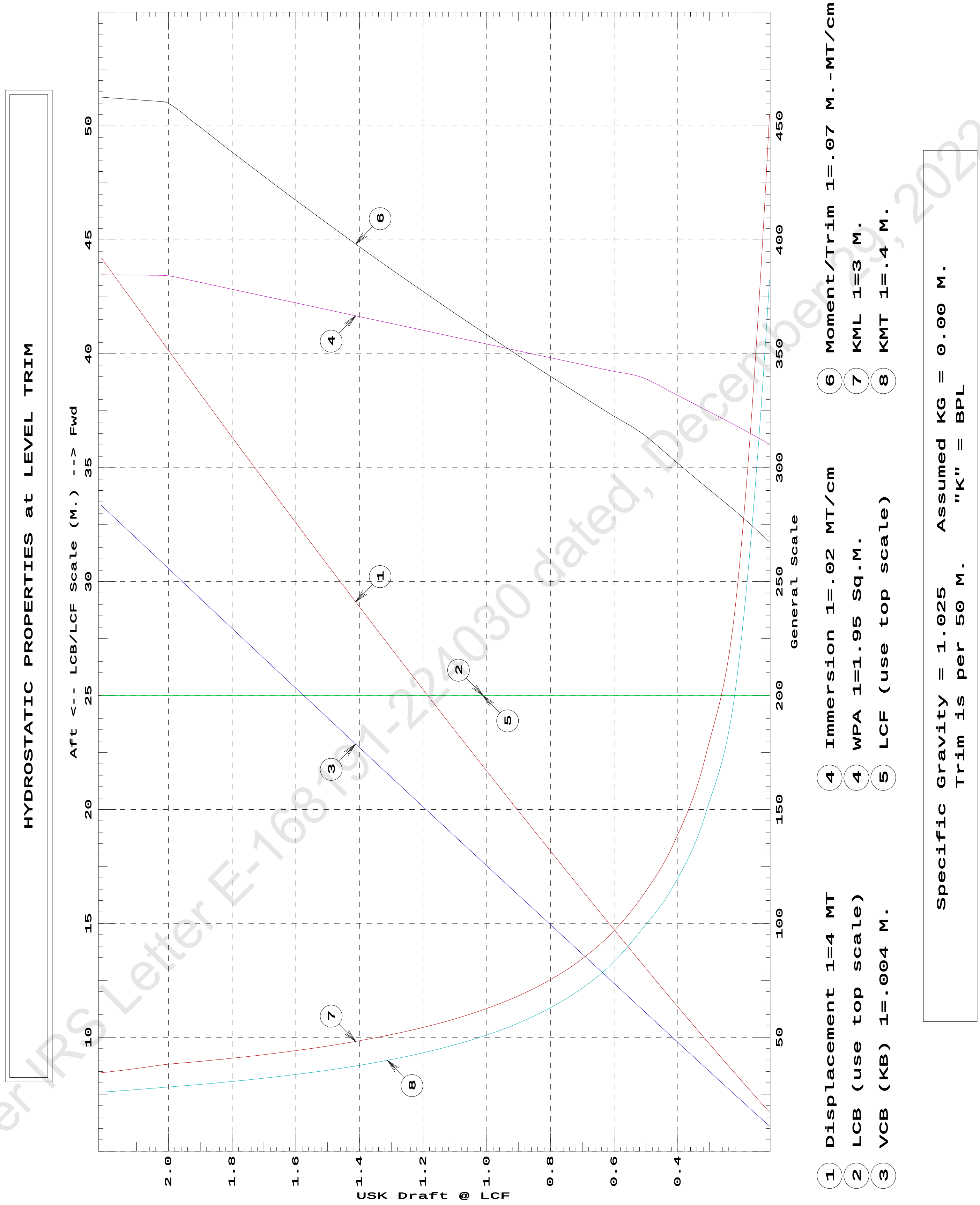
GHS 16.50

HYDROSTATIC PROPERTIES

No Trim, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	68.53	25.000f	0.044	6.21	25.000f	18.72	1365.62	152.943
0.212	131.38	25.000f	0.095	6.36	25.000f	19.61	746.26	84.110
0.312	195.72	25.000f	0.146	6.51	25.000f	20.41	521.46	59.400
0.412	261.52	25.000f	0.197	6.65	25.000f	21.24	406.00	46.750
0.512	328.77	25.000f	0.249	6.79	25.000f	22.05	335.40	38.975
0.612	396.99	25.000f	0.301	6.85	25.000f	22.65	285.27	32.654
0.712	465.83	25.000f	0.352	6.91	25.000f	23.27	249.73	28.175
0.812	535.26	25.000f	0.404	6.97	25.000f	23.88	223.09	24.822
0.912	605.29	25.000f	0.456	7.03	25.000f	24.52	202.53	22.237
1.012	675.92	25.000f	0.507	7.09	25.000f	25.16	186.15	20.181
1.112	747.15	25.000f	0.559	7.15	25.000f	25.82	172.79	18.506
1.212	818.99	25.000f	0.611	7.21	25.000f	26.50	161.77	17.125
1.312	891.43	25.000f	0.663	7.27	25.000f	27.18	152.43	15.959
1.412	964.46	25.000f	0.715	7.33	25.000f	27.87	144.50	14.970
1.512	1,038.11	25.000f	0.767	7.39	25.000f	28.59	137.68	14.122
1.612	1,112.35	25.000f	0.819	7.45	25.000f	29.30	131.72	13.382
1.712	1,187.19	25.000f	0.872	7.51	25.000f	30.04	126.53	12.740
1.812	1,262.64	25.000f	0.924	7.57	25.000f	30.79	121.92	12.171
1.912	1,338.68	25.000f	0.977	7.63	25.000f	31.55	117.84	11.669
2.012	1,415.33	25.000f	1.029	7.69	25.000f	32.24	113.91	11.217
2.112	1,492.22	25.000f	1.082	7.69	25.000f	32.31	108.27	10.755
2.212	1,569.15	25.000f	1.134	7.69	25.000f	32.38	103.19	10.344

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK.


HYDROSTATIC PROPERTIES at LEVEL TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

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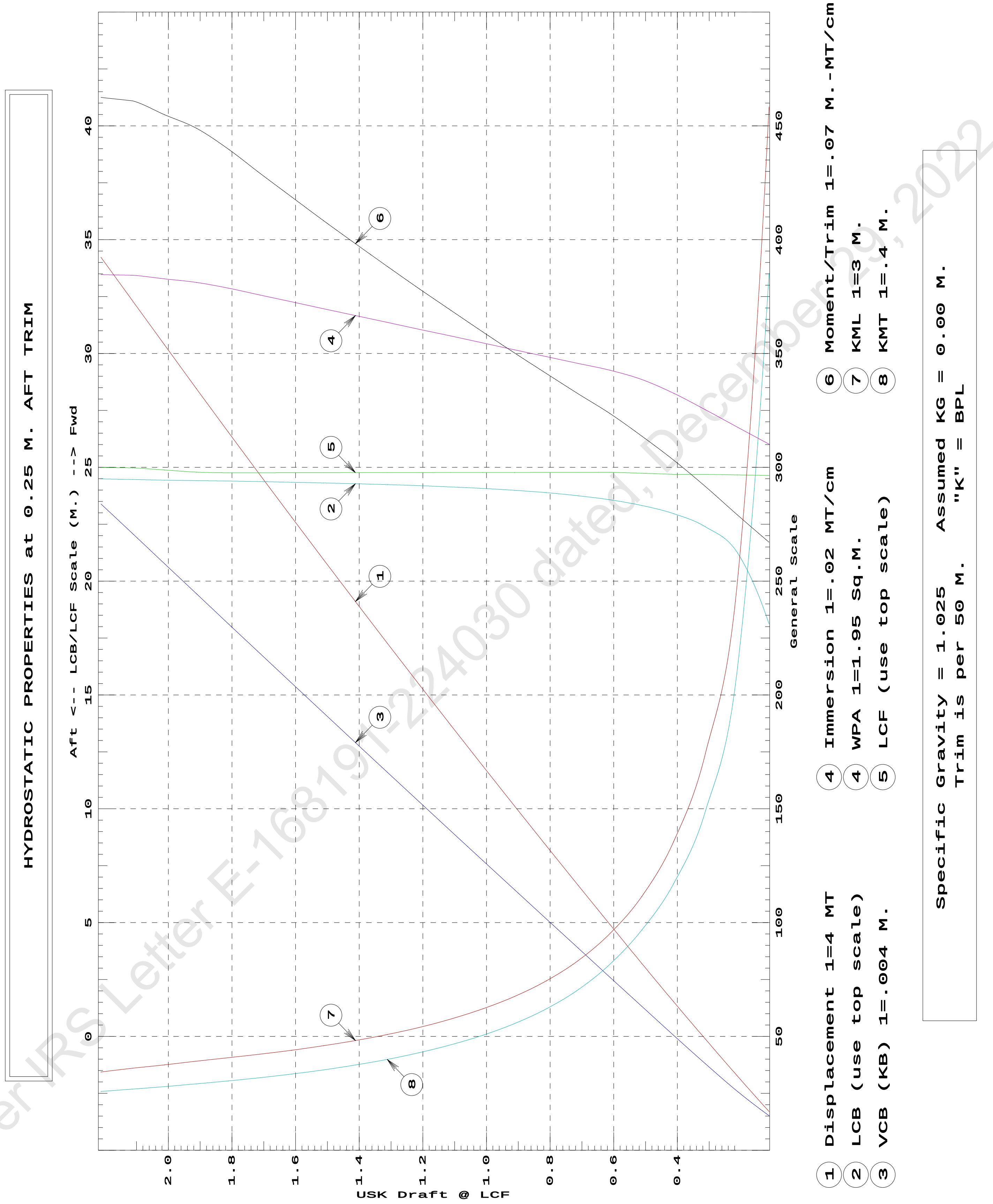
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Aft 0.250/50.000, No Heel, Fixed VCG = 0.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	67.99	18.121f	0.061	6.20	24.645f	18.70	1375.05	154.081
0.212	130.84	21.260f	0.104	6.35	24.667f	19.54	746.72	84.368
0.312	195.22	22.387f	0.152	6.51	24.691f	20.41	522.75	59.569
0.412	260.99	22.967f	0.202	6.65	24.691f	21.22	406.56	46.826
0.512	328.32	23.325f	0.253	6.77	24.742f	21.96	334.41	38.687
0.612	396.63	23.574f	0.304	6.85	24.782f	22.65	285.59	32.692
0.712	465.44	23.752f	0.355	6.91	24.781f	23.26	249.88	28.193
0.812	534.88	23.886f	0.406	6.97	24.779f	23.89	223.29	24.846
0.912	604.91	23.989f	0.458	7.03	24.777f	24.52	202.66	22.253
1.012	675.53	24.071f	0.509	7.09	24.775f	25.16	186.25	20.192
1.112	746.76	24.138f	0.561	7.15	24.773f	25.83	172.92	18.521
1.212	818.59	24.194f	0.613	7.21	24.771f	26.49	161.82	17.132
1.312	891.02	24.241f	0.664	7.27	24.770f	27.18	152.52	15.969
1.412	964.05	24.281f	0.716	7.33	24.767f	27.88	144.57	14.978
1.512	1,037.68	24.315f	0.768	7.39	24.766f	28.58	137.72	14.127
1.612	1,111.92	24.345f	0.821	7.45	24.764f	29.31	131.80	13.391
1.712	1,186.75	24.371f	0.873	7.51	24.762f	30.04	126.56	12.744
1.812	1,262.19	24.395f	0.925	7.57	24.760f	30.79	121.97	12.177
1.912	1,338.33	24.416f	0.978	7.62	24.791f	31.43	117.42	11.662
2.012	1,415.08	24.439f	1.031	7.66	24.883f	31.84	112.52	11.180
2.112	1,492.15	24.464f	1.083	7.69	24.976f	32.27	108.13	10.753
2.212	1,569.13	24.490f	1.136	7.69	24.991f	32.37	103.16	10.342

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK.


HYDROSTATIC PROPERTIES at 0.25 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=3 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

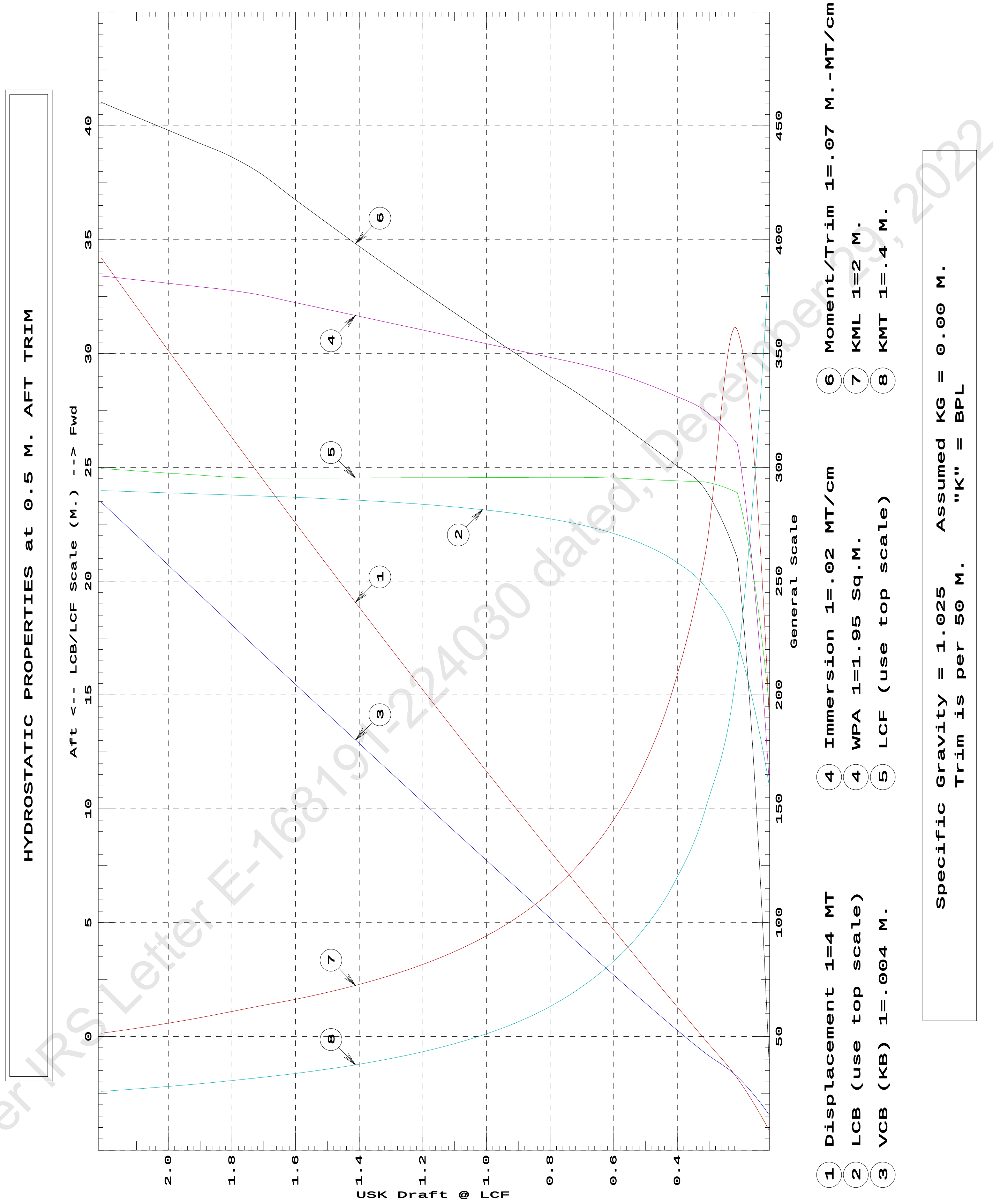
15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES
Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

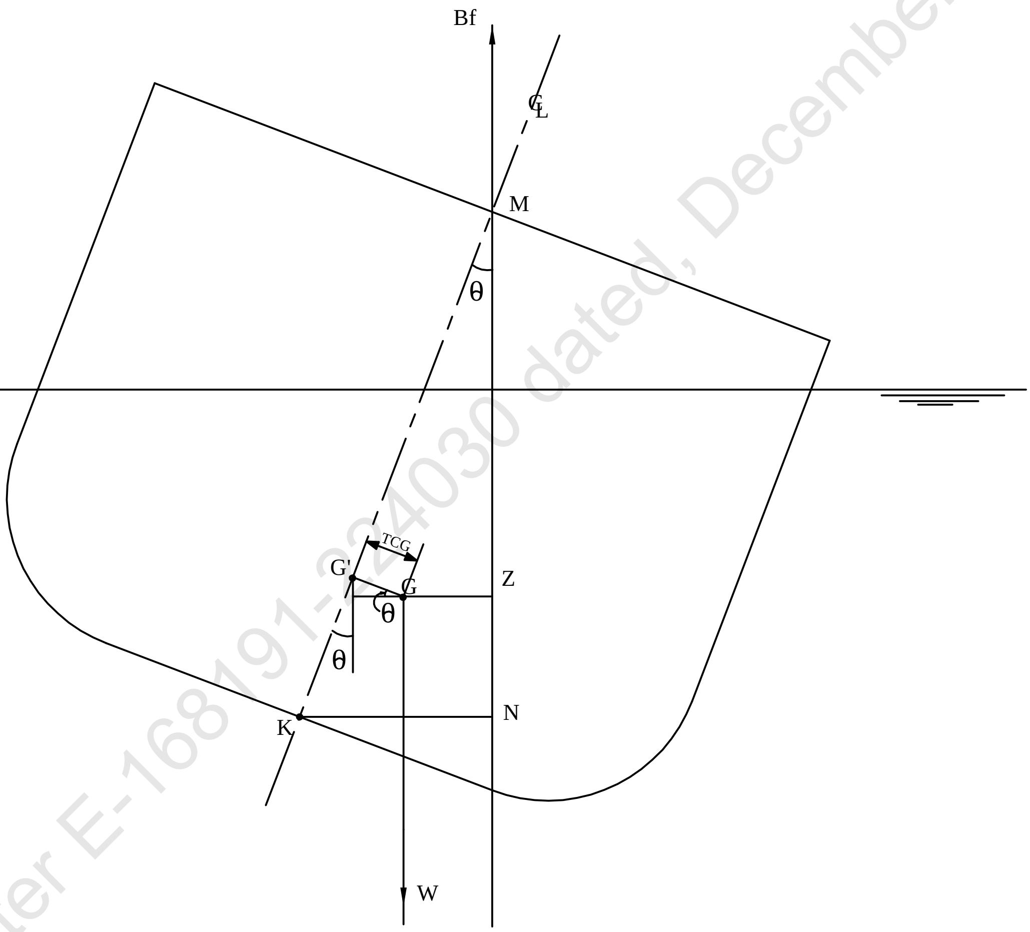
LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	KML	KMT
0.112	35.14	11.050f	0.062	3.25	14.553f	2.68	381.82	155.997
0.212	126.38	17.305f	0.129	6.21	23.886f	18.21	720.54	85.309
0.312	193.59	19.742f	0.171	6.49	24.356f	20.31	524.44	59.946
0.412	259.63	20.921f	0.216	6.63	24.413f	21.12	406.77	46.784
0.512	326.97	21.646f	0.264	6.75	24.475f	21.85	334.11	38.459
0.612	395.34	22.141f	0.314	6.84	24.535f	22.58	285.54	32.637
0.712	464.31	22.499f	0.364	6.91	24.561f	23.27	250.53	28.272
0.812	533.71	22.767f	0.414	6.97	24.557f	23.89	223.77	24.904
0.912	603.73	22.974f	0.465	7.03	24.554f	24.52	203.09	22.303
1.012	674.39	23.140f	0.515	7.09	24.549f	25.17	186.61	20.234
1.112	745.60	23.274f	0.567	7.15	24.546f	25.83	173.19	18.551
1.212	817.41	23.386f	0.618	7.21	24.543f	26.50	162.10	17.163
1.312	889.82	23.480f	0.669	7.27	24.538f	27.18	152.73	15.994
1.412	962.83	23.560f	0.721	7.33	24.535f	27.88	144.77	15.001
1.512	1,036.44	23.629f	0.773	7.39	24.530f	28.59	137.92	14.149
1.612	1,110.65	23.689f	0.825	7.45	24.527f	29.31	131.94	13.407
1.712	1,185.47	23.742f	0.877	7.51	24.524f	30.05	126.73	12.762
1.812	1,261.17	23.790f	0.929	7.56	24.573f	30.60	121.30	12.168
1.912	1,337.60	23.837f	0.982	7.59	24.665f	31.00	115.89	11.627
2.012	1,414.34	23.885f	1.034	7.62	24.757f	31.42	111.06	11.148
2.112	1,491.42	23.932f	1.087	7.65	24.852f	31.82	106.67	10.716
2.212	1,568.81	23.980f	1.139	7.68	24.947f	32.23	102.71	10.332

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

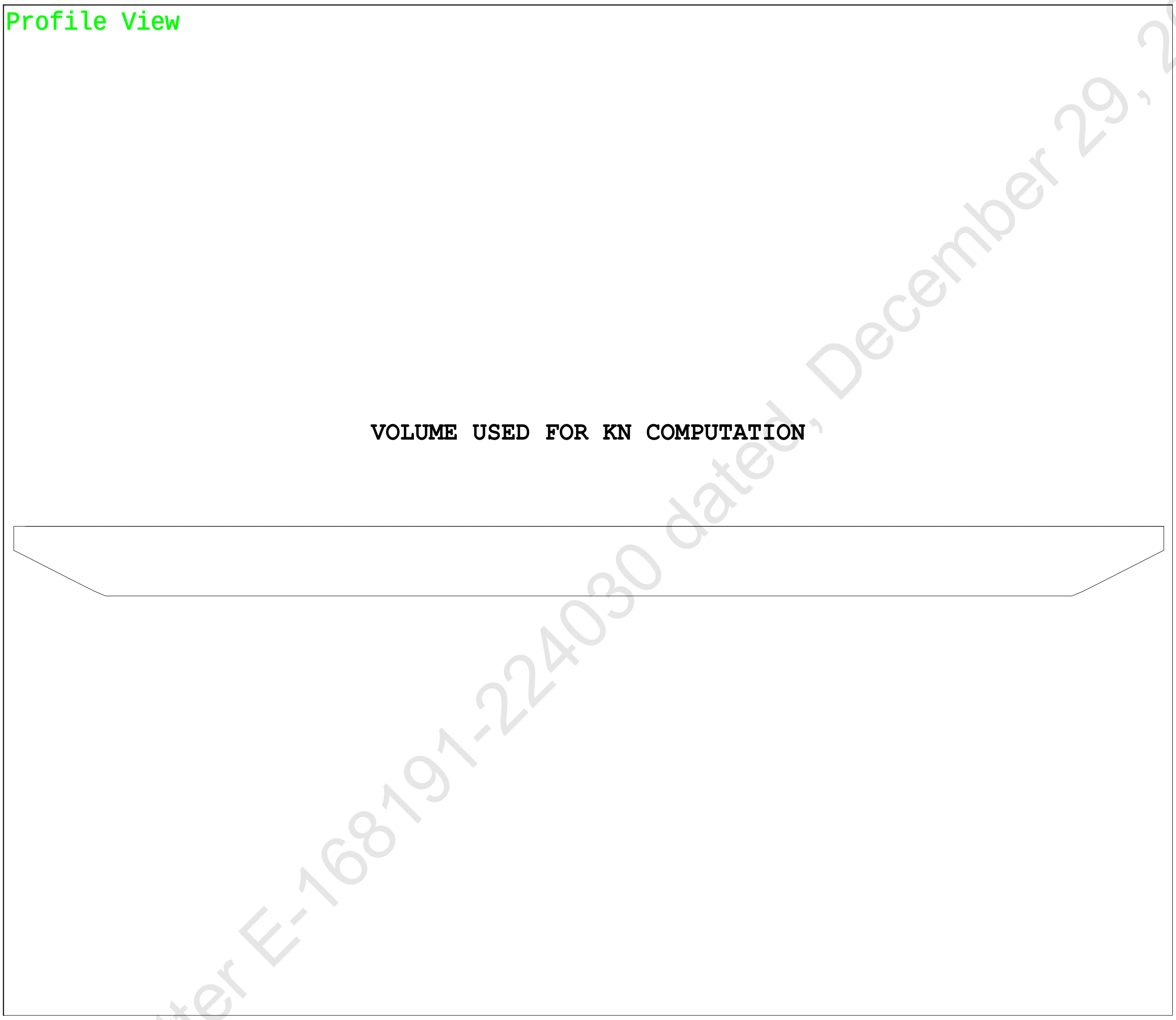
Draft is from USK.


HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM
Aft <-- LCB/LCF Scale (M.) --> Fwd
General Scale
USK Draft @ LCF
1 Displacement 1=4 MT
2 LCB (use top scale)
3 VCB (KB) 1=.004 M.
4 Immersion 1=.02 MT/cm
4 WPA 1=1.95 Sq.M.
5 LCF (use top scale)
6 Moment/Trim 1=.07 M.-MT/cm
7 KML 1=2 M.
8 KMT 1=.4 M.
Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

KN CONCEPT



$$GZ = KN - KG' \sin\theta - TCG \cos\theta$$

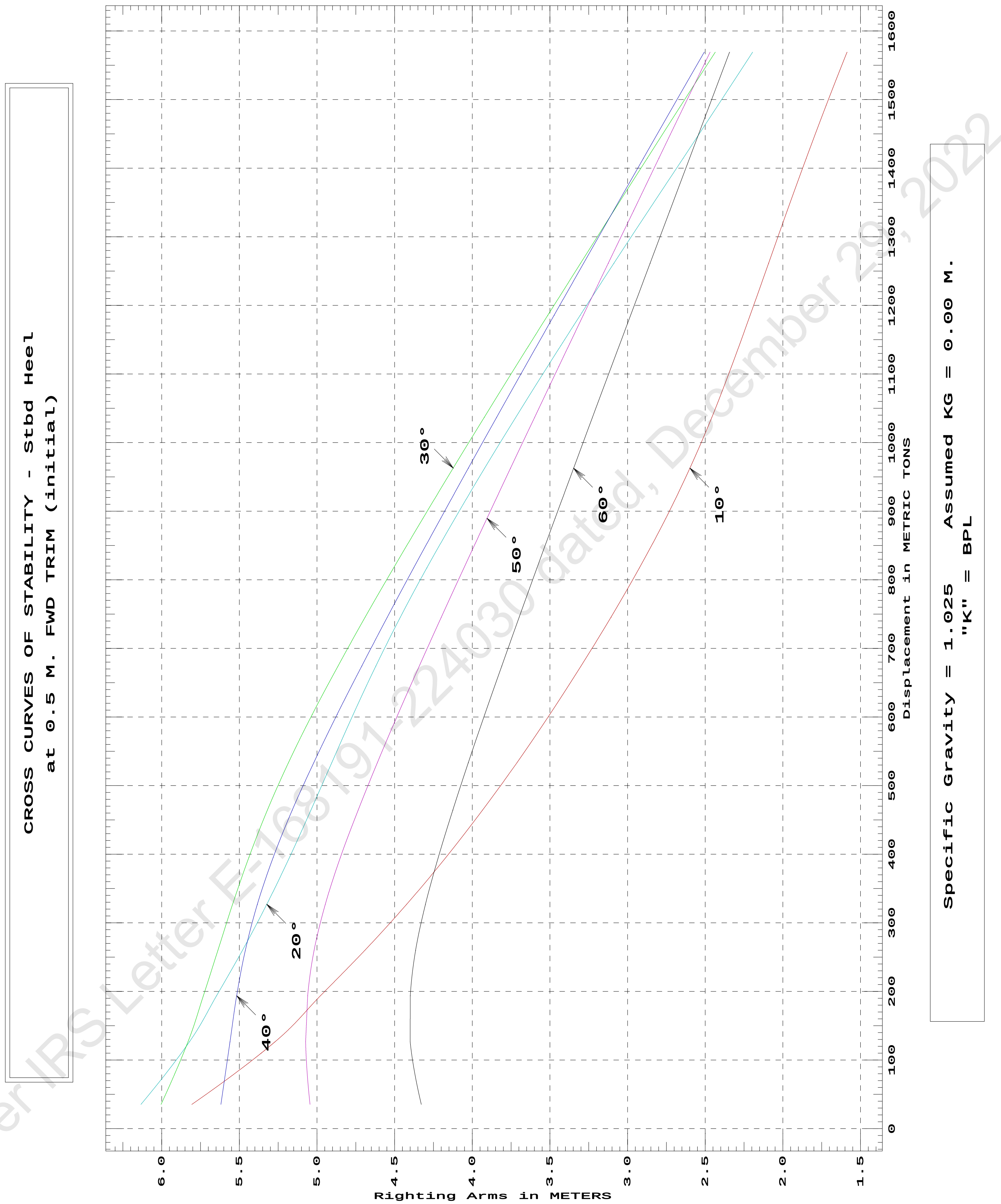

Profile View
VOLUME USED FOR KN COMPUTATION

CROSS CURVE TABLES

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.31	5.807s	6.134s	6.004s	5.619s	5.045s	4.329s
126.67	5.270s	5.821s	5.830s	5.560s	5.073s	4.400s
193.60	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.480s	5.639s	5.462s	5.019s	4.366s
326.98	4.423s	5.323s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.176s	5.439s	5.282s	4.849s	4.219s
464.24	3.934s	5.039s	5.319s	5.159s	4.735s	4.126s
533.64	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.65	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.25	3.300s	4.621s	4.865s	4.712s	4.342s	3.811s
745.46	3.107s	4.463s	4.690s	4.547s	4.200s	3.698s
817.28	2.924s	4.293s	4.507s	4.377s	4.054s	3.584s
889.68	2.754s	4.111s	4.318s	4.203s	3.905s	3.467s
962.69	2.599s	3.920s	4.123s	4.026s	3.754s	3.348s
1,036.38	2.458s	3.722s	3.924s	3.844s	3.600s	3.227s
1,110.58	2.331s	3.516s	3.721s	3.661s	3.444s	3.105s
1,185.40	2.211s	3.305s	3.514s	3.474s	3.286s	2.981s
1,261.19	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.62	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.37	1.850s	2.642s	2.873s	2.898s	2.799s	2.601s
1,491.44	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.83	1.588s	2.195s	2.436s	2.507s	2.469s	2.344s

Distances in METERS.---Specific Gravity = 1.025.-------------

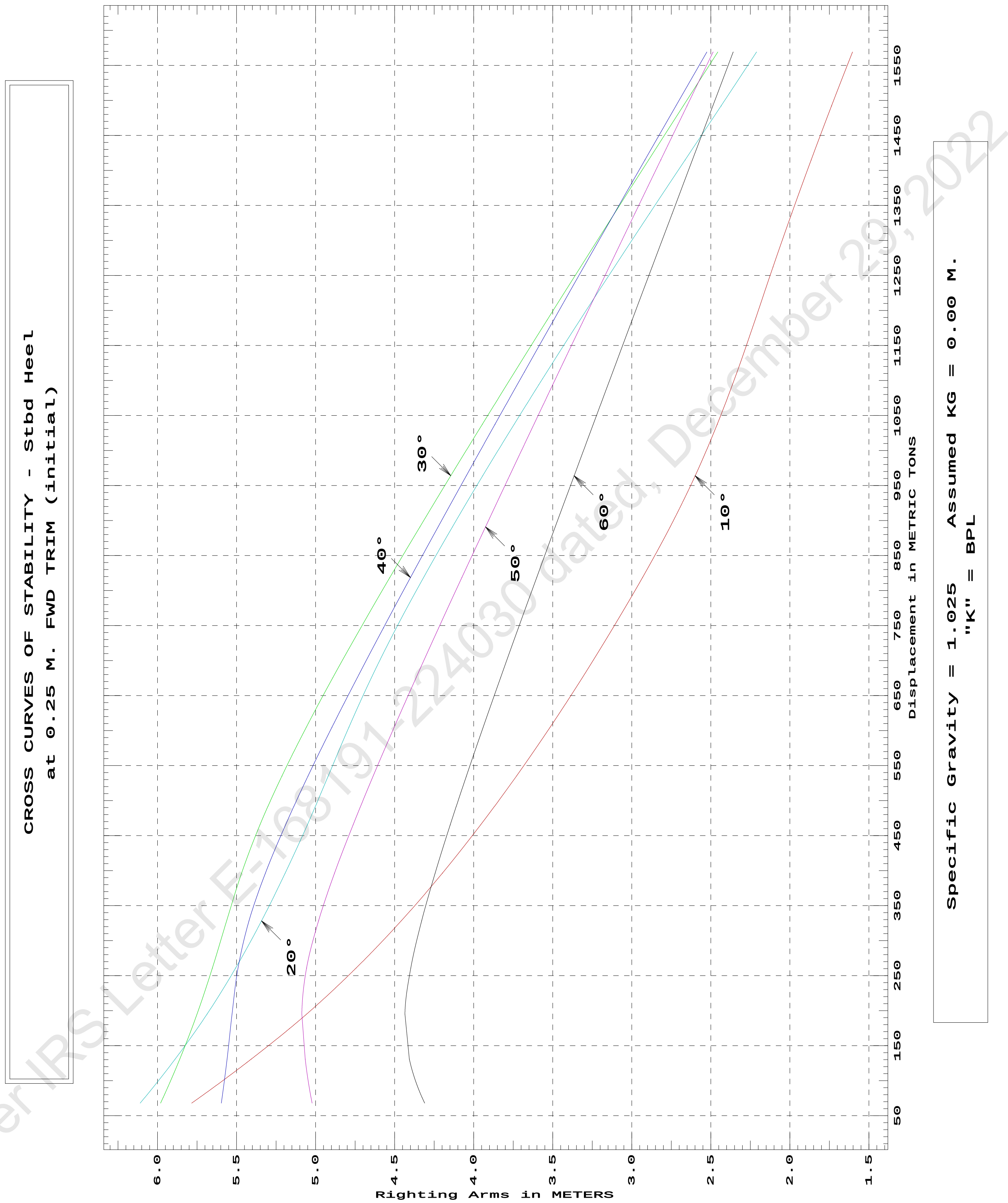

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
1500
1600
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.785s	6.110s	5.981s	5.596s	5.022s	4.310s
130.85	5.407s	5.889s	5.859s	5.560s	5.066s	4.407s
195.25	5.056s	5.686s	5.750s	5.529s	5.087s	4.435s
261.01	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.33	4.459s	5.342s	5.563s	5.421s	4.981s	4.329s
396.62	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.86	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.89	3.516s	4.789s	5.058s	4.896s	4.503s	3.940s
675.51	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.74	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.57	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.00	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.03	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.66	2.459s	3.744s	3.945s	3.865s	3.619s	3.244s
1,111.90	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.73	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.18	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.21	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.14	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s

Heel Angles in Degrees

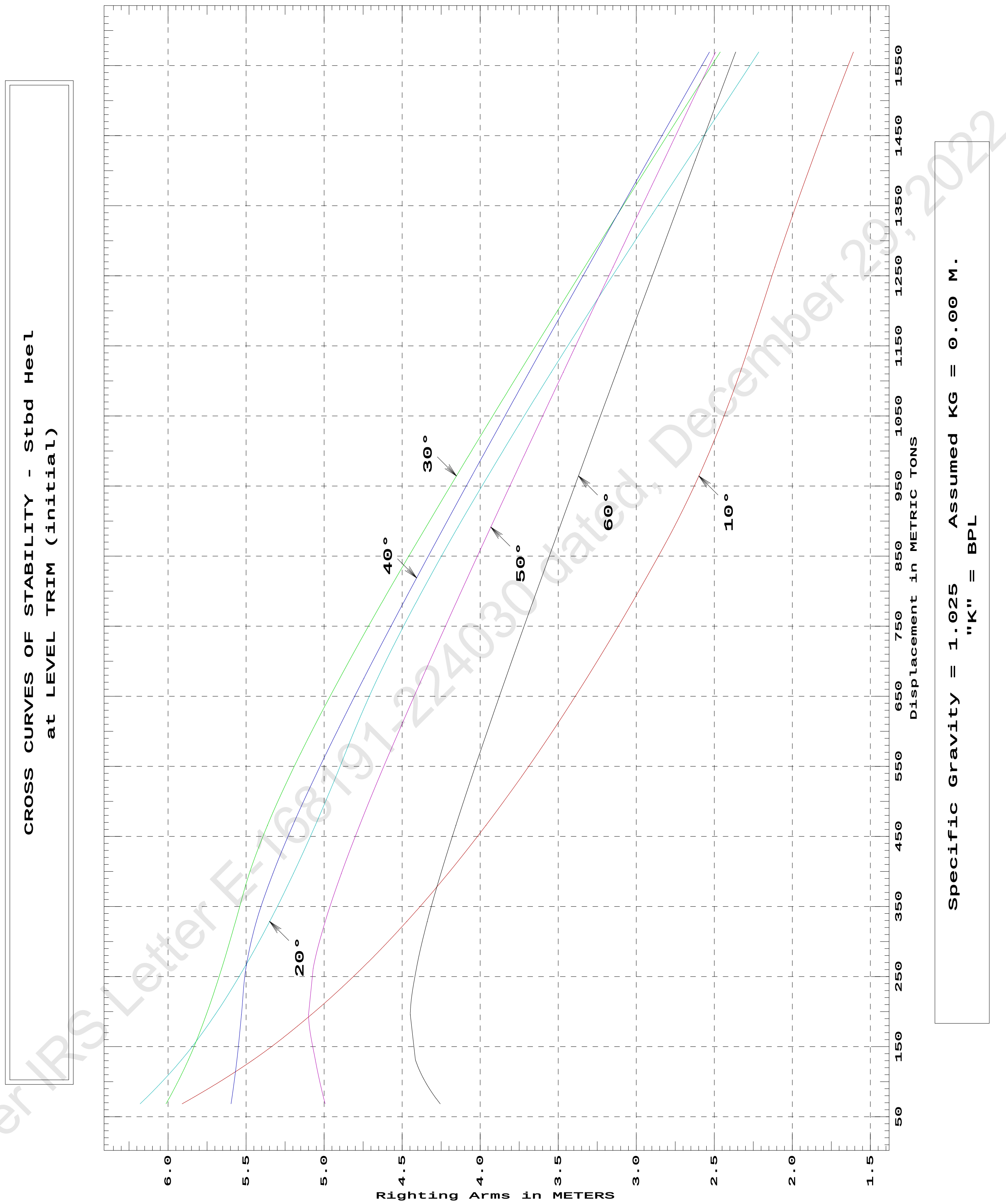
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. FWD TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: zero at zero heel (trim righting arm held at zero)

Displacement	Heel Angles in Degrees					
METRIC TONS	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
68.53	5.910s	6.179s	6.013s	5.596s	4.993s	4.256s
131.38	5.453s	5.913s	5.870s	5.558s	5.057s	4.416s
195.72	5.082s	5.699s	5.755s	5.529s	5.100s	4.448s
261.52	4.760s	5.514s	5.657s	5.502s	5.071s	4.409s
328.77	4.471s	5.349s	5.568s	5.434s	4.993s	4.339s
396.99	4.206s	5.197s	5.480s	5.328s	4.890s	4.252s
465.83	3.963s	5.057s	5.363s	5.200s	4.772s	4.156s
535.26	3.735s	4.925s	5.223s	5.058s	4.645s	4.054s
605.29	3.521s	4.798s	5.067s	4.905s	4.511s	3.947s
675.92	3.317s	4.657s	4.899s	4.745s	4.372s	3.836s
747.15	3.123s	4.498s	4.723s	4.579s	4.228s	3.723s
818.99	2.936s	4.326s	4.539s	4.407s	4.082s	3.607s
891.43	2.757s	4.143s	4.347s	4.232s	3.932s	3.490s
964.46	2.599s	3.950s	4.152s	4.053s	3.780s	3.370s
1,038.11	2.459s	3.751s	3.952s	3.871s	3.625s	3.249s
1,112.35	2.334s	3.545s	3.748s	3.687s	3.468s	3.126s
1,187.19	2.222s	3.333s	3.541s	3.499s	3.309s	3.002s
1,262.64	2.112s	3.116s	3.330s	3.310s	3.149s	2.877s
1,338.68	1.994s	2.895s	3.117s	3.117s	2.986s	2.750s
1,415.33	1.871s	2.670s	2.900s	2.923s	2.822s	2.621s
1,492.22	1.742s	2.441s	2.682s	2.728s	2.657s	2.493s
1,569.15	1.609s	2.216s	2.463s	2.531s	2.492s	2.363s

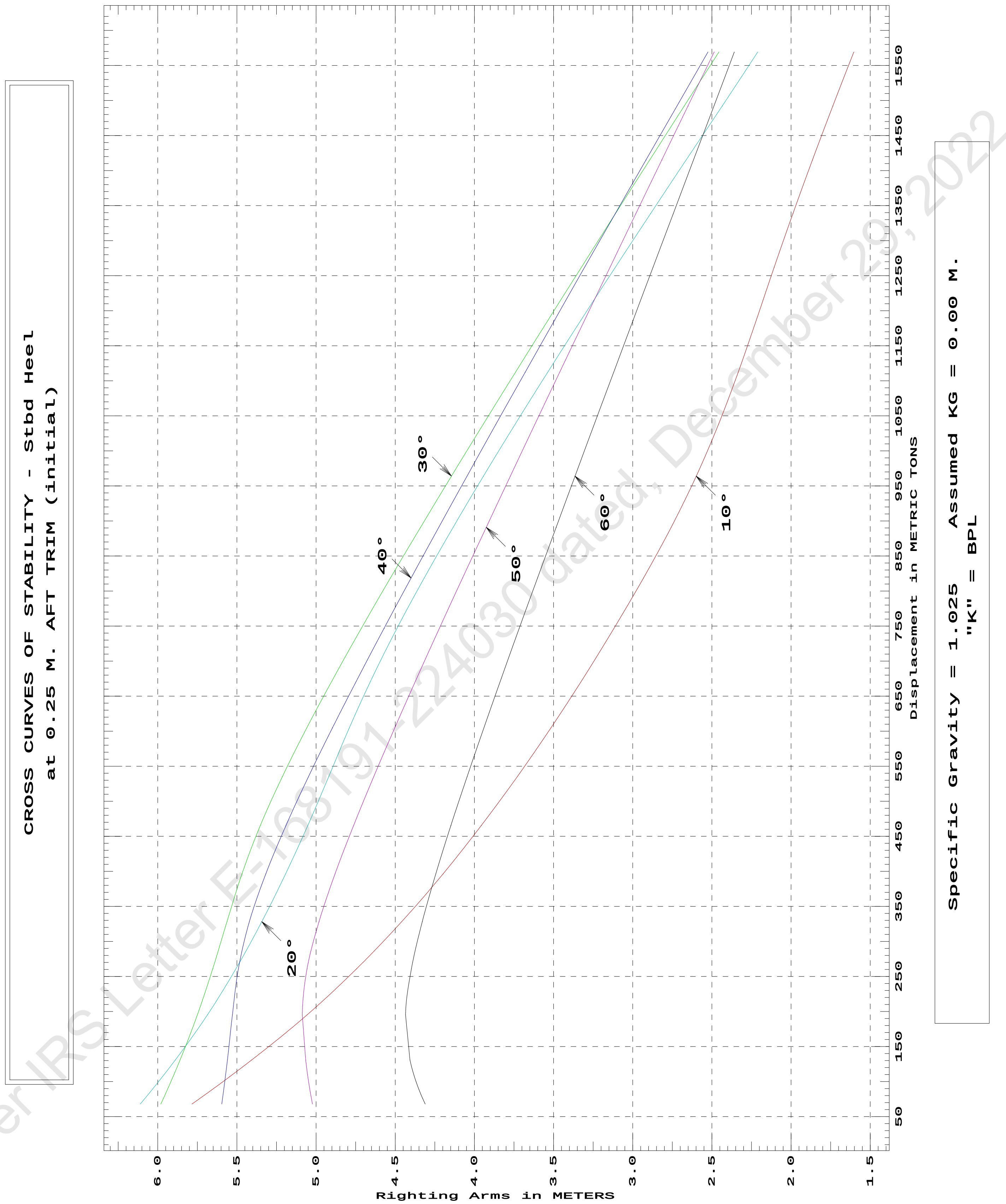
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550
Righting Arms in METERS
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
10°
20°
30°
40°
50°
60°

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.784s	6.111s	5.981s	5.596s	5.023s	4.311s
130.84	5.407s	5.889s	5.860s	5.560s	5.066s	4.407s
195.22	5.057s	5.686s	5.750s	5.529s	5.087s	4.435s
260.99	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.32	4.459s	5.343s	5.563s	5.421s	4.981s	4.329s
396.63	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.88	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.91	3.516s	4.790s	5.058s	4.896s	4.503s	3.940s
675.53	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.76	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.59	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.02	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.05	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.68	2.459s	3.743s	3.945s	3.865s	3.619s	3.243s
1,111.92	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.75	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.19	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.15	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.13	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s

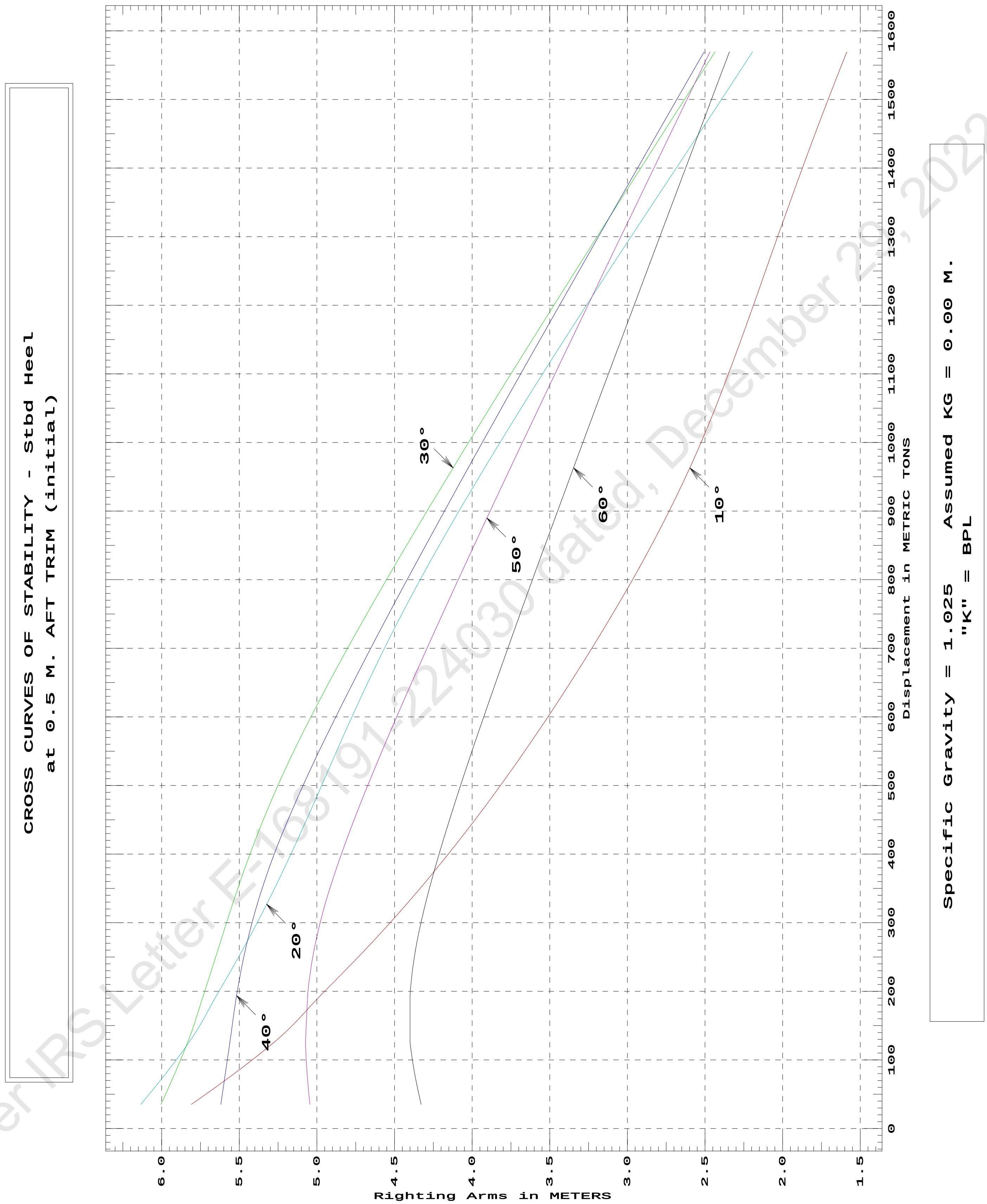
Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
50
150
250
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

CROSS CURVES OF STABILITY
Showing righting arms in heel at VCG = 0.00
Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees 10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.14	5.809s	6.135s	6.004s	5.619s	5.045s	4.328s
126.38	5.272s	5.822s	5.831s	5.560s	5.073s	4.400s
193.59	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.481s	5.639s	5.462s	5.019s	4.366s
326.97	4.423s	5.324s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.177s	5.439s	5.282s	4.849s	4.219s
464.31	3.934s	5.039s	5.319s	5.159s	4.735s	4.125s
533.71	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.73	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.39	3.300s	4.621s	4.864s	4.711s	4.341s	3.810s
745.60	3.107s	4.463s	4.689s	4.547s	4.199s	3.698s
817.41	2.924s	4.292s	4.506s	4.377s	4.054s	3.583s
889.82	2.754s	4.111s	4.317s	4.203s	3.905s	3.466s
962.83	2.598s	3.920s	4.123s	4.025s	3.753s	3.347s
1,036.44	2.458s	3.721s	3.924s	3.844s	3.600s	3.227s
1,110.65	2.331s	3.516s	3.721s	3.660s	3.443s	3.105s
1,185.47	2.211s	3.305s	3.514s	3.474s	3.285s	2.981s
1,261.17	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.60	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.34	1.850s	2.643s	2.873s	2.898s	2.799s	2.601s
1,491.42	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.81	1.588s	2.195s	2.437s	2.507s	2.469s	2.344s

Distances in METERS.---Specific Gravity = 1.025.-------------


CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)
Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL
Displacement in METRIC TONS
Righting Arms in METERS
10°
20°
30°
40°
50°
60°
0
100
200
300
400
500
600
700
800
900
1000
1100
1200
1300
1400
1500
1600
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5

DEADWEIGHT TABLE

Draft M	Disp T	Dwt T	
0.612	396.99	23.62	
0.712	465.83	92.46	
0.812	535.26	161.89	
0.912	605.29	231.92	
1.012	675.92	302.55	
1.112	747.15	373.78	
1.212	818.99	445.62	
1.312	891.43	518.06	
1.412	964.46	591.09	
1.512	1038.11	664.74	
1.612	1112.35	738.98	
1.712	1187.19	813.82	
1.812	1262.64	889.27	
1.912	1338.68	965.31	
2.012	1415.33	1041.96	
2.112	1492.22	1118.85	
2.212	1569.15	1195.78	**Full Load Displacement**

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind
Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.372	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.916	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.807	0%	6d	2d
1,750.00	~~3.468~~ 3.343	0%	15d	1d

Heeling moment is present from: wind
Trim = Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.445	95%	0d	14d
500.00	14.325	51%	0d	10d
750.00	12.852	14%	0d	7d
1,000.00	11.011	0%	0d	5d
1,250.00	8.477	0%	2d	4d
1,500.00	5.833	0%	6d	3d
1,750.00	~~3.494~~ 3.235	0%	15d	2d

Heeling moment is present from: wind
Trim = zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.471	95%	0d	17d
500.00	14.340	51%	0d	11d
750.00	12.878	13%	0d	8d
1,000.00	11.044	0%	0d	6d
1,250.00	8.477	0%	2d	4d
1,500.00	5.832	0%	6d	3d
1,750.00	~~3.467~~ 3.161	0%	LARGE	2d

Heeling moment is present from: wind
Trim = Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.445	95%	0d	14d
500.00	14.325	51%	0d	10d
750.00	12.852	14%	0d	7d
1,000.00	11.011	0%	0d	5d
1,250.00	8.462	0%	2d	4d
1,500.00	5.832	0%	6d	3d
1,750.00	~~3.493~~ 3.225	0%	LARGE	2d

MAX VCG DOES NOT COVER THE CRANE OPERATION / LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW WITH THE CRANE IN STOWED POSITION.

MAX VCG IS VERIFIED FOR MAXIMUM DECK CARGO HEIGHT OF 1M ABOVE MAIN DECK.

Heeling moment is present from: wind

Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

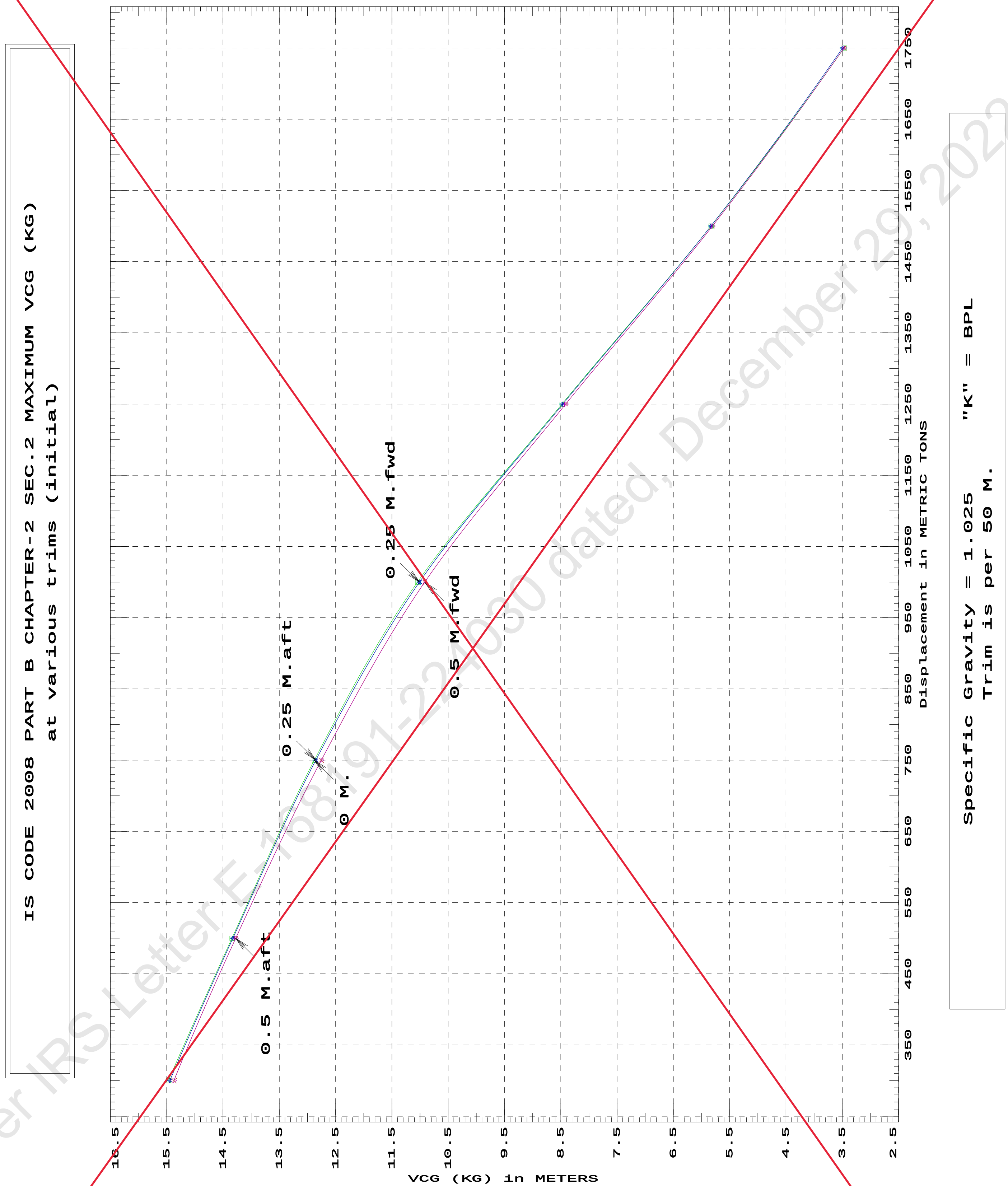
Displacement METRIC TONS	Max VCG	--- Margins --- LIM1	LIM2	LIM3
300.00	15.373	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.917	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.805	0%	6d	2d
1,750.00	~~3.469~~ 3.336	0%	15d	1d

Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM--------IS CODE 2008 PART B CHAPTER-2 SEC.2--------		Min/Max
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg

MAX VCG DOES NOT COVER THE CRANE OPERATION / LIFTING OF WEIGHTS AND IS VALID ONLY FOR UNMANNED TOW WITH THE CRANE IN STOWED POSITION.

MAX VCG IS VERIFIED FOR MAXIMUM DECK CARGO HEIGHT OF 1M ABOVE MAIN DECK.


IS CODE 2008 PART B CHAPTER-2 SEC.2 MAXIMUM VCG (KG)
at various trims (initial)
Specific Gravity = 1.025 "K" = BPL
Trim is per 50 M.
Displacement in METRIC TONS
VCG (KG) in METERS
0.25 M.fwd
0.5 M.fwd
0.25 M.aft
0 M.
0.5 M.aft
350
450
550
650
750
850
950
1050
1150
1250
1350
1450
1550
1650
1750
16.5
15.5
14.5
13.5
12.5
11.5
10.5
9.5
8.5
7.5
6.5
5.5
4.5
3.5
2.5

LOADING CONDITIONS

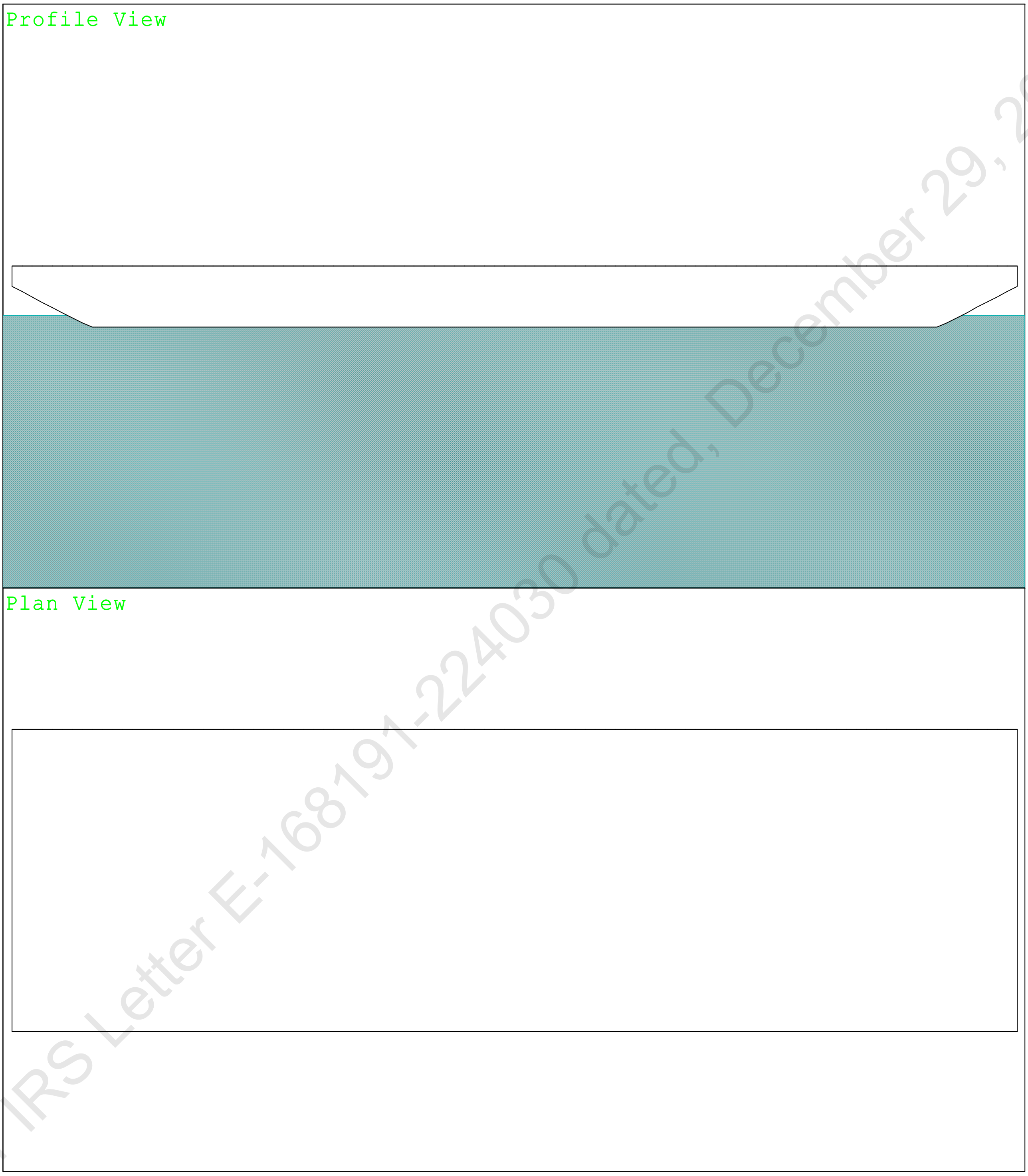
SUMMARY OF CONDITIONS

CONDITION NO.	0	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Per. Ballast	Condition 2 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 3 : Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd	Condition 5 : Lightship + Crane + Per. Ballast + Deck Cargo (Towing Condition)
TOTAL TANK LOAD (T)	0.000	264.600	264.600	264.600	264.600	264.600
DWT CONST.(T)	0.000	250.000	286.600	286.600	286.600	931.180
DEADWEIGHT (T)	0.000	514.600	551.200	551.200	551.200	1195.780
LIGHT WEIGHT (T)	373.370	373.370	373.370	373.370	373.370	373.370
DISPLACEMENT (T)	373.370	887.970	924.570	924.570	924.570	1569.150
FWD DRAFT (M)	0.575	1.305	1.230	1.350	1.480	2.210
AFT DRAFT (M)	0.580	1.309	1.484	1.355	1.234	2.214
MEAN DRAFT (M)	0.578	1.307	1.357	1.353	1.357	2.212
TRIM (M)	0.005a	0.004a	0.255a	0.004a	0.246f	0.004a
L.C.G. (FROM AP) (M)	24.970f	24.987f	24.275f	24.988f	25.700f	24.993f
L.C.B. (FROM AP) (M)	24.970f	24.987f	24.247f	24.987f	25.728f	24.993f
M.C.T. (T-CM)	22.220	26.530	26.330	26.400	26.330	31.300
KG (M)	3.000	3.431	6.341	6.341	6.341	3.461
GM_T (M)	31.578	12.372	8.956	9.082	8.956	6.766
MAX. GZ (M)	4.197	2.897	1.946	1.284	1.946	1.049

~~USK draft refers to the line:~~
~~0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000~~

Condition 0 : Lightship Condition

Condition Graphic - Draft: 0.575 @ 50.000f, 0.580 @ 0.000 Heel: zero



PONTOON

Condition 0 : Lightship Condition

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 0.575 @ 50.00f, 0.577 @ 25.00f, 0.580 @ 0.00
                     Trim: Aft 0.005/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG-------FSM
WEIGHT                                   373.37  24.970f  0.000   3.000
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG
  Total Tanks--------->                    0.00                                0.00
                                    Displ(MT)----LCB-----TCB-----VCB
HULL                        1.025        373.37  24.970f  0.000   0.283
-------------------------------------------------------------------------------
              Righting Arms:                        0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            666.5  24.996f  0.000    300.25  34.295
                                          MT/cm--------m.-MT/cm-------GML-----GMT
                                           6.83           22.22    297.54  31.578
Distances in METERS.-----------------------------------------Moments in m.-MT.
```

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.575

DRAFT AT MS (M) = 0.577

DRAFT AT AP (M) = 0.580

DRAFT AT FWD MARK (M) = 0.575

DRAFT AT AFT MARK (M) = 0.580

PONTOON

Condition 0 : Lightship Condition

HYDROSTATIC PROPERTIES
Trim: Aft 0.005/50.000, No Heel, VCG = 3.000

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	Buoyancy-Ctr. VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
0.577	373.37	24.970f	0.283	6.83	24.996f	22.22	297.54	31.578

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 24.970f TCG = 0.000 VCG = 3.000

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
0.568	0.01a	0.00	373.37	0.000	-0.025	0.0000	14.712
0.568	0.01a	0.04s	373.37	0.000	0.000	-0.0000	14.668
0.547	0.01a	5.04s	373.36	0.000	2.605	0.1137	9.668
0.374	0.01a	10.04s	373.38	0.000	3.755	0.4015	4.668
0.099	0.01a	14.71s	373.37	0.000	4.100	0.7262	0.000
0.077	0.01a	15.04s	373.37	0.000	4.113	0.7500	-0.332
-0.270	0.01a	19.83s	373.18	0.000	4.197	1.0988	-5.117
-0.287	0.01a	20.04s	373.37	0.000	4.197	1.1145	-5.332
-0.692	0.01a	25.04s	373.16	0.000	4.137	1.4792	-10.332
-1.123	0.01a	30.04s	373.17	0.000	3.986	1.8344	-15.332
-1.557	0.01a	35.04s	373.20	0.000	3.739	2.1721	-20.332
-1.978	0.02a	40.04s	373.22	0.000	3.412	2.4847	-25.332
-2.385	0.02a	45.04s	373.23	0.000	3.028	2.7661	-30.332
-2.774	0.02a	50.04s	373.24	0.000	2.601	3.0120	-35.332
-3.141	0.02a	55.04s	373.25	0.000	2.141	3.2191	-40.332
-3.485	0.02a	60.04s	373.26	0.000	1.656	3.3850	-45.332

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

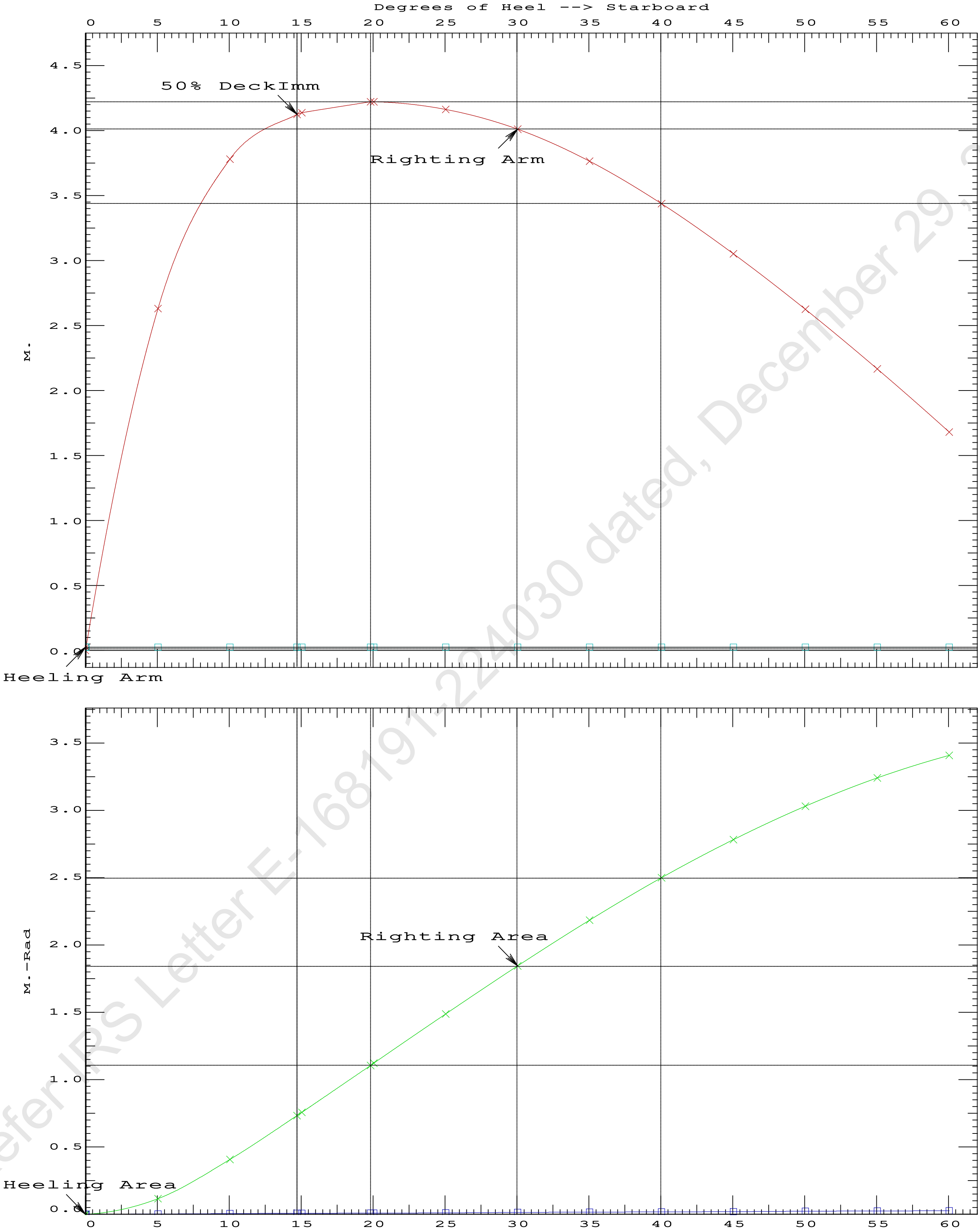
Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.32 (constant)

LIM------------INTACT CRITERIA-IS CODE 2008PART B		----Min/Max		--------Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	1.0988 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	14.67 P

PONTOON

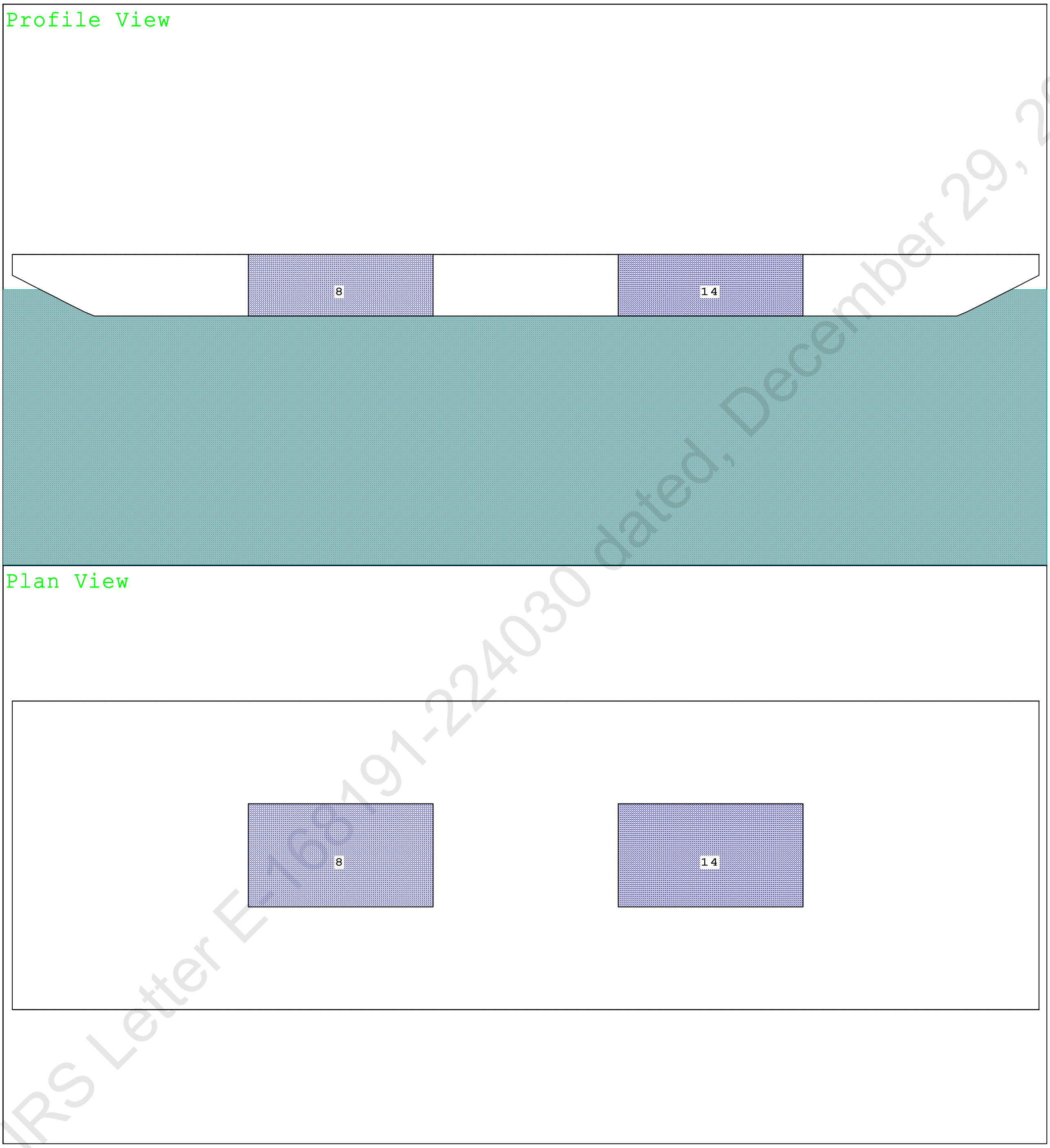
Condition 0 : Lightship Condition



USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 1.305 @ 50.000f, 1.309 @ 0.000 Heel: zero



Tanks
8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

PONTOON

Condition 1 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00
Trim: Aft 0.004/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Total Fixed--------->			623.37	24.982f	0.000	4.251	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			887.97	24.987f	0.000	3.431	
			Displ(MT)	**LCB**	**TCB**	**VCB**	
HULL		1.025	887.97	24.987f	0.000	0.660	
Righting Arms:				0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	709.3	24.996f	0.000	152.18	15.143*
		MT/cm		**m.-MT/cm**	**GML**	**GMT**
		7.27		26.53	149.41	12.372

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane

DRAFT AT FP (M) = 1.305

DRAFT AT MS (M) = 1.307

DRAFT AT AP (M) = 1.309

DRAFT AT FWD MARK (M) = 1.305

DRAFT AT AFT MARK (M) = 1.309

PONTOON

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES
Trim: Aft 0.004/50.000, No Heel, VCG = 3.431

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.307	887.97	24.987f	0.660	7.27	24.996f	26.53	149.41	12.372

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.987f TCG = 0.000 VCG = 3.431
Free Surface Adjustment: 0.207
Adjusted CG: LCG = 24.987f TCG = 0.000 VCG = 3.638

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.297	0.00a	0.00	887.97	0.000	-0.015	0.0000	6.751
1.297	0.00a	0.07s	887.97	0.000	0.000	-0.0000	6.682
1.286	0.00a	5.07s	887.97	0.000	1.085	0.0473	1.682
1.277	0.00a	6.75s	887.97	0.000	1.451	0.0846	0.000
1.246	0.00a	10.07s	887.96	0.000	2.132	0.1884	-3.318
1.126	0.01a	15.07s	887.97	0.000	2.789	0.4052	-8.318
0.999	0.01a	19.12s	887.97	0.000	2.897	0.6083	-12.367
0.968	0.01a	20.07s	887.97	0.000	2.891	0.6563	-13.318
0.803	0.01a	25.07s	887.97	0.000	2.758	0.9053	-18.318
0.631	0.01a	30.07s	887.77	0.000	2.519	1.1363	-23.318
0.456	0.01a	35.07s	887.96	0.000	2.220	1.3436	-28.318
0.276	0.02a	40.07s	887.97	0.000	1.882	1.5228	-33.318
0.095	0.02a	45.07s	887.97	0.000	1.516	1.6713	-38.318
-0.088	0.02a	50.07s	887.97	0.000	1.131	1.7870	-43.318
-0.269	0.02a	55.07s	887.97	0.000	0.732	1.8684	-48.318
-0.449	0.02a	60.07s	887.97	0.000	0.324	1.9145	-53.318

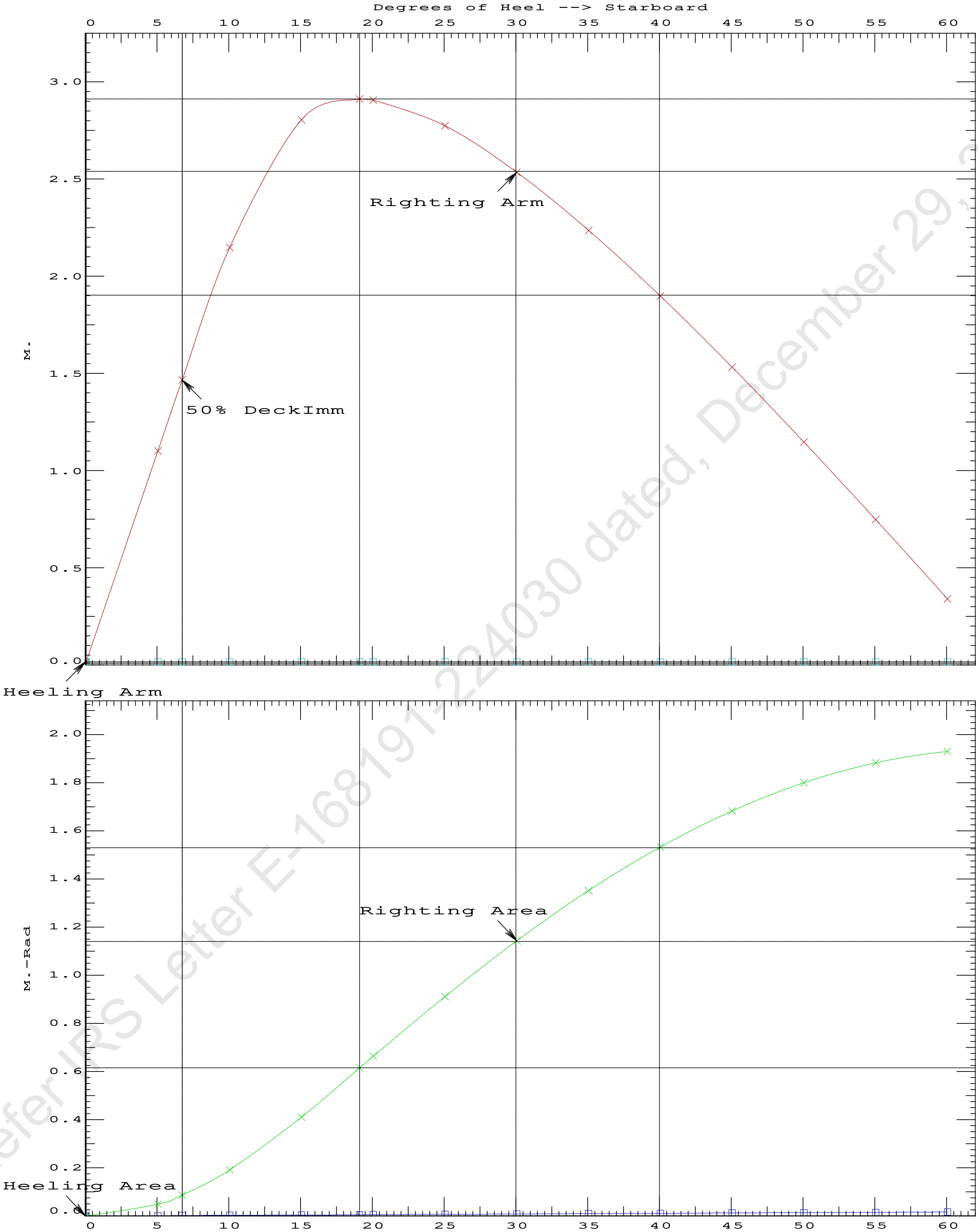
Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.65 (constant)

LIM	INTACT CRITERIA-IS CODE 2008PART B		Min/Max		Attained
(1)	Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.6083 P
(2)	Angle from Equilibrium to RAzero or Flood	>	20.00	deg	LARGE
(3)	Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	6.68 P

Condition 1 : Lightship + Crane + Ballast

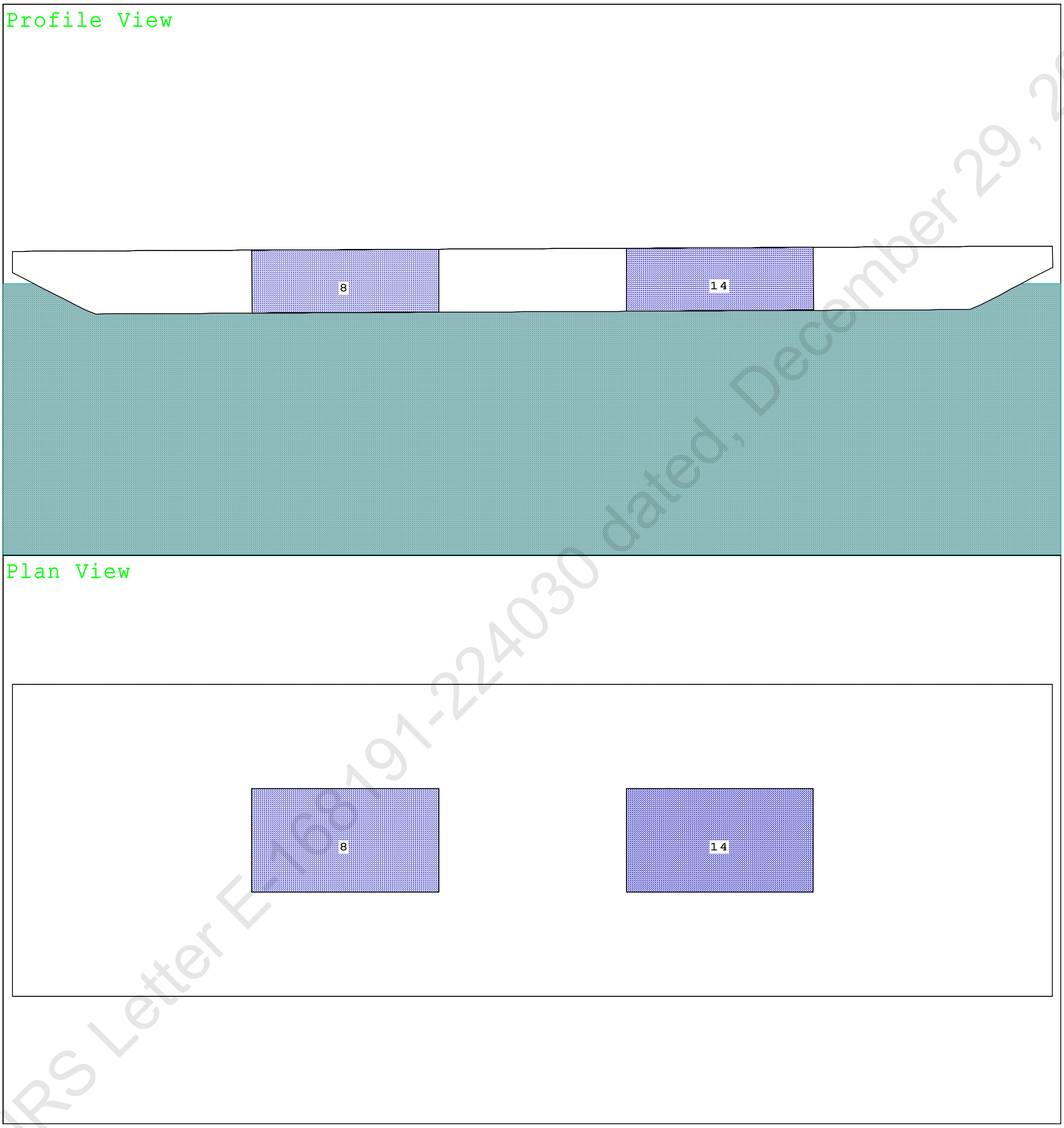


USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.230 @ 50.000f, 1.484 @ 0.000 Heel: zero



Tanks	
	8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.230 @ 50.00f, 1.357 @ 25.00f, 1.484 @ 0.00
Trim: Aft 0.255/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	7.000f	0.000	76.930	
Total Fixed--------->			659.97	23.985f	0.000	8.282	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			924.57	24.275f	0.000	6.341	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	924.57	24.247f	0.000	0.688

			LCF	TCF		
Righting Arms:			0.000	0.000		

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	712.3	24.764f	0.000	148.04	14.609*
		MT/cm		m.-MT/cm	GML	GMT
		7.30		26.33	142.39	8.956

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane 36.60 MT Crane Load

DRAFT AT FP (M) = 1.230

DRAFT AT MS (M) = 1.357

DRAFT AT AP (M) = 1.484

DRAFT AT FWD MARK (M) = 1.250

DRAFT AT AFT MARK (M) = 1.464

HYDROSTATIC PROPERTIES
Trim: Aft 0.255/50.000, No Heel, VCG = 6.341

LCF Draft	Displacement Weight(MT)	Buoyancy-Ctr. LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
1.358	924.57	24.247f	0.688	7.30	24.764f	26.33	142.39	8.956

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
     Total CG:  LCG = 24.275f  TCG = 0.000   VCG =  6.341
                         Free Surface Adjustment:   0.199
  Adjusted CG:  LCG = 24.276f  TCG = 0.000   VCG =  6.540
```

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.472	0.29a	0.00	924.57	0.000	-0.014	0.0000	5.923
1.472	0.29a	0.08s	924.57	0.000	0.000	-0.0000	5.841
1.460	0.29a	5.08s	924.57	0.000	0.786	0.0343	0.841
1.456	0.29a	5.92s	924.57	0.000	0.919	0.0468	0.000
1.425	0.30a	10.08s	924.57	0.000	1.544	0.1363	-4.159
1.353	0.38a	15.08s	924.57	0.000	1.943	0.2903	-9.159
1.346	0.39a	15.55s	924.57	0.000	1.946	0.3061	-9.624
1.272	0.51a	20.08s	924.57	0.000	1.791	0.4569	-14.159
1.185	0.64a	25.08s	924.57	0.000	1.420	0.5994	-19.159
1.090	0.78a	30.08s	924.57	0.000	0.958	0.7038	-24.159
0.989	0.91a	35.08s	924.57	0.000	0.449	0.7656	-29.159
0.899	1.02a	39.30s	924.57	0.000	0.000	0.7822	-33.377
0.882	1.04a	40.08s	924.57	0.000	-0.084	0.7816	-34.159
0.768	1.16a	45.08s	924.57	0.000	-0.630	0.7506	-39.159
0.649	1.27a	50.08s	924.57	0.000	-1.178	0.6717	-44.159
0.524	1.37a	55.08s	924.57	0.000	-1.722	0.5452	-49.159
0.394	1.46a	60.08s	924.57	0.000	-2.257	0.3715	-54.159

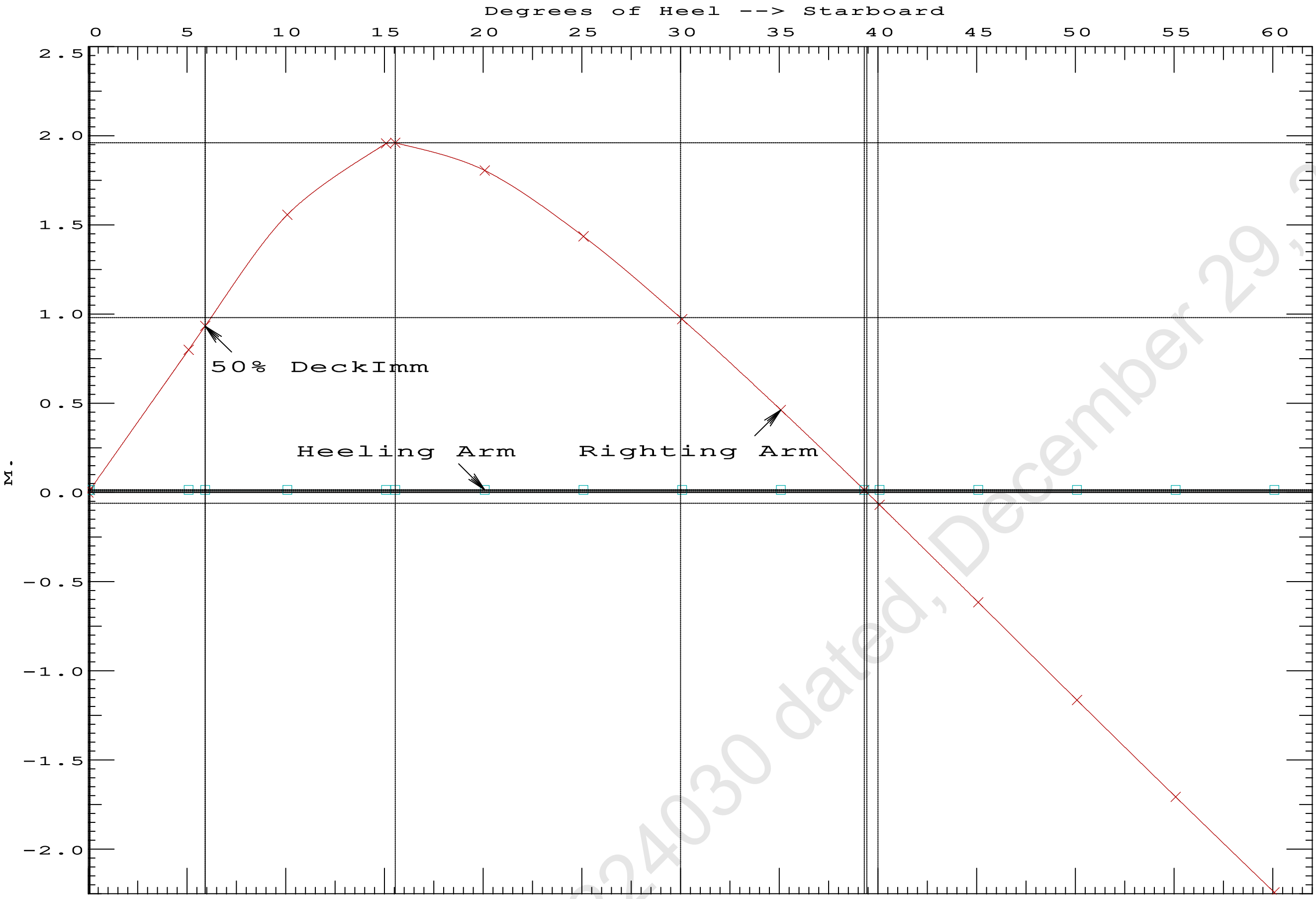
Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

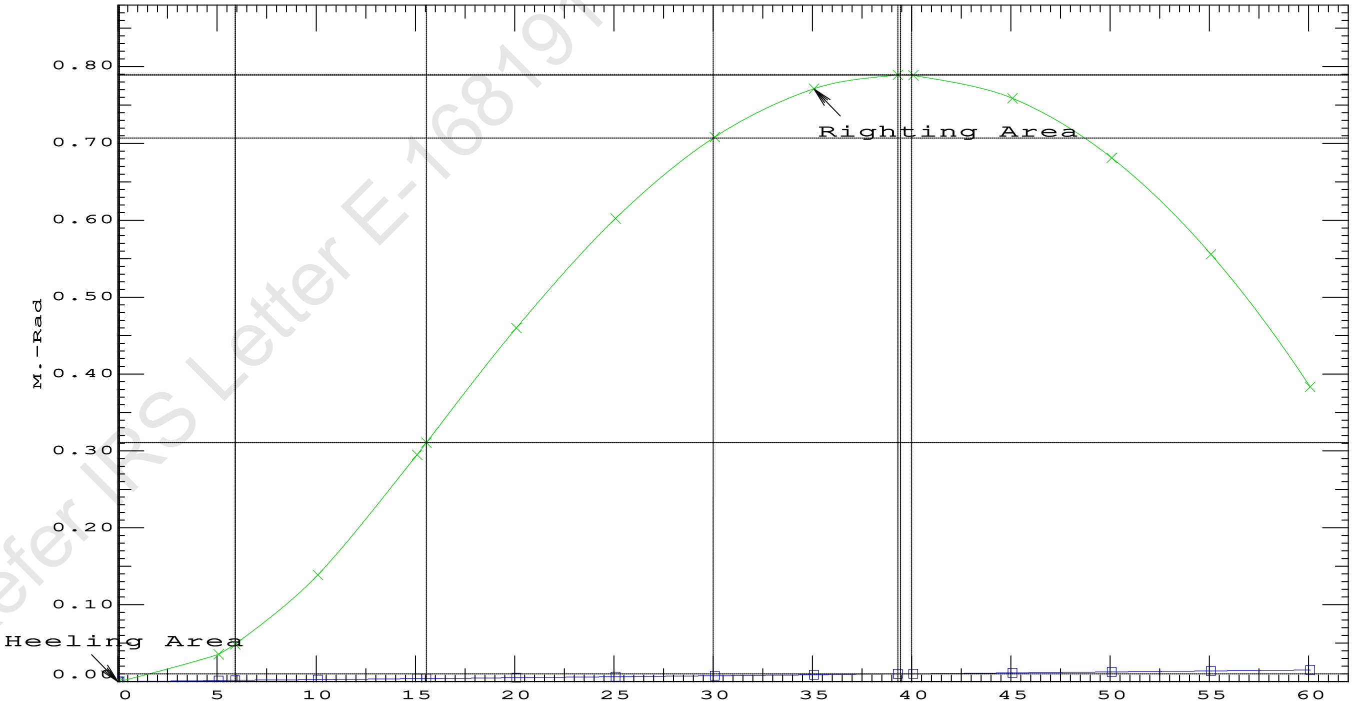
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.29 (constant)

```
LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max--------Attained
(1) Residual Area from abs 0 deg to MaxRA          >   0.0800 m.-Rad 0.3061 P
(2) Angle from Equilibrium to RAzero or Flood      >    20.00  deg    39.22 P
(3) Angle from Equilibrium to 50% Dk Imm Angle     >     0.00  deg     5.84 P
------------------------------------------------------------------------------
```

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft





Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.275f TCG = 0.000 VCG = 6.341
Free Surface Adjustment: 0.199
Adjusted CG: LCG = 24.276f TCG = 0.000 VCG = 6.540

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.472	0.29a	0.00	924.57	0.000	0.000	0.0000
1.460	0.29a	5.00p	924.57	0.000	0.786	0.0343
1.425	0.30a	10.00p	924.57	0.000	1.545	0.1362
1.354	0.38a	15.00p	924.57	0.000	1.956	0.2915
1.346	0.39a	15.56p	924.57	0.000	1.959	0.3107
1.274	0.51a	20.00p	924.57	0.000	1.809	0.4594
1.186	0.64a	25.00p	924.57	0.000	1.440	0.6036
1.091	0.78a	30.00p	924.57	0.000	0.979	0.7098
0.990	0.91a	35.00p	924.57	0.000	0.470	0.7734
0.896	1.02a	39.42p	924.57	0.000	0.000	0.7917
0.883	1.04a	40.00p	924.57	0.000	-0.063	0.7913
0.770	1.16a	45.00p	924.57	0.000	-0.608	0.7622
0.651	1.27a	50.00p	924.57	0.000	-1.156	0.6853
0.526	1.37a	55.00p	924.57	0.000	-1.700	0.5606
0.396	1.46a	60.00p	924.57	0.000	-2.235	0.3889

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

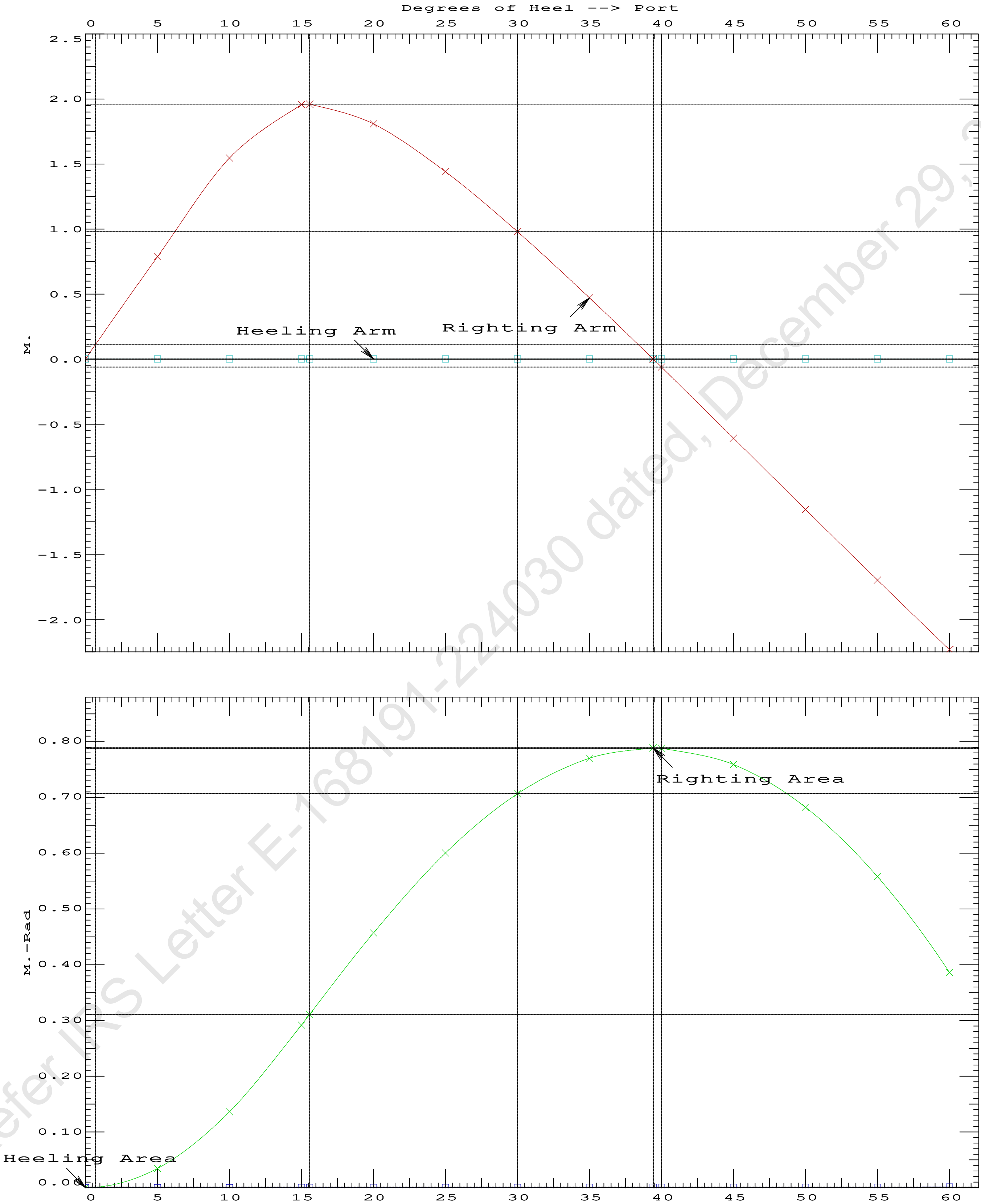
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM---STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	181117 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1064.88 P

PONTOON

Condition 2 : Lightship + Load at Aft + Ballast

Crane Lifting 36.6T @ 18m Radius Aft



USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition 2 : Lightship + Load at Aft + Ballast
Crane Lifting 36.60T @ 18m Radius Aft

Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.229 @ 50.00f, 1.357 @ 25.00f, 1.484 @ 0.00
Trim: Aft 0.254/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	7.000f	0.000	76.930	
Total Fixed--------->			659.97	23.985f	0.000	8.282	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			924.57	24.275f	0.000	6.341	
			Displ(MT)	LCB	TCB	VCB	
HULL		1.025	924.46	24.247f	0.001p	0.688	

Righting Arms:	0.000	-0.001s
External Arms:	0.000	-0.001p
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	709.1	24.873f	0.017p	146.04	14.563*
		MT/cm		m.-MT/cm	GML	GMT
		7.27		25.96	140.38	8.910

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

```
                    After hook load drop
           WEIGHT and DISPLACEMENT and WATERPLANE STATUS
       USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00
              Trim: Aft 0.004/50.000,  Heel: 0.00 deg.
Part------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                          373.37  24.970f  0.000   3.000
Crane                               250.00  25.000f  0.000   6.120
  Total Fixed--------->             623.37  24.982f  0.000   4.251
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000       132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000       132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->             264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->             887.97  24.987f  0.000   3.431
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025       887.97  24.987f  0.001   0.660
-----------------------------------------------------------------------------------
               Righting Arms:                0.000   0.001s
               External Arms:                0.000   0.001
      Residual Righting Arms:                0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025        708.7  24.997f  0.005s   151.80  15.119*
                                     MT/cm--------m.-MT/cm-------GML-----GMT
                                      7.26           26.47    149.03  12.348
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

               Theta3 is 0.07 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

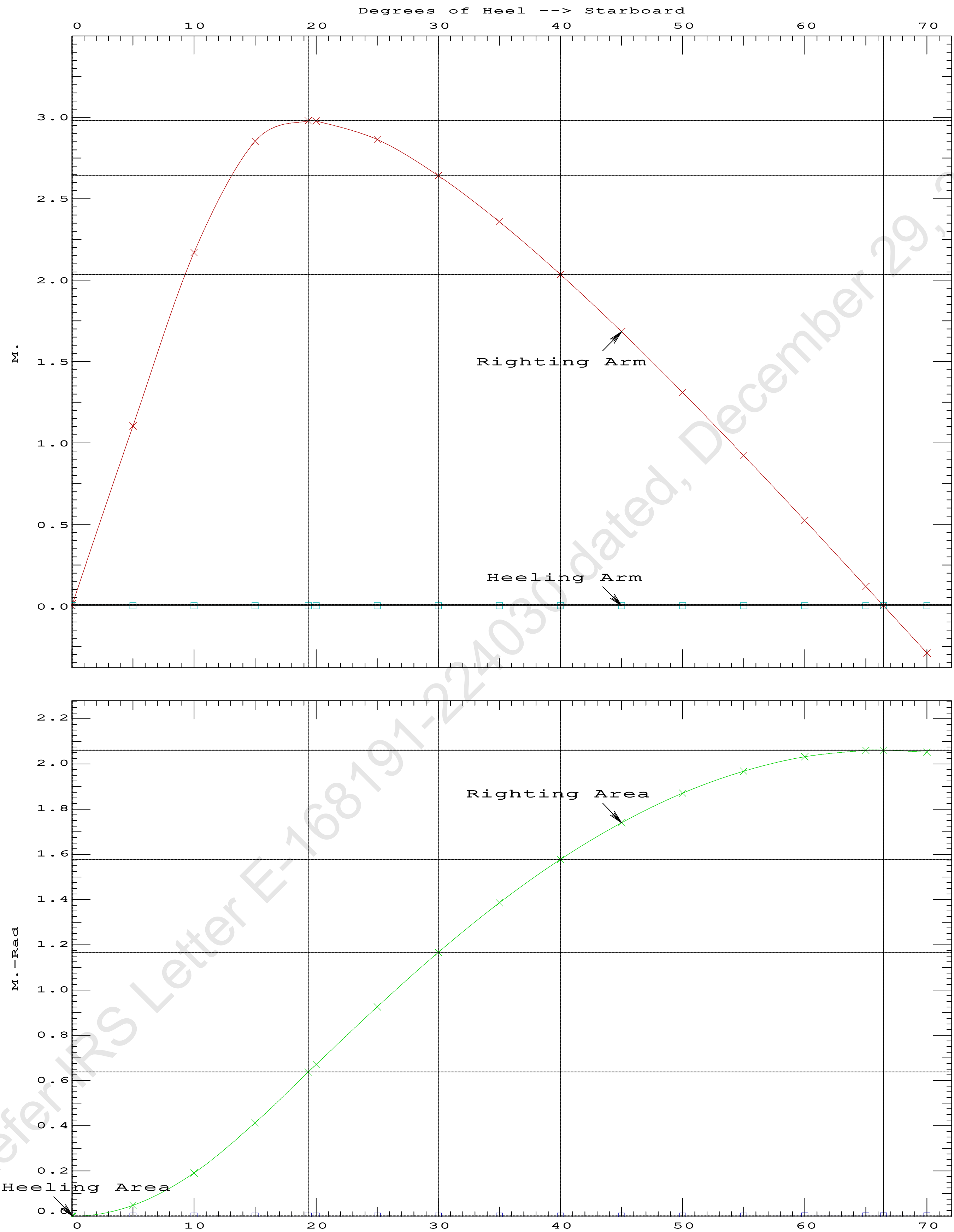
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693
0.458	0.01a	35.00s	887.97	0.000	2.357	1.3879
0.279	0.02a	40.00s	887.97	0.000	2.034	1.5798
0.097	0.02a	45.00s	887.97	0.000	1.682	1.7422
-0.085	0.02a	50.00s	887.96	0.000	1.310	1.8729
-0.267	0.02a	55.00s	887.97	0.000	0.922	1.9704
-0.446	0.02a	60.00s	887.97	0.000	0.523	2.0335
-0.622	0.02a	65.00s	887.97	0.000	0.118	2.0615
-0.672	0.02a	66.45s	887.97	0.000	0.000	2.0629
-0.794	0.03a	70.00s	887.97	0.000	-0.290	2.0540

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM--------------RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2) Angle from abs 0.1 deg to RAzero	>	20.00	deg	66.38 P

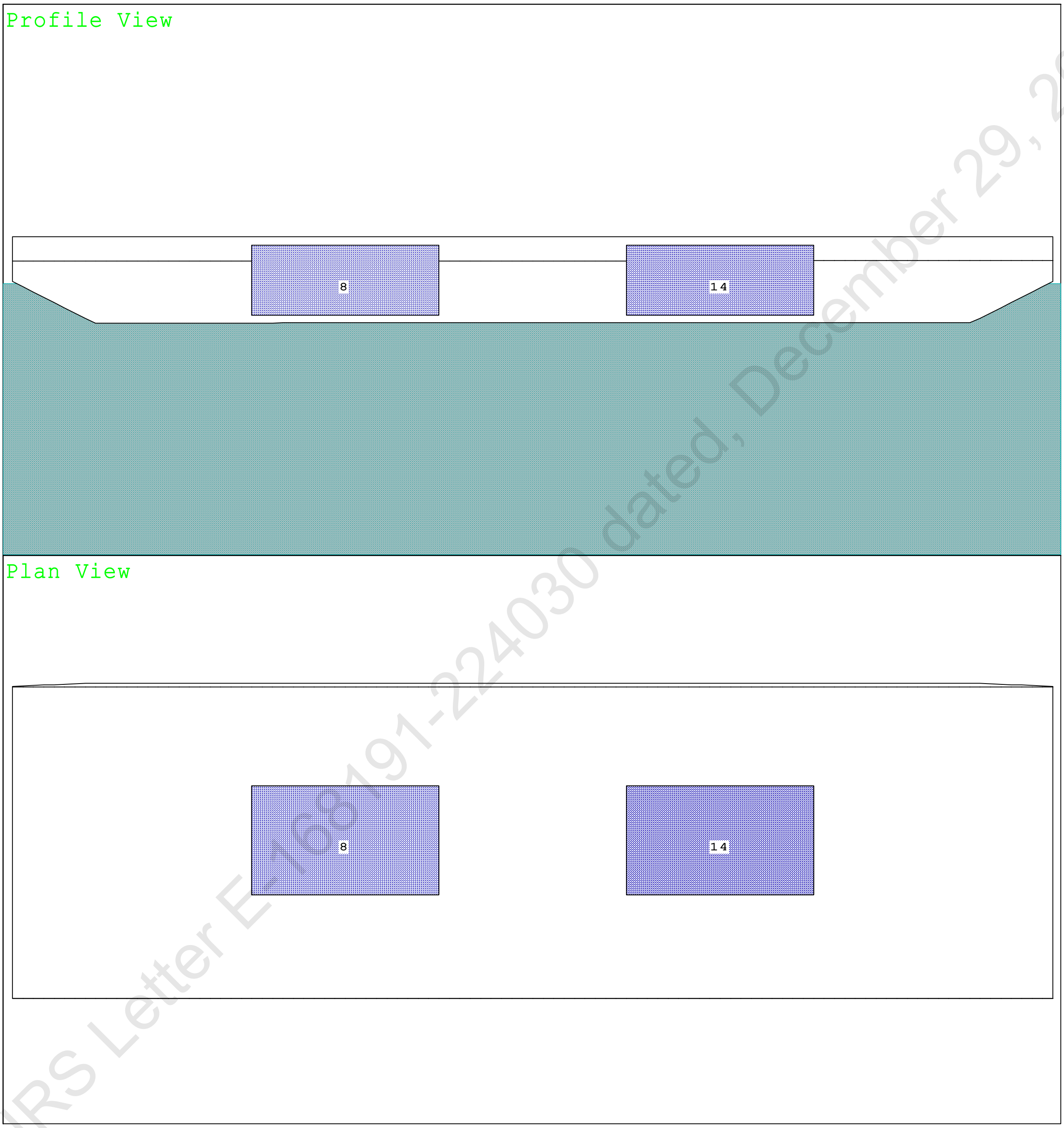

Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
M.
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Righting Arm
Heeling Arm
M.-Rad
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Righting Area
Heeling Area

USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

CG - Draft: 1.350 @ 50.000f, 1.355 @ 0.000 Heel: stbd 4.42 deg.



Tanks
8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

```
                 WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.350 @ 50.00f, 1.352 @ 25.00f, 1.355 @ 0.00
                 Trim: Aft 0.004/50.000,  Heel: Stbd 4.42 deg.
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                               373.37  24.970f  0.000   3.000
Crane                                    250.00  25.000f  0.000   6.120
Crane Load                                36.60  25.000f 18.000s 76.930
  Total Fixed--------->                  659.97  24.983f  0.998s  8.282
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500    91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500    91.87*
  Total Tanks--------->                  264.60  25.000f  0.000   1.500   183.75
  Total Weight-------->                  924.57  24.988f  0.713s  6.341
                                      Displ(MT)----LCB-----TCB-----VCB
HULL                     1.025           924.57  24.987f  1.146s  0.731
-------------------------------------------------------------------------------
               Righting Arms:                   0.000    0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            713.7  24.996f  0.125s   148.39  14.709*
                                         MT/cm--------m.-MT/cm-------GML-----GMT
                                          7.32           26.40    142.76   9.082
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       264.6 Cu.M.  (13%) FRESH WATER

      250.00 MT  Crane                       36.60 MT  Crane Load

DRAFT AT FP (M) =    1.350

DRAFT AT MS (M) =    1.352

DRAFT AT AP (M) =    1.355

DRAFT AT FWD MARK (M) =    1.351

DRAFT AT AFT MARK (M) =    1.354

                          HYDROSTATIC PROPERTIES
          Trim: Aft 0.004/50.000, Heel: Stbd 4.42 deg., VCG = 6.341

   LCF   Displacement   Buoyancy-Ctr.  Weight/         Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.352       924.57  24.987f  0.731     7.32  24.996f   26.40  142.76   9.082
   Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                             Trim is per 50.00m.
   Draft is from USK.                               Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 183.7 m.-MT
```

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.988f TCG = 0.713s VCG = 6.341
Free Surface Adjustment: 0.199
Adjusted CG: LCG = 24.988f TCG = 0.697s VCG = 6.539

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area	50% DeckImm Angle
1.348	0.00a	0.00	924.57	0.000	-0.711	0.0000	6.496
1.338	0.00a	4.51s	924.57	0.000	0.000	-0.0280	1.987
1.329	0.00a	6.50s	924.57	0.000	0.317	-0.0225	0.000
1.304	0.00a	9.51s	924.56	0.000	0.776	0.0063	-3.013
1.205	0.01a	14.51s	924.57	0.000	1.268	0.0969	-8.013
1.170	0.01a	15.82s	924.57	0.000	1.284	0.1262	-9.327
1.069	0.01a	19.51s	924.57	0.000	1.176	0.2069	-13.013
0.925	0.01a	24.51s	924.57	0.000	0.844	0.2961	-18.013
0.774	0.01a	29.51s	924.57	0.000	0.417	0.3518	-23.013
0.635	0.01a	33.93s	924.57	0.000	0.000	0.3681	-27.436
0.617	0.01a	34.51s	924.57	0.000	-0.056	0.3678	-28.013
0.455	0.02a	39.51s	924.57	0.000	-0.551	0.3415	-33.013
0.290	0.02a	44.51s	924.57	0.000	-1.055	0.2715	-38.013
0.123	0.02a	49.51s	924.57	0.000	-1.559	0.1574	-43.013
-0.045	0.02a	54.51s	924.57	0.000	-2.057	-0.0004	-48.013
-0.213	0.02a	59.51s	924.57	0.000	-2.543	-0.2013	-53.013
-0.380	0.02a	64.51s	924.57	0.000	-3.013	-0.4438	-58.013

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

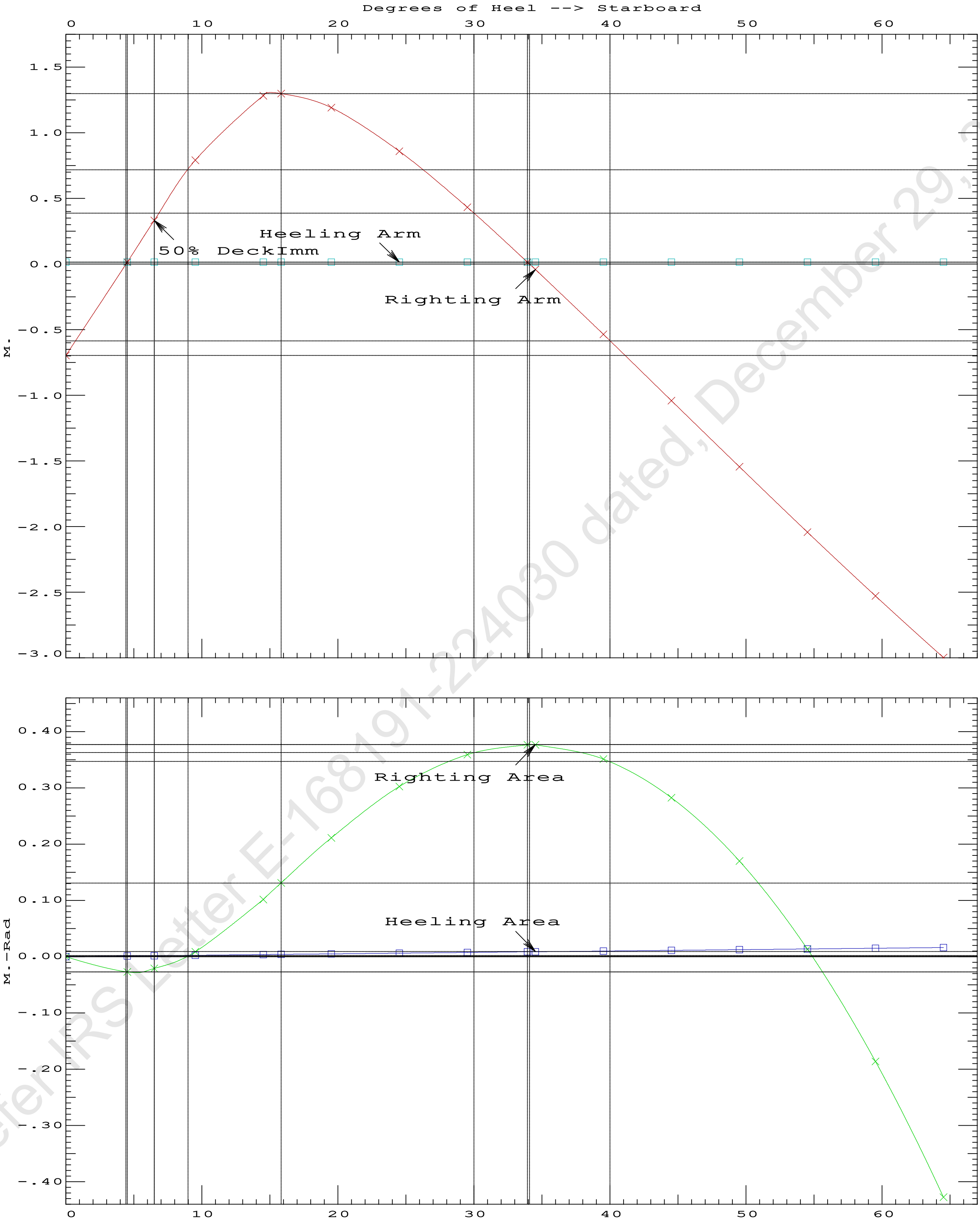
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.41 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1262 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	29.42 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	1.99 P

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast

Crane Lifting 36.6T @ 18m Radius Stbd



Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.988f TCG = 0.713s VCG = 6.341
Free Surface Adjustment: 0.199
Adjusted CG: LCG = 24.988f TCG = 0.697s VCG = 6.539

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.339	0.00a	4.42s	924.57	0.000	0.000	0.0000
1.347	0.00a	0.58p	924.57	0.000	0.789	0.0344
1.333	0.00a	5.58p	924.57	0.000	1.573	0.1375
1.291	0.01a	10.58p	924.56	0.000	2.313	0.3074
1.181	0.01a	15.42p	924.57	0.000	2.641	0.5191
1.177	0.01a	15.58p	924.57	0.000	2.641	0.5267
1.039	0.01a	20.58p	924.57	0.000	2.437	0.7522
0.893	0.01a	25.58p	924.57	0.000	2.030	0.9486
0.741	0.01a	30.58p	924.57	0.000	1.533	1.1047
0.583	0.02a	35.58p	924.57	0.000	0.987	1.2150
0.420	0.02a	40.58p	924.57	0.000	0.414	1.2764
0.304	0.02a	44.11p	924.57	0.000	0.000	1.2892
0.255	0.02a	45.58p	924.57	0.000	-0.173	1.2870
0.087	0.02a	50.58p	924.57	0.000	-0.767	1.2460
-0.081	0.02a	55.58p	924.57	0.000	-1.360	1.1531

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.10

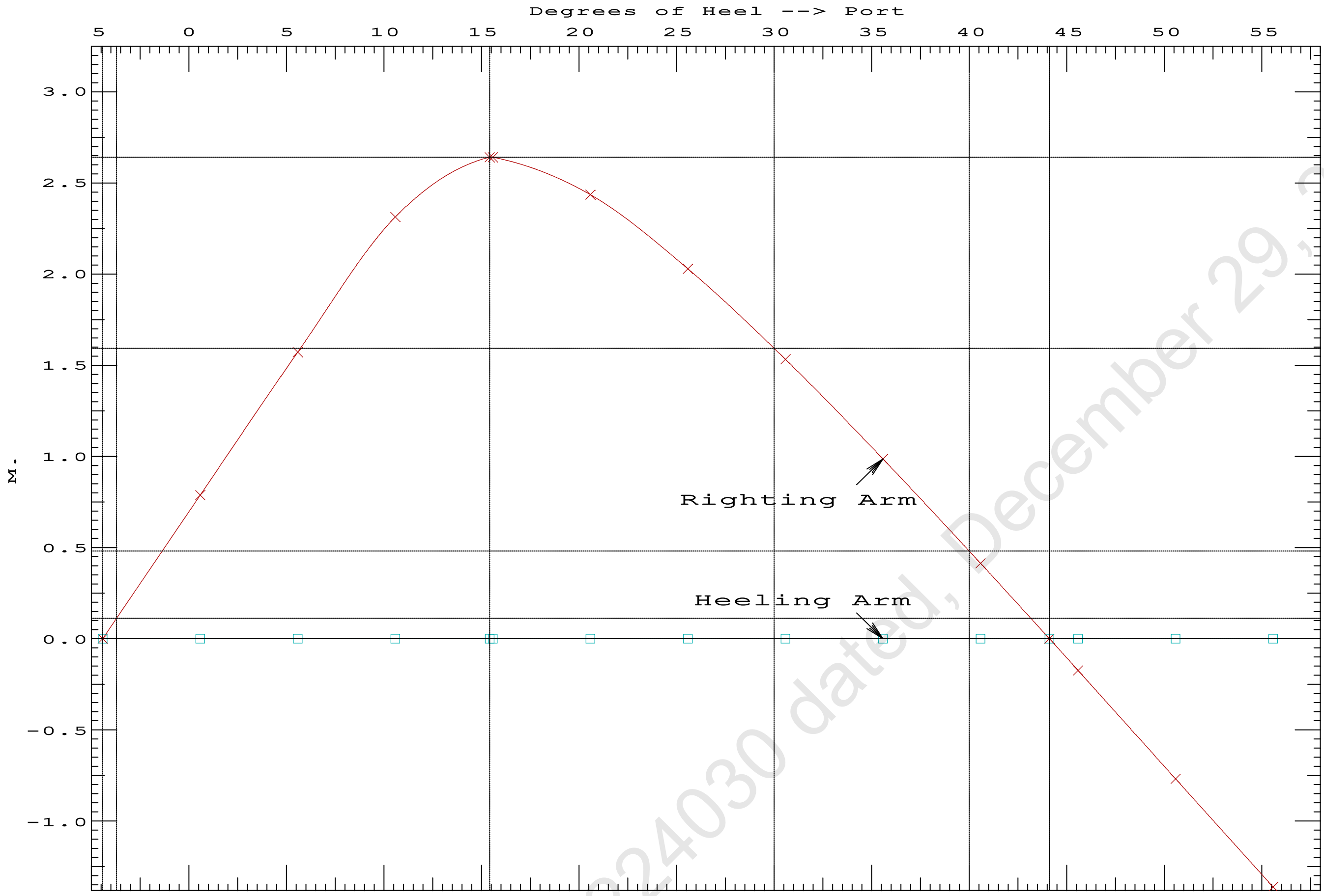
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

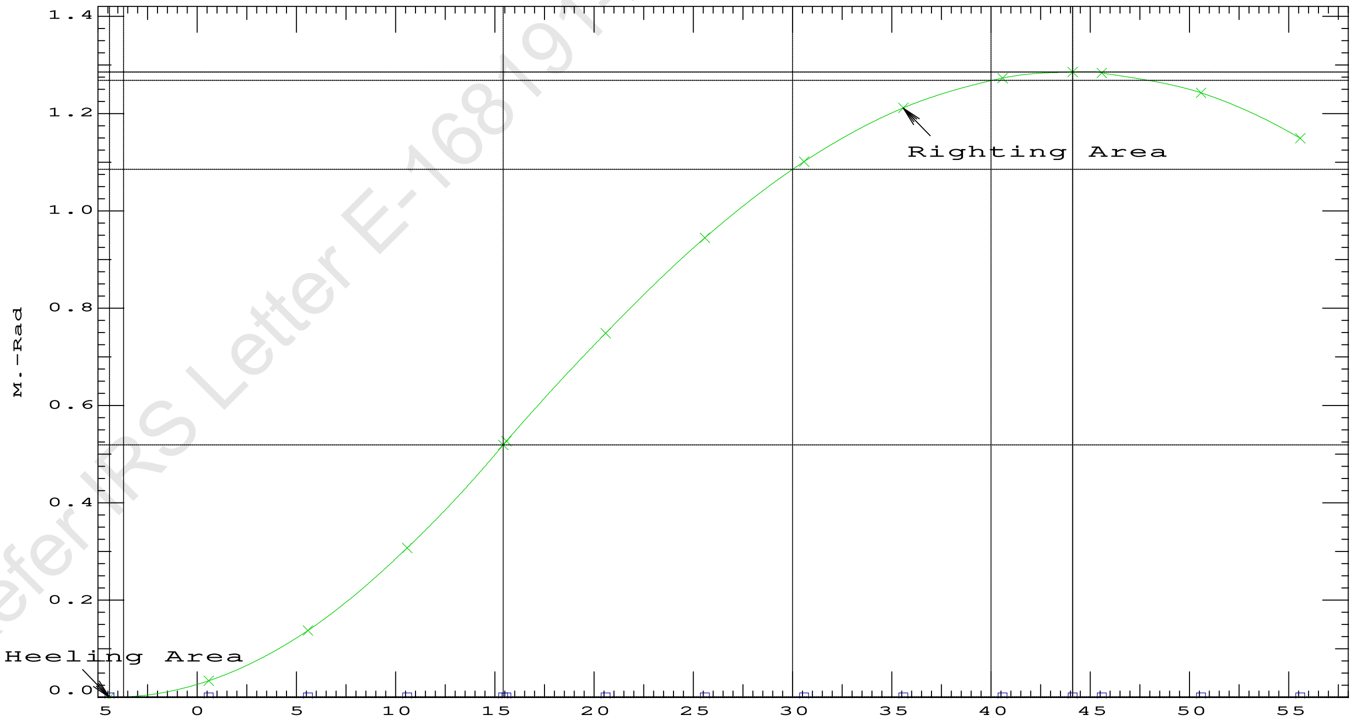
LIM------------STABILITY CRITERIA FOR CRANE OPERAT----		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	4.42 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	2442210 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		14072.5 P

PONTOON

Condition 3 : Lightship + Load at Stbd + Ballast

Crane Lifting 36.6T @ 18m Radius Stbd





USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition 3 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

```
                    Crane in operation with load in place
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.350 @ 50.00f, 1.352 @ 25.00f, 1.355 @ 0.00
                Trim: Aft 0.004/50.000,  Heel: Stbd 4.41 deg.
Part---------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              373.37  24.970f  0.000   3.000
Crane                                   250.00  25.000f  0.000   6.120
Crane Load                               36.60  25.000f 18.000s 76.630
  Total Fixed--------->                 659.97  24.983f  0.998s  8.265
               Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000           132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000           132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->                 264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->                 924.57  24.988f  0.713s  6.329
                                     Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025           924.57  24.987f  1.145s  0.731
-----------------------------------------------------------------------------
              Righting Arms:                     0.000    0.000s
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            713.7  24.996f  0.125s   148.38  14.708*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                          7.32            26.40   142.77   9.093
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                           Angle Theta1 = 4.41
```

```
                         After hook load drop
             WEIGHT and DISPLACEMENT and WATERPLANE STATUS
        USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00
               Trim: Aft 0.004/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           373.37  24.970f  0.000   3.000
Crane                                250.00  25.000f  0.000   6.120
  Total Fixed--------->              623.37  24.982f  0.000   4.251
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->              264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->              887.97  24.987f  0.000   3.431
                                  Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        887.86  24.987f  0.001   0.660
------------------------------------------------------------------------------
               Righting Arms:                  0.000  -0.001s
               External Arms:                  0.000  -0.001
      Residual Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025          708.7  24.997f  0.005p   151.81  15.121*
                                       MT/cm--------m.-MT/cm-------GML-----GMT
                                        7.26            26.47   149.04  12.350
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.
```

```
limmarg: 3014.26; t3: 30.00; t: 15.00
limmarg: 1000.30; t3: 15.00; t: 7.50
limmarg: 188.14; t3: 7.50; t: 3.75
limmarg: -27.99; t3: 3.75; t: 1.88
limmarg: 65.33; t3: 5.68; t: 0.94
limmarg: 15.09; t3: 4.74; t: 0.47
limmarg: -6.63; t3: 4.27; t: 0.23
limmarg: 5.10; t3: 4.53; t: 0.12
limmarg: -0.40; t3: 4.41; t: 0.06
limmarg: 1.87; t3: 4.46; t: 0.03
limmarg: 0.51; t3: 4.43; t: 0.01
```

Theta3 is 4.43 degrees, as demonstrated below.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.288	0.00a	4.41s	887.97	0.000	-0.974	0.0000
1.297	0.00a	0.00	887.97	0.000	0.000	-0.0375
1.288	0.00a	4.43p	887.97	0.000	0.977	0.0002
1.286	0.00a	5.00p	887.97	0.000	1.103	0.0106
1.247	0.00a	10.00p	887.96	0.000	2.169	0.1536
1.128	0.01a	15.00p	887.97	0.000	2.853	0.3755
0.991	0.01a	19.36p	887.97	0.000	2.979	0.6000
0.970	0.01a	20.00p	887.97	0.000	2.977	0.6334
0.805	0.01a	25.00p	887.97	0.000	2.863	0.8908
0.633	0.01a	30.00p	887.78	0.000	2.641	1.1317

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM	STABILITY CRITERION	Min/Max	Attained
(1)	Res. Area Ratio from abs -4.4 deg to abs 4.4	> 1.000	1.005 P

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

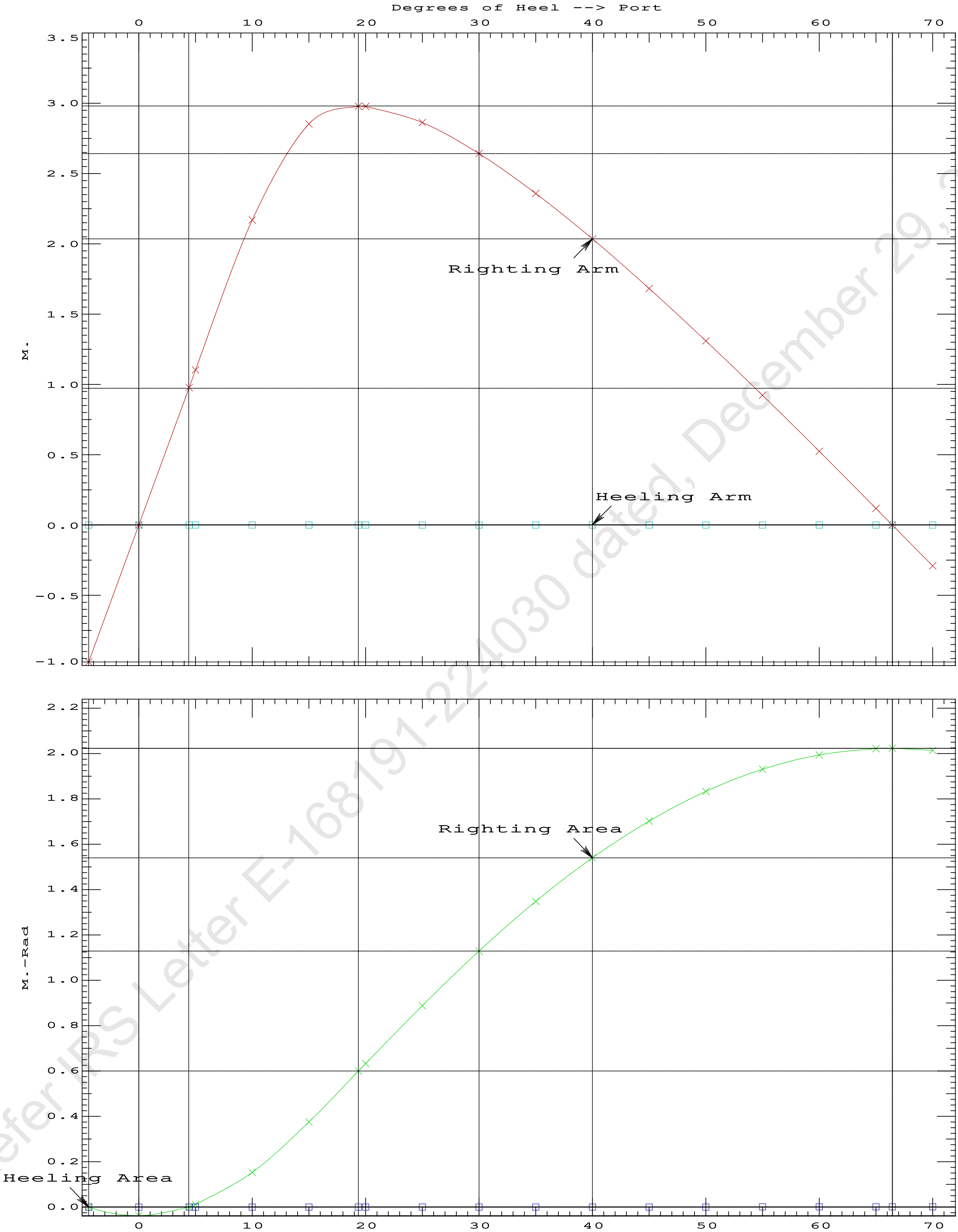
Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.288	0.00a	4.41s	887.97	0.000	-0.974	0.0000
1.297	0.00a	0.00	887.97	0.000	0.000	-0.0375
1.288	0.00a	4.43p	887.97	0.000	0.977	0.0002
1.286	0.00a	5.00p	887.97	0.000	1.103	0.0106
1.247	0.00a	10.00p	887.96	0.000	2.169	0.1536
1.128	0.01a	15.00p	887.97	0.000	2.853	0.3755
0.991	0.01a	19.36p	887.97	0.000	2.979	0.6000
0.970	0.01a	20.00p	887.97	0.000	2.977	0.6334
0.805	0.01a	25.00p	887.97	0.000	2.863	0.8908
0.633	0.01a	30.00p	887.78	0.000	2.641	1.1317
0.458	0.01a	35.00p	887.97	0.000	2.357	1.3502
0.279	0.02a	40.00p	887.97	0.000	2.034	1.5422
0.097	0.02a	45.00p	887.97	0.000	1.682	1.7045
-0.085	0.02a	50.00p	887.97	0.000	1.310	1.8352
-0.267	0.02a	55.00p	887.97	0.000	0.922	1.9327
-0.446	0.02a	60.00p	887.97	0.000	0.523	1.9958
-0.622	0.02a	65.00p	887.97	0.000	0.118	2.0238
-0.672	0.02a	66.45p	887.97	0.000	0.000	2.0253
-0.794	0.03a	70.00p	887.97	0.000	-0.290	2.0163

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM---RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1) Res. Area Ratio from abs -4.4 deg to RAzero	>	1.000		54.939 P
(2) Angle from abs 4.4 deg to RAzero	>	20.00	deg	62.02 P


Degrees of Heel --> Port
0
10
20
30
40
50
60
70
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
M.
Righting Arm
Heeling Arm
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
M.-Rad
Righting Area
Heeling Area

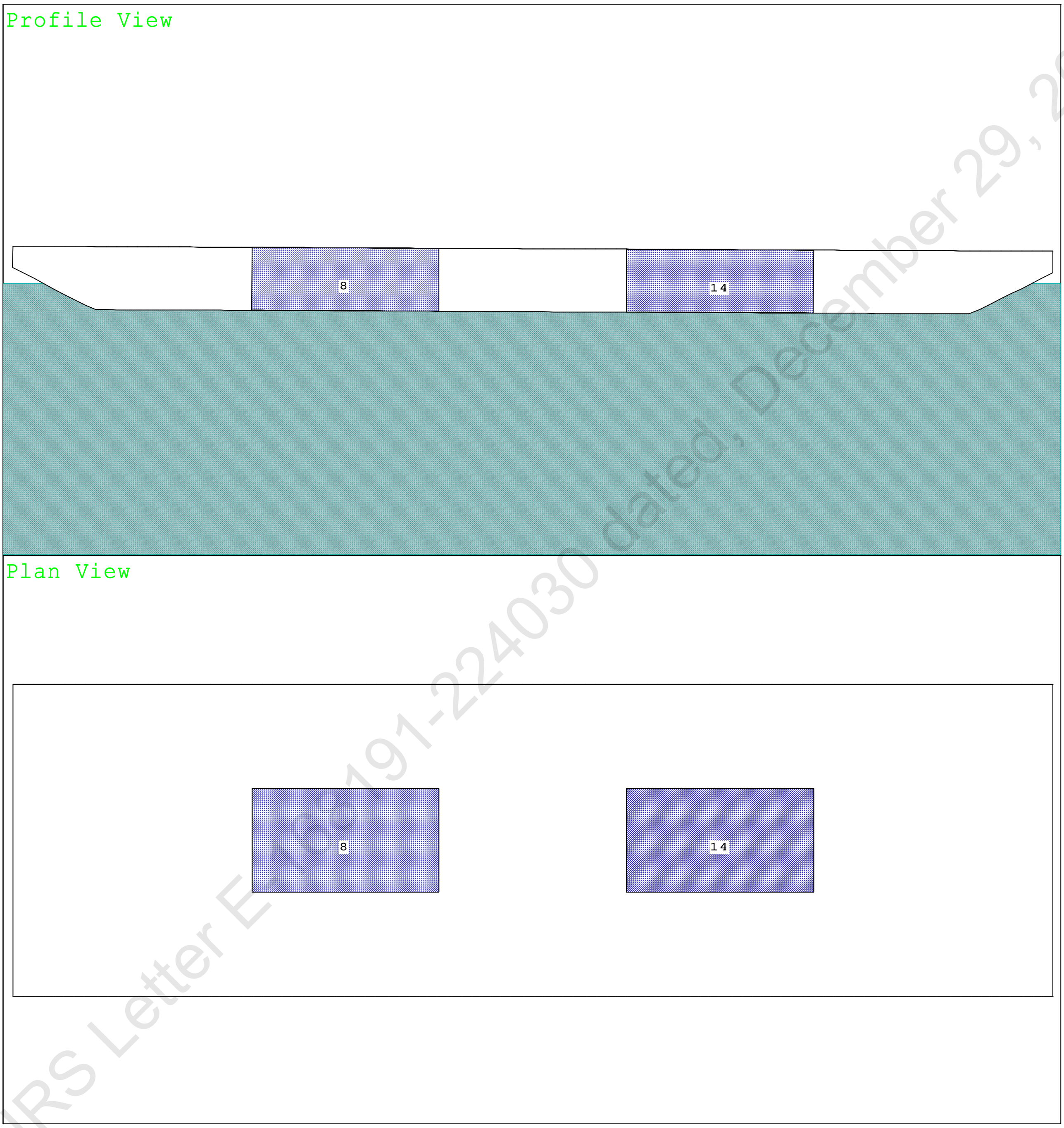
USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 1.480 @ 50.000f, 1.234 @ 0.000 Heel: zero



Tanks	
	8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

```
                WEIGHT and DISPLACEMENT and WATERPLANE STATUS
           USK draft: 1.480 @ 50.00f, 1.357 @ 25.00f, 1.234 @ 0.00
                     Trim: Fwd 0.246/50.000,  Heel: zero
Part----------------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                              373.37  24.970f  0.000   3.000
Crane                                   250.00  25.000f  0.000   6.120
Crane Load                               36.60  43.000f  0.000  76.930
  Total Fixed--------->                 659.97  25.981f  0.000   8.282
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000           132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000           132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->                 264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->                 924.57  25.700f  0.000   6.341
                                     Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025           924.53  25.728f  0.000   0.688
------------------------------------------------------------------------------
               Righting Arms:                    0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025            712.3  25.228f  0.000    148.04  14.609*
                                        MT/cm--------m.-MT/cm-------GML-----GMT
                                          7.30             26.33    142.39   8.956
Distances in METERS.-----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


                            SUMMARY OF LOADING
       264.6 Cu.M.  (13%) FRESH WATER

      250.00 MT  Crane                          36.60 MT  Crane Load

DRAFT AT FP (M) =     1.480

DRAFT AT MS (M) =     1.357

DRAFT AT AP (M) =     1.234

DRAFT AT FWD MARK (M) =     1.460

DRAFT AT AFT MARK (M) =     1.254

                          HYDROSTATIC PROPERTIES
                 Trim: Fwd 0.246/50.000, No Heel, VCG = 6.341

   LCF   Displacement   Buoyancy-Ctr.  Weight/          Moment/
  Draft----Weight(MT)----LCB-----VCB-------cm-----LCF---cm trim----GML-----GMT
  1.358       924.53  25.728f  0.688     7.30  25.228f   26.33  142.39   8.956
 Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
                            Trim is per 50.00m.
 Draft is from USK.                              Formal Free Surface included.

      Note:  GMT includes the formal free surface moment 183.7 m.-MT
```

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.700f TCG = 0.000 VCG = 6.341
Free Surface Adjustment: 0.199
Adjusted CG: LCG = 25.699f TCG = 0.000 VCG = 6.540

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area	50% DeckImm Angle
1.222	0.28f	0.00	924.53	0.000	-0.014	0.0000	5.942
1.222	0.28f	0.08s	924.57	0.000	0.000	-0.0000	5.857
1.211	0.28f	5.08s	924.57	0.000	0.786	0.0343	0.857
1.207	0.28f	5.94s	924.57	0.000	0.922	0.0471	0.000
1.169	0.29f	10.08s	924.57	0.000	1.544	0.1363	-4.143
1.027	0.37f	15.08s	924.57	0.000	1.944	0.2903	-9.143
1.010	0.38f	15.55s	924.57	0.000	1.946	0.3060	-9.604
0.834	0.49f	20.08s	924.57	0.000	1.791	0.4569	-14.143
0.632	0.62f	25.08s	924.57	0.000	1.421	0.5995	-19.143
0.423	0.75f	30.08s	924.57	0.000	0.958	0.7039	-24.143
0.208	0.88f	35.08s	924.57	0.000	0.449	0.7656	-29.143
0.026	0.98f	39.30s	924.64	0.000	-0.000	0.7823	-33.361
-0.009	1.00f	40.08s	924.57	0.000	-0.084	0.7817	-34.143
-0.226	1.12f	45.08s	924.57	0.000	-0.630	0.7507	-39.143
-0.442	1.23f	50.08s	924.57	0.000	-1.178	0.6718	-44.143
-0.654	1.33f	55.08s	924.57	0.000	-1.722	0.5453	-49.143
-0.859	1.41f	60.08s	924.57	0.000	-2.257	0.3716	-54.143

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

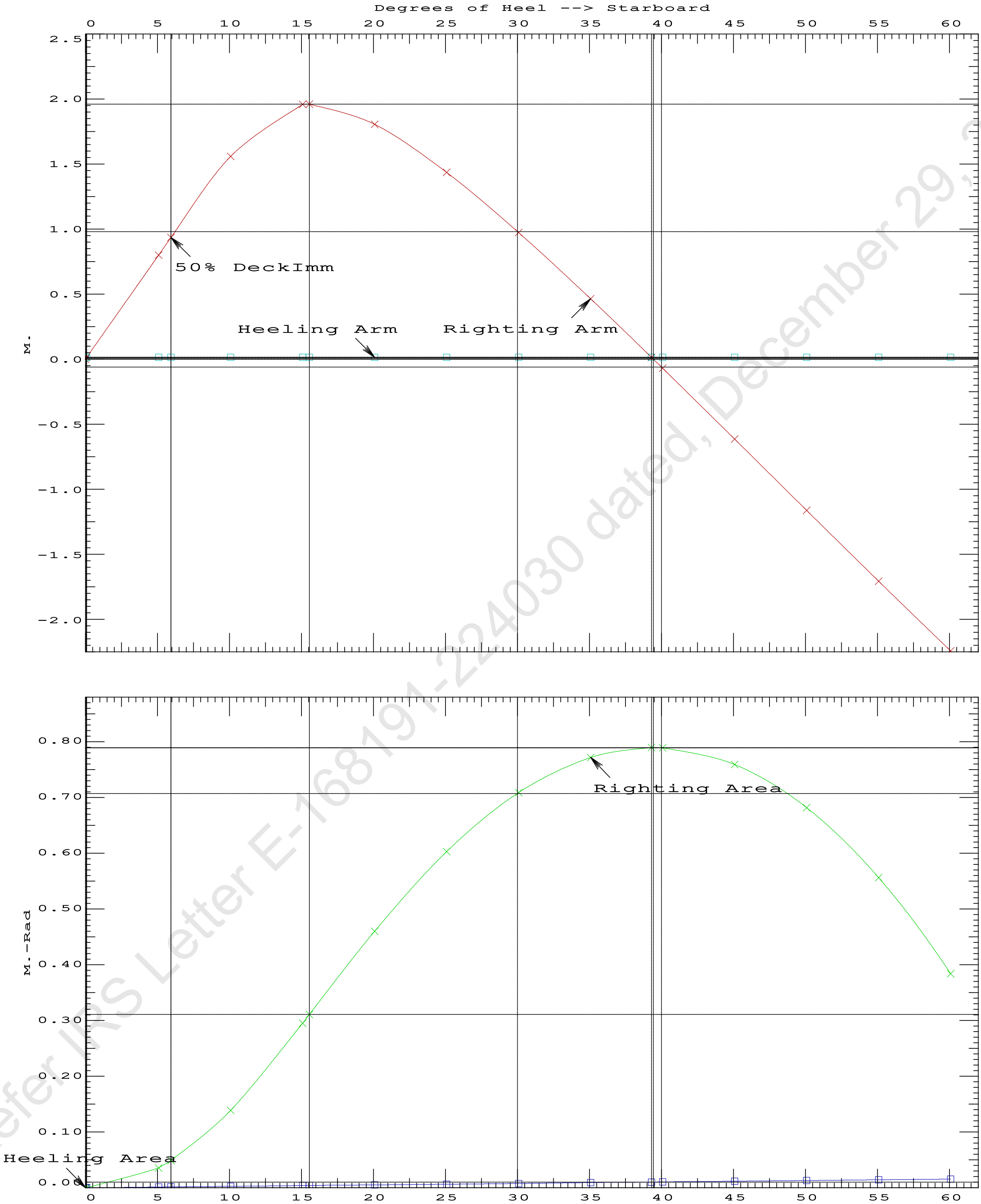
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.58 (constant)

LIM------------IS CODE 2008 PART B CHAPTER-2 SEC.2		Min/Max		Attained	
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.3060	P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	39.22	P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	5.86	P

Condition 4 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd



Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

```
   Total CG:  LCG = 25.700f  TCG = 0.000   VCG =  6.341
                        Free Surface Adjustment:   0.199
Adjusted CG:  LCG = 25.699f  TCG = 0.000   VCG =  6.540
```

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.222	0.28f	0.00	924.45	0.000	0.000	0.0000
1.211	0.28f	5.00p	924.50	0.000	0.786	0.0343
1.170	0.29f	10.00p	924.57	0.000	1.545	0.1363
1.030	0.36f	15.00p	924.57	0.000	1.956	0.2916
1.009	0.38f	15.56p	924.57	0.000	1.960	0.3107
0.837	0.49f	20.00p	924.57	0.000	1.809	0.4595
0.635	0.62f	25.00p	924.57	0.000	1.441	0.6037
0.426	0.75f	30.00p	924.57	0.000	0.979	0.7100
0.212	0.88f	35.00p	924.57	0.000	0.471	0.7737
0.020	0.99f	39.43p	924.62	0.000	0.000	0.7920
-0.005	1.00f	40.00p	924.57	0.000	-0.062	0.7917
-0.223	1.12f	45.00p	924.57	0.000	-0.607	0.7626
-0.438	1.23f	50.00p	924.57	0.000	-1.155	0.6857
-0.650	1.33f	55.00p	924.57	0.000	-1.700	0.5611
-0.856	1.41f	60.00p	924.57	0.000	-2.235	0.3893

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

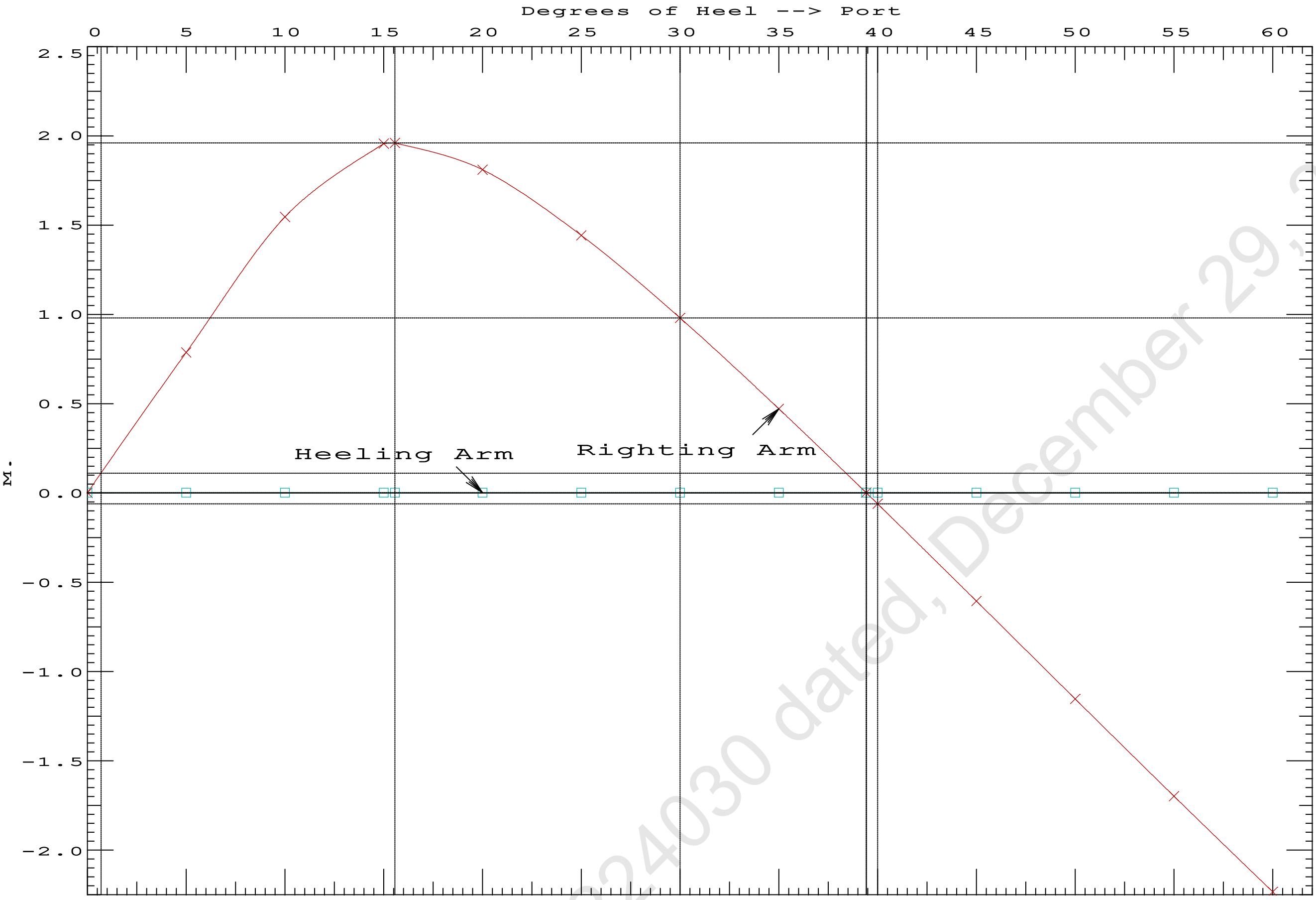
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

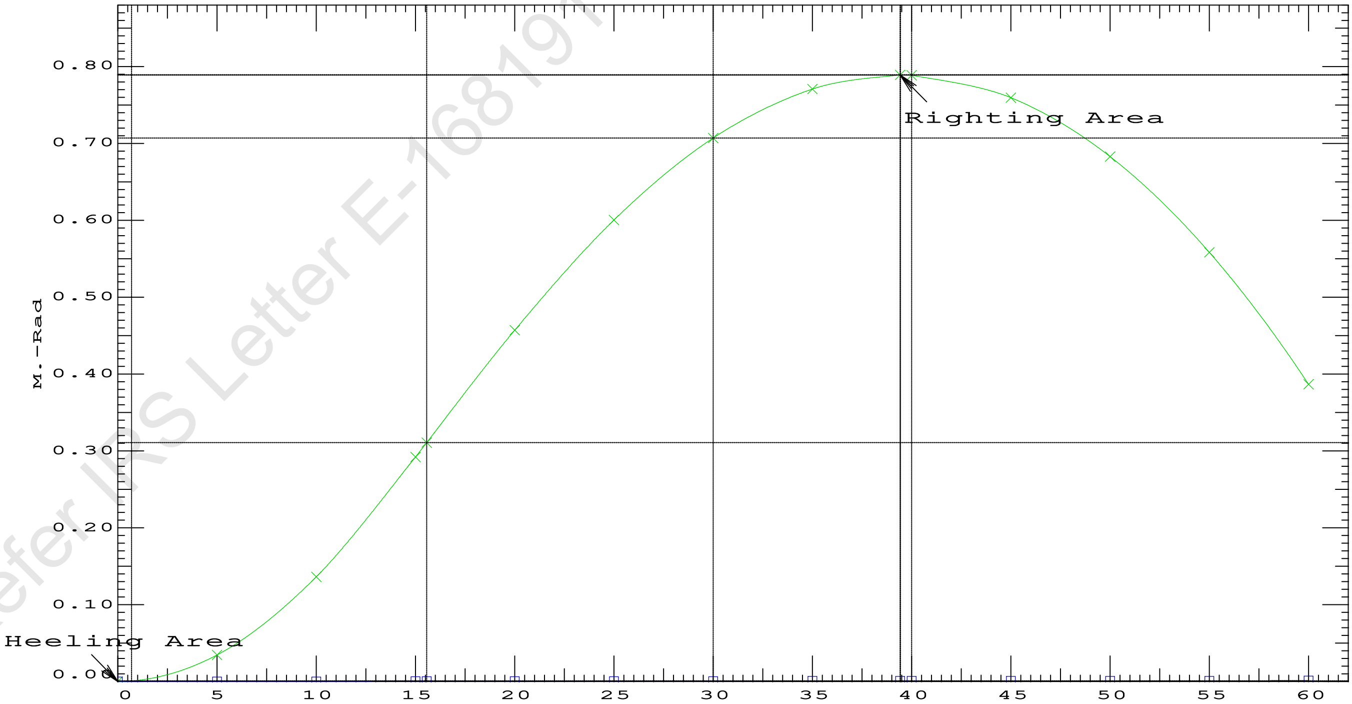
LIM------------STABILITY CRITERIA FOR CRANE OPERAT		Min/Max		Attained
(1) Absolute Angle at Equilibrium	<	15.00	deg	0.00 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7	%	181152 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667		1065.13 P

PONTOON

Condition 4 : Lightship + Load at Fwd + Ballast

Crane Lifting 36.60T @ 18m Radius Fwd





USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

Condition 4 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd
Crane in operation with load in place
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.480 @ 50.00f, 1.357 @ 25.00f, 1.234 @ 0.00
Trim: Fwd 0.246/50.000, Heel: 0.00 deg.

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Crane Load			36.60	43.000f	0.000	76.930	
Total Fixed--------->			659.97	25.981f	0.000	8.282	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			924.57	25.700f	0.000	6.341	

Part	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	924.45	25.728f	0.001p	0.688

Righting Arms:	0.000	-0.001s
External Arms:	0.000	-0.001p
Residual Righting Arms:	0.000	0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	709.2	25.123f	0.017p	146.10	14.562*

MT/cm	m.-MT/cm	GML	GMT
7.27	25.97	140.45	8.909

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 0.00

```
                    After hook load drop
          WEIGHT and DISPLACEMENT and WATERPLANE STATUS
     USK draft: 1.305 @ 50.00f, 1.307 @ 25.00f, 1.309 @ 0.00
            Trim: Aft 0.004/50.000,  Heel: 0.00 deg.
Part-----------------------------Weight(MT)----LCG-----TCG-----VCG
LIGHT SHIP                           373.37  24.970f  0.000   3.000
Crane                                250.00  25.000f  0.000   6.120
  Total Fixed--------->              623.37  24.982f  0.000   4.251
                Load-----SpGr-----Weight(MT)----LCG-----TCG-----VCG-------FSM
03_PFWBTK.C    1.000    1.000        132.30  16.000f  0.000   1.500     91.87*
05_PFWBTK.C    1.000    1.000        132.30  34.000f  0.000   1.500     91.87*
  Total Tanks--------->              264.60  25.000f  0.000   1.500    183.75
  Total Weight-------->              887.97  24.987f  0.000   3.431
                                   Displ(MT)----LCB-----TCB-----VCB
HULL                    1.025        887.97  24.987f  0.001   0.660
-----------------------------------------------------------------------------
               Righting Arms:                  0.000   0.001s
               External Arms:                  0.000   0.001
      Residual Righting Arms:                  0.000   0.000
Part---------------------SpGr-----------WPA-----LCF-----TCF-------BML-----BMT
  Total Waterplane----> 1.025         708.7  24.997f  0.005s   151.80  15.119*
                                      MT/cm--------m.-MT/cm-------GML-----GMT
                                       7.26             26.47    149.03  12.348
Distances in METERS.----------------------------------------Moments in m.-MT.

    Note:  FSM values marked with an asterisk (*) are formal values which are
           not the same as the true values in the present condition.


limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

              Theta3 is 0.07 degrees, as demonstrated below.
```

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM	STABILITY CRITERION		Min/Max	Attained
(1)	Residual Area Ratio from abs 0 deg to abs 0.1	>	1.000	LARGE

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Fixed CG: LCG = 24.982f TCG = 0.000 VCG = 4.251

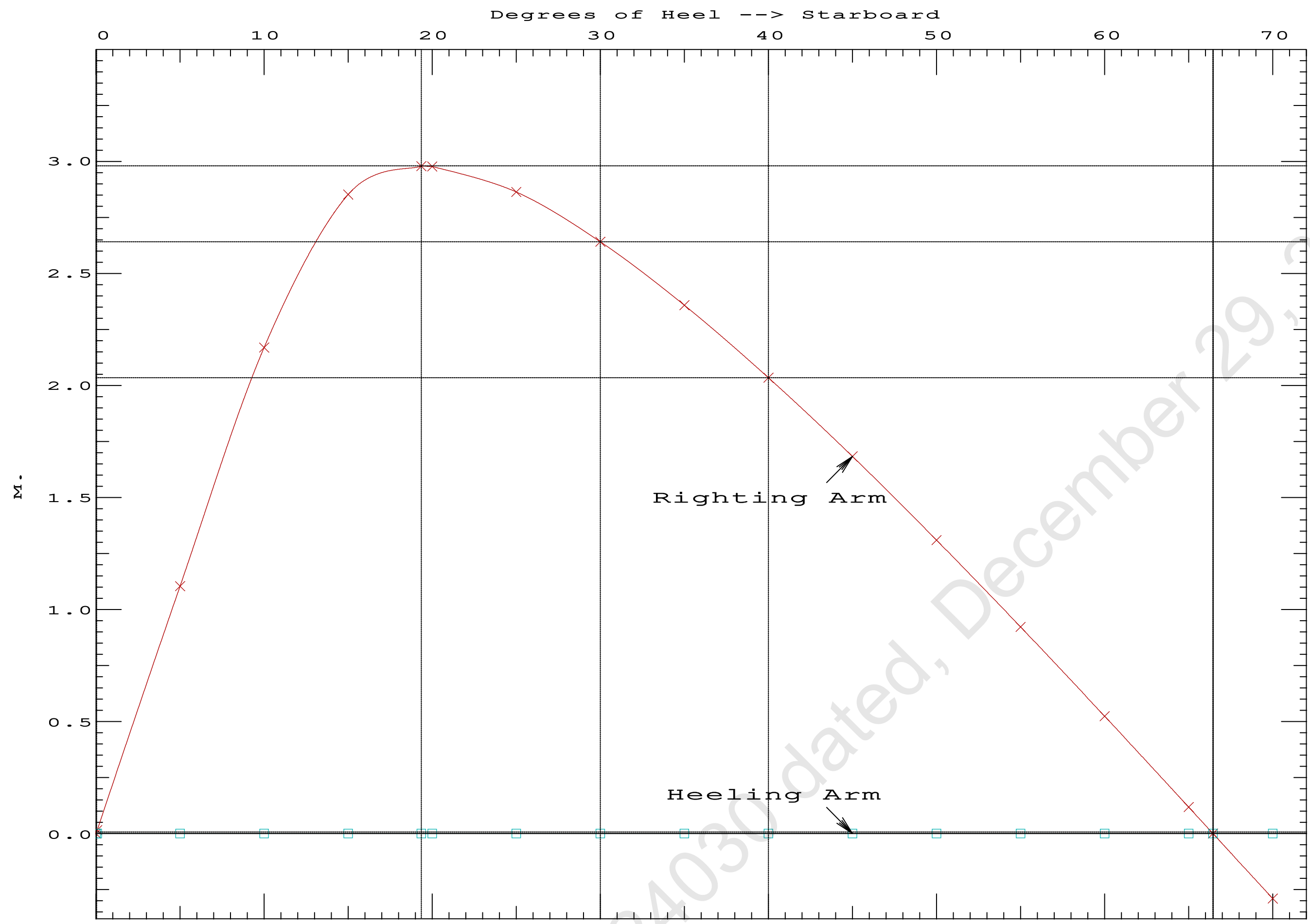
Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. Area
1.297	0.00a	0.00	887.97	0.000	0.000	0.0000
1.297	0.00a	0.07s	887.97	0.000	0.015	0.0000
1.286	0.00a	5.00s	887.97	0.000	1.103	0.0481
1.247	0.00a	10.00s	887.96	0.000	2.169	0.1912
1.128	0.01a	15.00s	887.97	0.000	2.853	0.4131
0.991	0.01a	19.36s	887.97	0.000	2.979	0.6377
0.970	0.01a	20.00s	887.97	0.000	2.977	0.6711
0.805	0.01a	25.00s	887.97	0.000	2.863	0.9284
0.633	0.01a	30.00s	887.78	0.000	2.641	1.1693
0.458	0.01a	35.00s	887.97	0.000	2.357	1.3879
0.279	0.02a	40.00s	887.97	0.000	2.034	1.5798
0.097	0.02a	45.00s	887.97	0.000	1.682	1.7422
-0.085	0.02a	50.00s	887.96	0.000	1.310	1.8729
-0.267	0.02a	55.00s	887.97	0.000	0.922	1.9704
-0.446	0.02a	60.00s	887.97	0.000	0.523	2.0335
-0.622	0.02a	65.00s	887.97	0.000	0.118	2.0615
-0.672	0.02a	66.45s	887.97	0.000	0.000	2.0629
-0.794	0.03a	70.00s	887.97	0.000	-0.290	2.0540

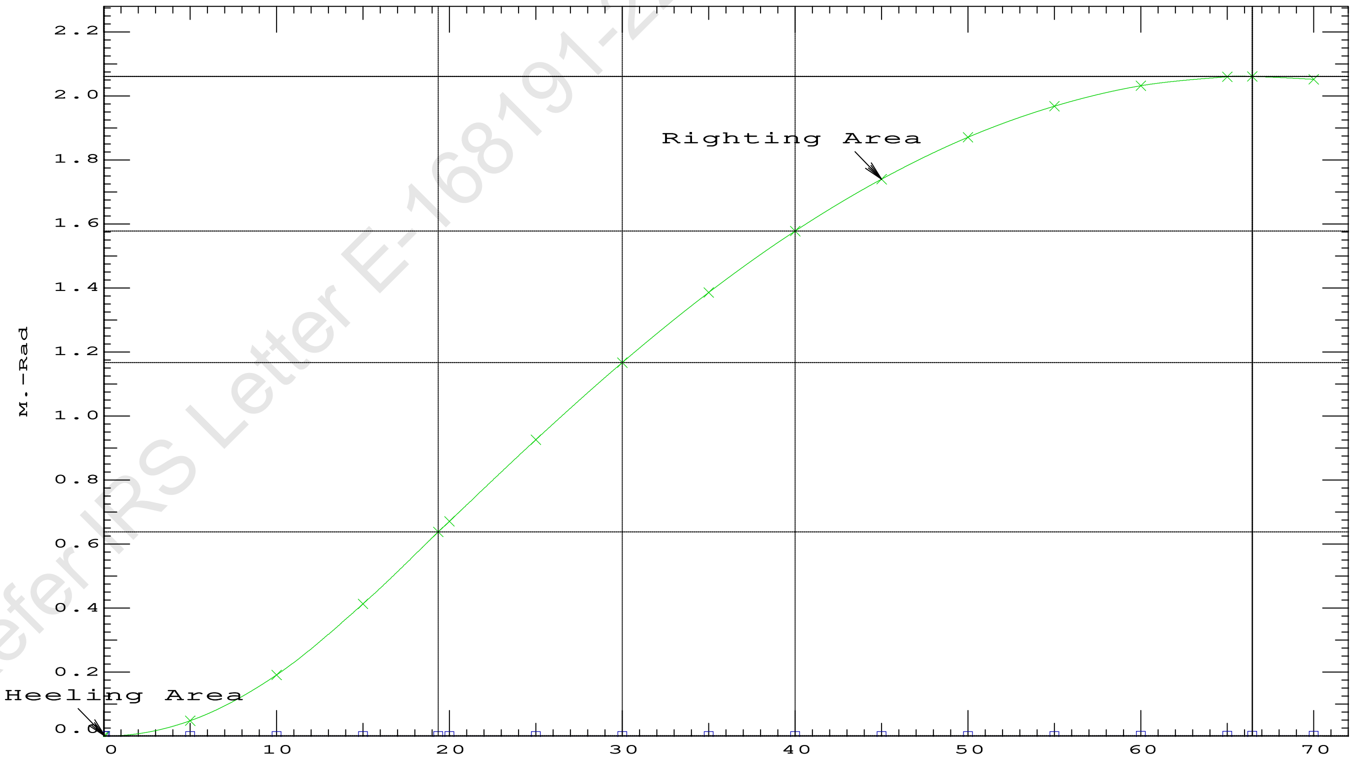
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 623.37 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM	RINA PT E, CH 19, SEC 2: 2.2.3		Min/Max		Attained
(1)	Residual Area Ratio from abs 0 deg to RAzero	>	1.000		LARGE
(2)	Angle from abs 0.1 deg to RAzero	>	20.00	deg	66.38 P


Degrees of Heel --> Starboard
0
10
20
30
40
50
60
70
M.
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Righting Arm
Heeling Arm


M.-Rad
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Righting Area
Heeling Area
0
10
20
30
40
50
60
70

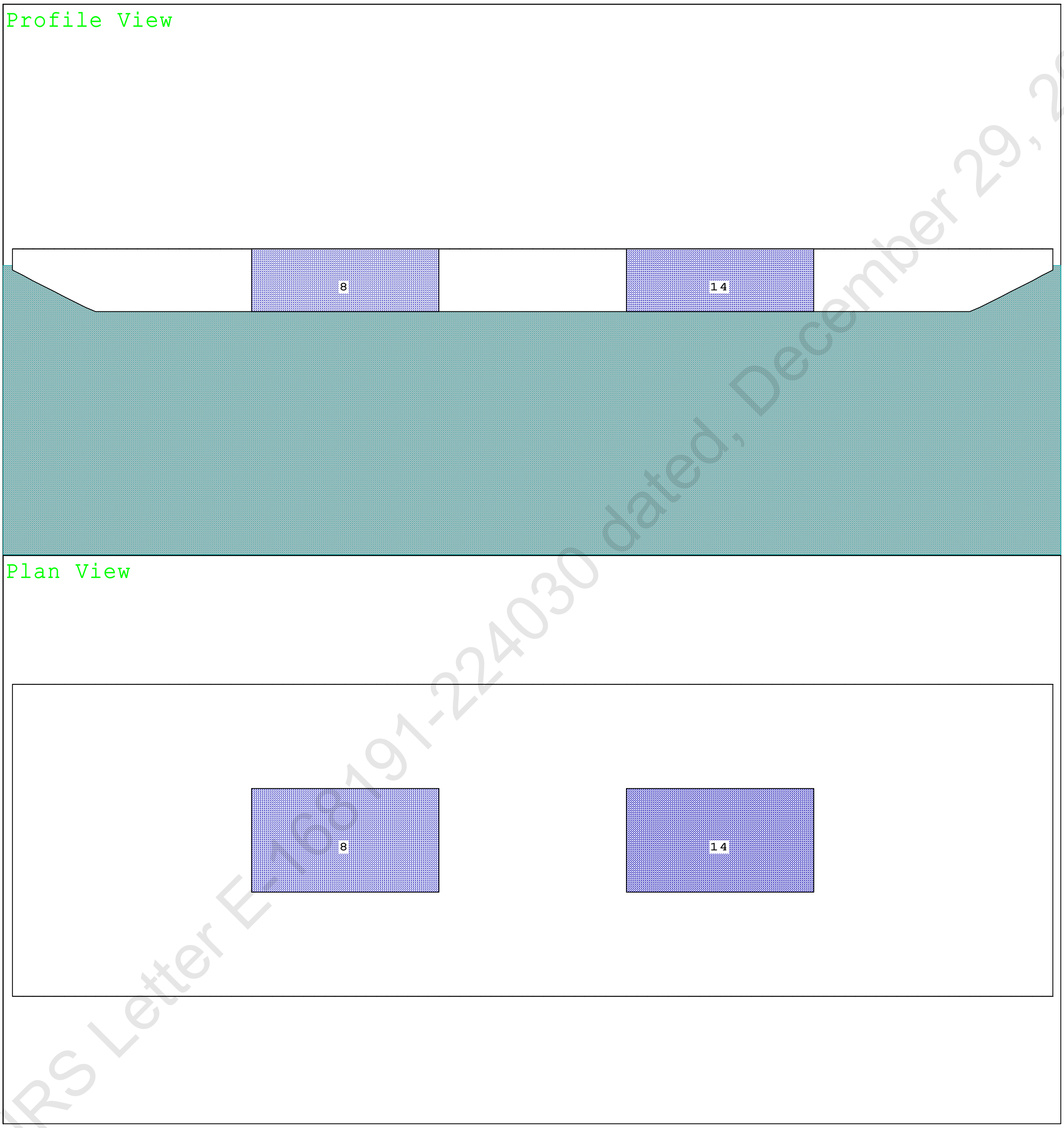
USK draft refers to the line:
0.012 below baseline @ 50.000f and 0.012 below baseline @ 0.000

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition

Condition Graphic - Draft: 2.210 @ 50.000f, 2.214 @ 0.000 Heel: zero



Tanks
8 03_PFWBTK.C.100% FRESH WATER 14 05_PFWBTK.C.100% FRESH WATER

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.210 @ 50.00f, 2.212 @ 25.00f, 2.214 @ 0.00
Trim: Aft 0.004/50.000, Heel: zero

Part	Load	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
LIGHT SHIP			373.37	24.970f	0.000	3.000	
Crane			250.00	25.000f	0.000	6.120	
Deck Cargo			681.18	25.000f	0.000	3.500	
Total Fixed--------->			1,304.55	24.991f	0.000	3.859	
03_PFWBTK.C	1.000	1.000	132.30	16.000f	0.000	1.500	91.87*
05_PFWBTK.C	1.000	1.000	132.30	34.000f	0.000	1.500	91.87*
Total Tanks--------->			264.60	25.000f	0.000	1.500	183.75
Total Weight-------->			1,569.15	24.993f	0.000	3.461	

	SpGr	Displ(MT)	LCB	TCB	VCB
HULL	1.025	1,569.15	24.993f	0.000	1.134

Righting Arms: 0.000 0.000

Part	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	750.6	25.000f	0.000	102.05	9.092*
		MT/cm	m.-MT/cm		GML	GMT
		7.69		31.30	99.73	6.766

Distances in METERS.--Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

264.6 Cu.M. (13%) FRESH WATER

250.00 MT Crane 681.18 MT Deck Cargo

DRAFT AT FP (M) = 2.210

DRAFT AT MS (M) = 2.212

DRAFT AT AP (M) = 2.214

DRAFT AT FWD MARK (M) = 2.210

DRAFT AT AFT MARK (M) = 2.214

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition

HYDROSTATIC PROPERTIES
Trim: Aft 0.004/50.000, No Heel, VCG = 3.461

Draft	Displacement Weight(MT)	LCB	VCB	Weight/ cm	LCF	Moment/ cm trim	GML	GMT
2.212	1,569.15	24.993f	1.134	7.69	25.000f	31.30	99.73	6.766

Distances in METERS.-----Specific Gravity = 1.025.-----------Moment in m.-MT.
Trim is per 50.00m.
Draft is from USK. Formal Free Surface included.

Note: GMT includes the formal free surface moment 183.7 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
Total CG: LCG = 24.993f TCG = 0.000 VCG = 3.461
Free Surface Adjustment: 0.117
Adjusted CG: LCG = 24.993f TCG = 0.000 VCG = 3.578

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel--->	Res. Area	50% DeckImm Angle
2.202	0.00a	0.00	1,569.1	0.000	-0.006	0.0000	3.079
2.202	0.00a	0.05s	1,569.1	0.000	0.000	-0.0000	3.030
2.198	0.00a	3.08s	1,569.1	0.000	0.359	0.0095	0.000
2.192	0.00a	5.05s	1,569.1	0.000	0.591	0.0258	-1.970
2.223	0.01a	10.05s	1,569.1	0.000	0.984	0.0959	-6.970
2.353	0.01a	14.58s	1,569.1	0.000	1.049	0.1781	-11.500
2.369	0.01a	15.05s	1,569.1	0.000	1.049	0.1867	-11.970
2.562	0.01a	20.05s	1,569.1	0.000	0.985	0.2768	-16.970
2.768	0.01a	25.05s	1,569.1	0.000	0.851	0.3574	-21.970
2.955	0.01a	30.05s	1,569.2	0.000	0.665	0.4239	-26.970
3.120	0.01a	35.05s	1,569.1	0.000	0.452	0.4729	-31.970
3.262	0.02a	40.05s	1,569.1	0.000	0.223	0.5025	-36.970
3.372	0.02a	44.74s	1,569.1	0.000	0.000	0.5117	-41.657
3.378	0.02a	45.05s	1,569.1	0.000	-0.015	0.5116	-41.970
3.469	0.02a	50.05s	1,569.1	0.000	-0.258	0.4998	-46.970
3.534	0.02a	55.05s	1,569.1	0.000	-0.502	0.4666	-51.970
3.571	0.02a	60.05s	1,569.1	0.000	-0.744	0.4123	-56.970

Distances in METERS.----Specific Gravity = 1.025.---------Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 183.7 m.-MT was applied to artificially modify the CG.

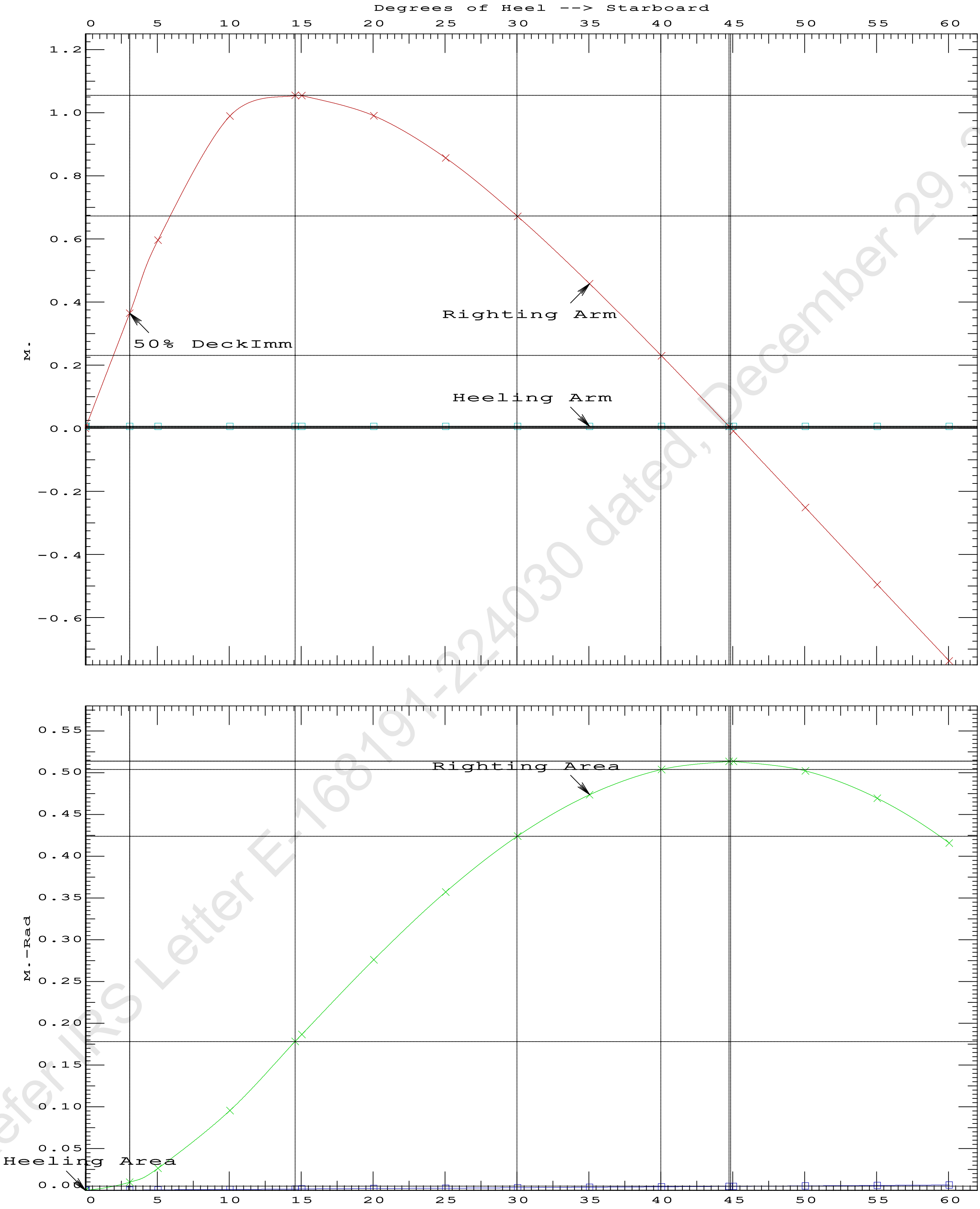
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.39 (constant)

LIM------------INTACT CRITERIA-IS CODE 2008PART B		----Min/Max		--------Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800	m.-Rad	0.1781 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00	deg	44.69 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00	deg	3.03 P

PONTOON

Condition 5 : Lightship + Crane + Ballast + Deck Cargo

Towing Condition



LIGHTSHIP PARTICULARS

Lightship Weight	=	373.37	T
VCG	=	3.000	M above Baseline
LCG	=	24.970	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved draft survey report.)

LATERAL PLANE STATUS
USK draft: 0.500 @ 50.00f, 0.500 @ 25.00f, 0.500 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	21.6	25.000f	-0.248	121.7	25.000f	1.289
CRANE.C				35.2	7.500f	4.712
DECKCARGO.C				40.0	25.000f	3.512
Total Lateral Plane->	21.6	25.000f	-0.248	196.9	21.871f	2.352

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	26.0	25.000f	-0.297	117.2	25.000f	1.236
CRANE.C				35.2	7.500f	4.612
DECKCARGO.C				40.0	25.000f	3.412
Total Lateral Plane->	26.0	25.000f	-0.297	192.4	21.799f	2.306

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	30.5	25.000f	-0.346	112.8	25.000f	1.183
CRANE.C				35.2	7.500f	4.512
DECKCARGO.C				40.0	25.000f	3.312
Total Lateral Plane->	30.5	25.000f	-0.346	188.0	21.723f	2.259

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	35.0	25.000f	-0.395	108.3	25.000f	1.130
CRANE.C				35.2	7.500f	4.412
DECKCARGO.C				40.0	25.000f	3.212
Total Lateral Plane->	35.0	25.000f	-0.395	183.5	21.642f	2.214

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	39.6	25.000f	-0.443	103.7	25.000f	1.077
CRANE.C				35.2	7.500f	4.312
DECKCARGO.C				40.0	25.000f	3.112
Total Lateral Plane->	39.6	25.000f	-0.443	178.9	21.557f	2.169

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	44.1	25.000f	-0.492	99.1	25.000f	1.025
CRANE.C				35.2	7.500f	4.212
DECKCARGO.C				40.0	25.000f	3.012
Total Lateral Plane->	44.1	25.000f	-0.492	174.3	21.467f	2.124

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	48.8	25.000f	-0.540	94.5	25.000f	0.972
CRANE.C				35.2	7.500f	4.112
DECKCARGO.C				40.0	25.000f	2.912
Total Lateral Plane->	48.8	25.000f	-0.540	169.7	21.370f	2.081

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	53.4	25.000f	-0.589	89.8	25.000f	0.920
CRANE.C				35.2	7.500f	4.012
DECKCARGO.C				40.0	25.000f	2.812
Total Lateral Plane->	53.4	25.000f	-0.589	165.0	21.267f	2.038

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	58.2	25.000f	-0.637	85.1	25.000f	0.868
CRANE.C				35.2	7.500f	3.912
DECKCARGO.C				40.0	25.000f	2.712
Total Lateral Plane->	58.2	25.000f	-0.637	160.3	21.158f	1.997

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	62.9	25.000f	-0.685	80.4	25.000f	0.817
CRANE.C				35.2	7.500f	3.812
DECKCARGO.C				40.0	25.000f	2.612
Total Lateral Plane->	62.9	25.000f	-0.685	155.6	21.040f	1.956

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	67.7	25.000f	-0.733	75.6	25.000f	0.765
CRANE.C				35.2	7.500f	3.712
DECKCARGO.C				40.0	25.000f	2.512
Total Lateral Plane->	67.7	25.000f	-0.733	150.8	20.915f	1.916

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	72.5	25.000f	-0.781	70.8	25.000f	0.714
CRANE.C				35.2	7.500f	3.612
DECKCARGO.C				40.0	25.000f	2.412
Total Lateral Plane->	72.5	25.000f	-0.781	146.0	20.780f	1.878

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	77.4	25.000f	-0.829	65.9	25.000f	0.663
CRANE.C				35.2	7.500f	3.512
DECKCARGO.C				40.0	25.000f	2.312
Total Lateral Plane->	77.4	25.000f	-0.829	141.1	20.634f	1.841

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	82.3	25.000f	-0.876	61.0	25.000f	0.612
CRANE.C				35.2	7.500f	3.412
DECKCARGO.C				40.0	25.000f	2.212
Total Lateral Plane->	82.3	25.000f	-0.876	136.2	20.477f	1.806

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	87.2	25.000f	-0.924	56.0	25.000f	0.561
CRANE.C				35.2	7.500f	3.312
DECKCARGO.C				40.0	25.000f	2.112
Total Lateral Plane->	87.2	25.000f	-0.924	131.2	20.307f	1.772

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	92.2	25.000f	-0.971	51.1	25.000f	0.511
CRANE.C				35.2	7.500f	3.212
DECKCARGO.C				40.0	25.000f	2.012
Total Lateral Plane->	92.2	25.000f	-0.971	126.3	20.123f	1.739

Distances in METERS.

LATERAL PLANE STATUS
USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00
Trim: zero, Heel: zero

Part	LPA	LCP	HCP	LPA	LCP	HCP
HULL	97.2	25.000f	-1.019	46.1	25.000f	0.461
CRANE.C				35.2	7.500f	3.112
DECKCARGO.C				40.0	25.000f	1.912
Total Lateral Plane->	97.2	25.000f	-1.019	121.3	19.922f	1.709

Distances in METERS.

IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244
52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-168026-223617

07-October-2022

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002816
Project name Specific Yard No(s):	Chisti Marine Engineering Works - CHISTI/023 and CHISTI/025 CHISTI/023
Project Manager	M. Saravanan
Project In-Charge	Srinivas Kundrapu

Document Details	
Plan No.	CM22-VI-112-0208
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	06-October-2022
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

   The following lightship particulars are approved based on the Draft survey for Yard No. CHISTI/023 conducted on 06th October 2022:

   Lightship Weight : 373.37 T
   Lightship L.C.G : 24.970 m forward of AP(Frame no. 0)
   Lightship V.C.G : 3.000 m above Baseline (VCG taken at the level of main deck)
   Lightship T.C.G : 0.000 m off Centreline.

   Final stability booklet prepared based on the approved lightship particulars should be submitted for approval.

   Copy of the approved Draft survey report should form part of the Final Stability booklet.

Thanking you,
Yours Sincerely,
For INDIAN REGISTER OF SHIPPING

Vimal R
Surveyor

This is an electronically published document and does not require signature.

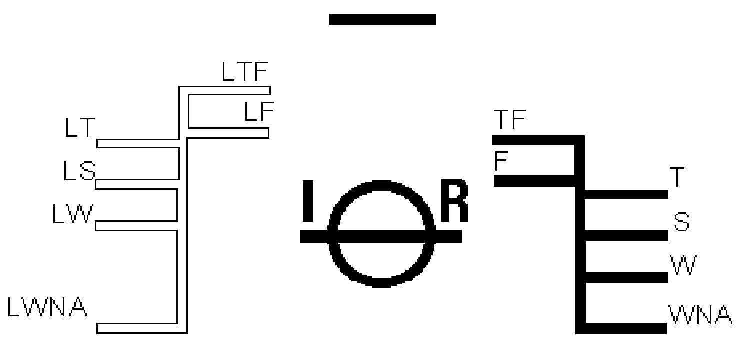
MEMORANDUM OF FREEBOARDS
5 September 2022

Chisti Marine Engineering Works.	Type B freeboard		
Yard No. 023	Freeboard Length	:	48.000 m
	Summer WL Extreme Draught - load line amidships	:	2.212 m

The upper edge of the deckline from which these freeboards are measured is 0 **mm below top of * ~~Wood~~/ * steel Deck * at side / *~~continued to side~~. (*Delete words not applicable).**

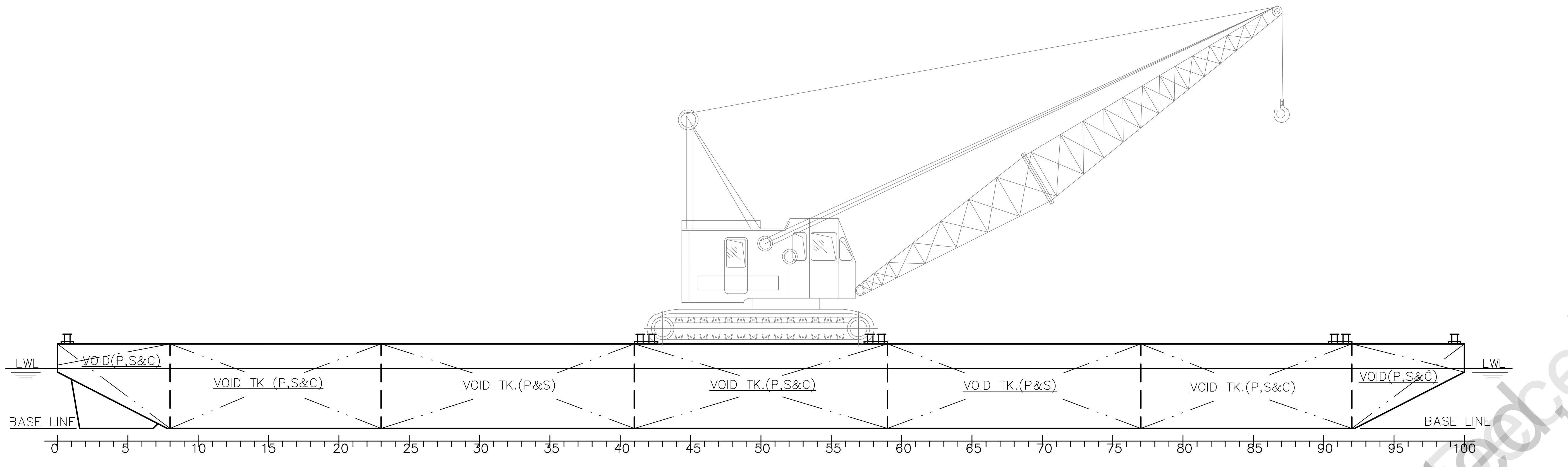
Freeboards from Deck Line			**Seasonal, Fresh Water, or Timber allowances**	
Tropical Fresh Water	713	**mm.**	97	**mm. above summer**
Fresh Water	759	**mm.**	51	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**	**Upper edge of line through centre**	
Winter	856	**mm.**	46	**mm. below summer**
Winter North Atlantic	906	**mm.**	96	**mm. below summer**
Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**mm.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic	—	**mm.**	—	**mm. below timber summer**

Valid for unmanned tow only.

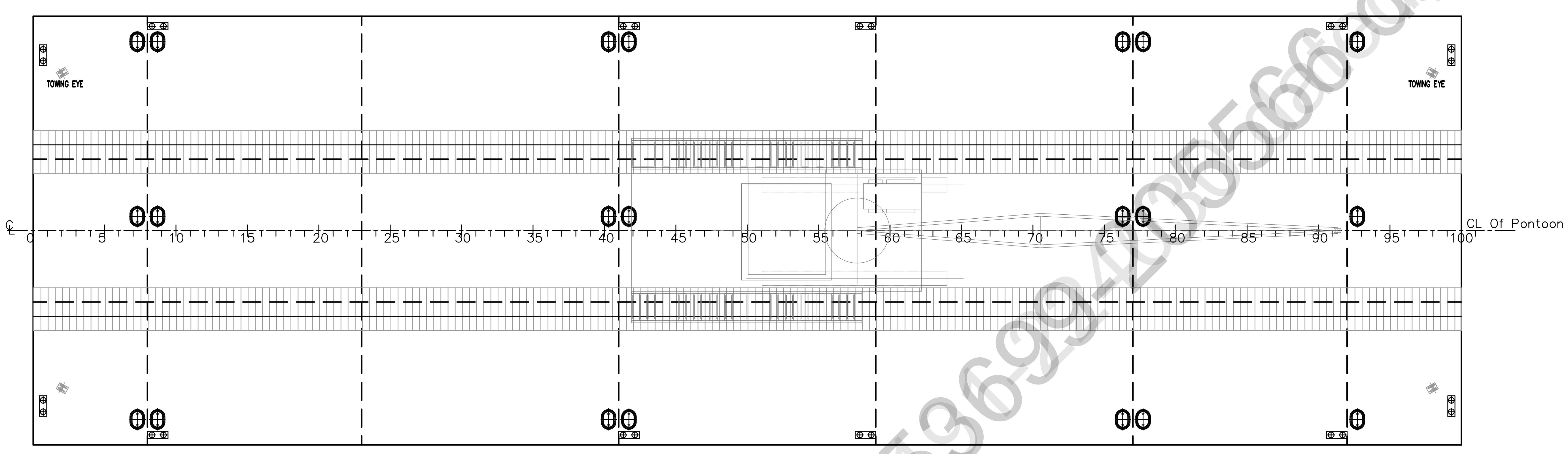


At Fr. 52

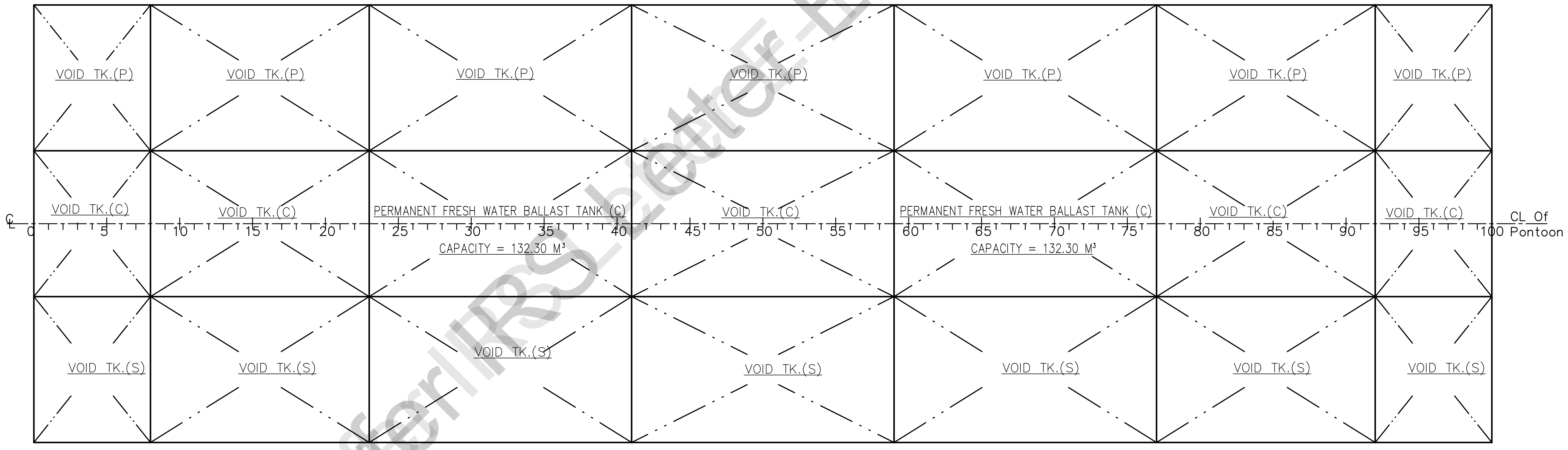
Form No. 4111 (07-83)



PROFILE



MAIN DECK PLAN



BELOW DECK PLAN

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	15.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M
SCANTLING DRAFT	:	2.50 M
FRAME SPG.	:	500 mm

CLASS NOTATION:-

卐 SU PONTOON, DECK STRENGTHNED FOR 10 t/m², FOR CRANE OPERATION

CRANE SPECIFICATION:-

CRANE WEIGHT = 250 T
LIFTING CAPACITY = 36.60 T
OPERATING RADIUS = 18 M
MAKE/MODEL : LIEBHERR LR 1250
BOOM/JIB : 74m MAIN BOOM, NO JIB
CRAWLERS : TWO 7520x1200 TRACKS, SEPARATED BY 7.0m

NOTE : ALL THE TANKS ARE DECLARED AS VOID TANKS HOWEVER THE STRUCTURE FOR THE SAME IS CHECKED FOR FULLY LOADED WITH LIQUIDS.
FOR STABILITY REQUIREMENT THE TWO TANKS DECLARED AS PERMANENT FRESH WATER BALLAST.

SEE LETTER E-153699-205566

INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 05 Jul 2022
NOTED FOR INFORMATION

Rev.#	Description	Date	Drawn	Review

Issued for : Approval

Client: KB SHIPPING & CO. (JAMNAGAR)	Project: PONTOON (YARD NO. CHISTI / 023) AND 025
Builder: CHISTI MARINE ENGINEERING WORKS	

Drg.#	Rev.#	Drawing Title	Drawn	ISHWAR
CM22-VI-112-0001	00	GENERAL ARRANGEMENT	Checked	OMKAR
Scale 1:150 / Size A2 / Date 29.04.2022	Sheet 1 of 1		Review-1	MOHIT
			Review-2	SANJAY

Cybermarine
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Navi Mumbai 400 710 India
Tel : +91 22 61462400 (10 Lines)

#	Drawing / document Name	Drawing #	Rev.
01	GENERAL ARRANGEMENT	CM22–VI–112–0001	00

List of Reference drgs / documents used for drg.

INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 14 Jun 2022
NOTED FOR INFORMATION

SEE LETTER E-153873-205767

Rev.#	Description	Date	Drawn	Review

Issued for : Approval

Client	Project
KB SHIPPING & CO. (JAMNAGAR)	PONTOON (YARD NO. CHISTI / 023)
Builder CHISTI MARINE ENGINEERING WORKS	

Drg.#			Rev.#	Drawing Title	Drawn	ISHWAR
CM22-VI-112-4001			00	DISPOSITION OF DRAFT MARK	Checked	OMKAR
Scale	Size	Date	Sheet		Review-1	SHAILESH
1:50	A4/A3	09.05.2021	1 of 2		Review-2	

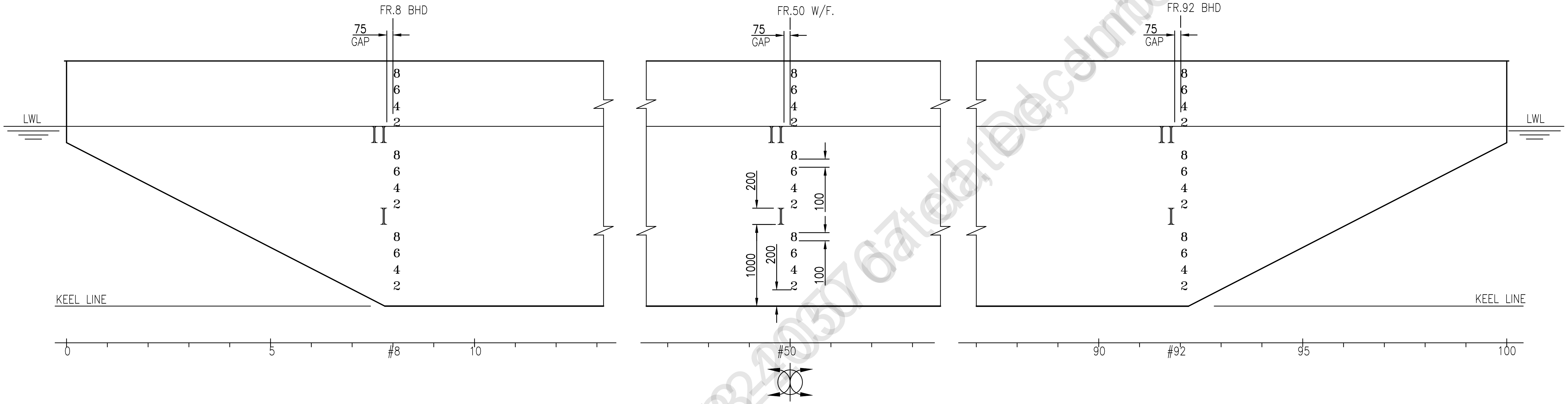
Cybermarine

www.cybermarine.net info@cybermarine.net

10, Bukit Batok Crescent,
#12-01, The Spire Building,
Singapore-658079.

501, Technocity IT Premises Co-op. Soc. Ltd,
Plot No. X-5/3, Mahape,
Maharashtra- 400710, India.

22503, Katy Freeway #7,
Katy, Texas,
USA 77450.



PROFILE
(PORT SHOWN AND STBD SIMILAR)

PRINCIPAL PARTICULARS:-

LENGTH O.A	:	50.00 M
BREADTH MLD.	:	15.00 M
DEPTH MLD.	:	3.00 M
DESIGN DRAFT	:	2.20 M

NOTE:-

1) DRAFT MARKS ARE TO BE CUT OF 6 MM PLT. AND TO BE WELDED TO THE HULL.
2) MARKS ARE TO BE PAINTED WITH COATS OF PRIMER TWO COATS OF ANTICORROSIVE PAINTS AND TWO COATS OF WHITE FINISH

Cybermarine		
Drg.Title :	DISPOSITION OF DRAFT MARK	Sheet No.
Drg.# :	CM22-VI-112-4001	2 OF 2